

SCOTT–VOGELIUS–NITSCHKE ON A POLYGONALLY APPROXIMATED BOUNDARY: CONVERGENCE, THE MISSING PRESSURE TRACTION, AND THE PENALTY THRESHOLD

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ABSTRACT. We analyse the Scott–Vogelius–Nitsche method of Gjerde and Scott for two-dimensional Stokes flow whose no-slip curve is replaced by an inscribed polygon: continuous P_k velocities that are exactly divergence-free, with Dirichlet data imposed weakly on the polygon by Nitsche’s method. This answers Scott’s zero-gradient prize question, which asks whether the H^1 error behaves like $h_\Gamma^{3/2} + h_\Omega^k$ for general Stokes data and, if not, why not.

For the method as printed, the answer is no as soon as the pressure is not constant on the boundary. The printed Nitsche form omits the traction $-pn$. We prove that the method converges, at the sharp rate h/μ in the H^1 seminorm and at the exact rate $(h/\mu)^{1/2}$ in the mesh-dependent energy norm. To leading order the error is an explicit field: the discrete velocity leaks through the wall with normal velocity $(h/\mu)(p - \bar{p})$, with constant exactly 1 in the slip and energy norms. The H^1 rate is 1 and not 1/2 only because incompressibility turns the normal part of the missing traction into a tangential derivative. We also identify the H^1 constant: for fixed γ and bounded ρ , (μ/h) times the error converges to the Stokes flow driven by the leak as Dirichlet data. This constant depends on the domain; it is 2.7565 for our test and tends to $3/\sqrt{7}$ as the outer box grows.

The naive pressure-consistent correction is singular. A mean-corrected variant is well posed and attains $h_\Gamma^{3/2} + h_\Omega^k$ for general data, for every fixed $k \geq 4$ on meshes satisfying (M0)–(M2) and every $k \geq 2$ on Clough–Tocher refinements, provided the boundary edges are not too short relative to the bulk mesh. Whenever the wall shear is not identically zero, we prove a matching lower bound $ch_\Gamma^{3/2}$ for the H^1 error, so the $h_\Gamma^{3/2}$ term of Scott’s rate is sharp. The lower bound is unconditional for $k \geq 7$; for $4 \leq k \leq 6$ it is conditional on a local condition on the boundary stars, which we verified numerically on all our meshes but have not proved.

The Nitsche penalty must scale like $\rho = h_\Omega / \min_e |e|$. Sufficiency is proved; necessity is certified in exact rational arithmetic for every mesh whose shortest boundary edge carries a triangle with apex angle at least 35° and base angles at least 5° . The convergence theory extends to general smooth curved boundaries in two dimensions (nonconvex and multiply connected, with a nonempty polygonal component carrying strongly imposed data), to steady Navier–Stokes under a linearised-stability assumption that we prove for small data, and to three dimensions for a spherical obstacle, under a uniform inf-sup assumption. That assumption is known on Alfeld splits for $k \geq 3$, modulo the uniformity of the published constants over the perturbed domains, which we have not re-audited.

A computational study up to $N = 256$ boundary chords, for $k = 4, 5, 6$ and for Clough–Tocher refinements with $k = 2, 3$, confirms the rates and constants. It includes a pure-pressure test that isolates the traction response exactly, and a penalty study in which the threshold is linear in h_Ω/h_Γ , with $\mu^* h_\Gamma/h_\Omega \approx 20$ for $k = 4$.

1. INTRODUCTION

Exactly divergence-free finite elements are sensitive to how a curved no-slip boundary is represented. On a polygon, a conforming, piecewise-smooth, divergence-free field that vanishes on the

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boundary satisfies extra homogeneous constraints on its gradient at the boundary vertices. When a curved wall is replaced by an inscribed polygon with more and more vertices, these constraints cause order-one gradient errors at the vertices. The H^1 error of strongly imposed Scott–Vogelius elements then decays only like $h^{1/2}$ [16, 8].

Gjerde and Scott [8] impose the Dirichlet condition on the polygon weakly, by Nitsche’s method [15], and observed H^1 errors of order $h^{3/2}$ on a benchmark with constant pressure. Scott’s zero-gradient prize [16] asks for a proof that Scott–Vogelius–Nitsche converges for general Stokes data, including a computational study of the Nitsche penalty. It states the expected error $h_\Gamma^{3/2} + h_\Omega^k$ and adds: *if this does not hold, why not?*

Answer. For the method as printed in [8, (27)–(28)] and bounded γ , the expected estimate holds if and only if the pressure is constant on the wall. The printed Nitsche form contains no traction term $-pn$. When p varies along Γ , the penalty, not a traction, balances the pressure. The discrete velocity then leaks through the wall with normal velocity $(h/\mu)(p - \bar{p})$, and the error is dominated by this leak (Theorems A–C, Corollary B’ and Theorem B’). A pressure-consistent correction, well posed for $\gamma \geq \gamma_2$, restores the expected rate (Theorem D), and its $h_\Gamma^{3/2}$ term cannot be improved whenever the wall shear is not identically zero (Theorem D’). The penalty threshold must grow with the ratio of bulk to boundary mesh size (Proposition E).

Main results. The notation is fixed in Section 2. Throughout, $h = h_\Omega$ is the bulk mesh size used in Scott’s penalty μ/h , h_Γ is the length of the longest boundary edge, $\rho = h / \min_e |e|$, $\gamma = \mu h_\Gamma / h$ and $G = \|p - \bar{p}\|_{L^2(\Gamma)}$. Constants do not depend on h, h_Γ, μ, γ or ρ .

Theorem A (Theorem 6.1). *The printed method GS converges, and its error in the mesh-dependent energy norm is*

$$\|\tilde{u} - u_h\| \asymp (h/\mu)^{1/2} G + O((1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h),$$

with matching upper and lower bounds.

Theorem C (Theorem 6.4). $\|\nabla(\tilde{u} - u_h)\| \leq C_p h/\mu + C((1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h)$. *The rate 1 (rather than 1/2) holds because $\operatorname{div} v = 0$ turns the normal load into a tangential derivative (Lemma 6.3).*

Theorem B (Theorem 6.6). *In the regime where the pressure term dominates, the slip satisfies $\|\tilde{u} - u_h\|_{L^2(\Gamma_h)} \asymp (h/\mu)G$ and $\|\nabla(\tilde{u} - u_h)\| \asymp h/\mu$. So Theorem C is sharp.*

Corollary B’ (Corollary 6.7). $\tilde{u} - u_h = (h/\mu)v_\varphi + o((h/\mu)^{1/2})$ *in the energy norm, where v_φ is an explicit divergence-free field with $v_\varphi \approx -(p - \bar{p})n$ on Γ_h . In particular*

$$\frac{\|\tilde{u} - u_h\|_{L^2(\Gamma_h)}}{(h/\mu)G} \rightarrow 1, \quad \frac{\|\tilde{u} - u_h\|}{(h/\mu)^{1/2}G} \rightarrow 1.$$

Theorem B’’ (Theorem 6.10, Corollary 6.11). *Let U be the Stokes flow in Ω with Dirichlet data $-(p - \bar{p})n$ on Γ and 0 on ∂R . For fixed γ and bounded ρ , $(\mu/h)(\tilde{u} - u_h) \rightarrow U$ in H^1 , so*

$$\frac{\|\nabla(\tilde{u} - u_h)\|}{(h/\mu)G} \rightarrow K := \frac{\|\nabla U\|_{L^2(\Omega)}}{G}.$$

The constant K is not universal: it depends on the domain. For the pressure of our test P it is $K = 2.7565157$ (series computation, inside a rigorous bracket $[2.0746, 3.0526]$), and $K \rightarrow 3/\sqrt{7}$ as the outer box grows. Convergence in the regime $\gamma h_\Gamma \rightarrow \infty$ is observed but not proved.

Theorem D (Theorem 7.2). *The naive consistent form, which adds $\langle p_h, v \cdot n \rangle$, is singular (Proposition 7.1). The corrected method CNS*, which adds $\langle p_h - \bar{p}_\Gamma(p_h), v \cdot n \rangle$, is well posed for $\gamma \geq \gamma_2$ and satisfies*

$$\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k)$$

for general Stokes data, every fixed $k \geq 4$ and bounded γ , provided the boundary edges are not too short relative to the bulk ($h_\Gamma \geq h_\Omega^4$ for even k , $h_\Gamma \geq h_\Omega^2$ for odd k). This is Scott's expected rate. Without that proviso a single lower-order term, due to the net flux of the interpolated outer data, remains. On Clough–Tocher refinements the same holds for every $k \geq 2$ (Theorem F).

Theorem D' (Theorem 7.12, Corollary 7.13). *Suppose the wall shear $W = \partial_n u \cdot \tau$ is not identically zero. For CNS*, and for GS with any G ,*

$$\|\nabla(\tilde{u} - u_h)\| \geq c \|W\|_{L^2(\Gamma)} h_\Gamma^{3/2} - Ch_\Gamma^2,$$

with c independent of γ, μ, ρ and the pressure. So, whenever $\varepsilon_h \leq Ch_\Gamma^{3/2}$ and γ is bounded, the rate $h_\Gamma^{3/2}$ of Theorem D (and of Theorem A at $G = 0$) is exact; it is the $h_\Gamma^{3/2}$ term, not the h_Ω^k term, that is shown to be sharp. This is unconditional for $k \geq 7$. For $4 \leq k \leq 6$ it is conditional on a local, scale-invariant condition (M3) on the boundary stars, which we have not proved; we verified it numerically at every boundary vertex of our meshes, for N up to 1024.

Proposition E (Proposition 8.1, Theorem 8.2, Corollary 8.4). *The coercivity threshold of the Nitsche penalty scales like $\rho = h / \min_{e \in \mathcal{E}_h^\Gamma} |e|$. The condition $\mu \geq 4\kappa C_I^2 \rho$ is sufficient. On every mesh, coercivity requires $\mu \geq \lambda_T h / |e|$ for each triangle T with an edge e on Γ_h (and not touching ∂R), with an explicit λ_T depending only on the shape of T . In exact rational arithmetic, $\lambda_T > 1$ whenever the apex angle is at least 35° and $\lambda_T > 9.3$ when it is at least 60° (base angles at least 5°). So $\mu \gtrsim \rho$ is necessary on a broad class of meshes. Necessity is also certified on stars close to two reference stars.*

Theorem A with $G = 0$ gives $\|\tilde{u} - u_h\| \leq C((1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h)$. This recovers, for the printed method, the constant-pressure benchmark studied numerically in [8] and analysed in [5], which covers $k \geq 4$ and also Clough–Tocher splits with $k \geq 2$. Our proof does not use [5]. Theorems A–D' hold for every fixed $k \geq 4$ on meshes satisfying (M0)–(M2) (Theorem D' with the proviso above).

Extensions.

Theorem F (Theorem 9.8, Proposition 9.10). *On the Clough–Tocher refinement of any shape-regular mesh whose boundary edges have comparable lengths, Theorems A–D, Corollary B' and the sufficiency part of Proposition E hold for every $k \geq 2$, with no angle or singular-vertex condition. For example, CNS* satisfies $\|\tilde{u} - u_h\|_{H^1} \leq C(h_\Gamma^{3/2} + h^2)$ for $k = 2$ if $h_\Gamma \geq h^4$. The penalty necessity is recertified exactly on split stars.*

Theorem G (Section 10). *In two dimensions, Theorems A–D and Corollary B' extend to a bounded connected domain whose curved boundary consists of finitely many closed smooth curves, convex or not, with a nonempty polygonal component Σ on which the data are imposed strongly. The geometric lemmas use only C^3 curves; the results assume the regularity (R1) of the solution, which for generic f requires $\Gamma \in C^{k+1, \alpha}$, and the results that use the leak test field assume $\Gamma \in C^{k+3}$. With several curved components, the correct pressure constant is the global boundary mean. CNS* needs one global correction, and a per-component correction is inconsistent.*

Theorem H (Section 11). *For steady Navier–Stokes, assume the discrete linearisation at the exact solution is uniformly stable; we prove this under a small-data condition. Then GS converges at H^1 rate h/μ with the same leak (slip and energy constants 1), and CNS* attains $h_\Gamma^{3/2} + h_\Omega^k$. No inflow/outflow correction of the convective term is needed for these statements.*

Theorem I (Theorem 12.13). *In three dimensions, with an inscribed triangulated polyhedron, Theorems A–D and Corollary B' hold with the same rates, assuming a uniform discrete inf-sup constant. That assumption is known for Alfeld (barycentric) splits with $k \geq 3$, by the published*

split-mesh theory combined with a uniform Bogovskiĭ bound that we prove; we have not re-audited the published proofs for hidden domain dependence. Only a spherical obstacle is treated. Penalty necessity, sharpness and unsplit meshes remain open in 3D.

Computations. Section 13 reports computations with up to $N = 256$ boundary chords, mostly for $k = 4$, with $k = 5, 6$ and Clough–Tocher $k = 2, 3$ in Section 13.8. They show the following.

- The CNS* H^1 rate decreases monotonically to $3/2$, independently of the pressure, for $k = 4, 5, 6$.
- The printed method has rate 1 whenever $G > 0$.
- On a pure-pressure test that isolates the traction response exactly, the slip and energy constants of Corollary B' approach 1 for every element tested. For $k = 4$ the H^1 constants approach the K of Theorem B''; that they do so also for $k = 5, 6$ and on Clough–Tocher refinements is observed, not proved.
- In the penalty study, the coercivity threshold μ^* is linear in h/h_Γ over more than a decade, with $\gamma^* = \mu^* h_\Gamma / h \approx 18\text{--}21$ for $k = 4$. It is larger for higher k and on Clough–Tocher refinements.

All code, raw outputs and logs are in the accompanying repository (Appendix A).

Organisation. Section 2 fixes the setting. Sections 3–5 collect geometric and discrete tools and the error equation. Section 6 treats the printed method and identifies its H^1 constant. Section 7 treats the pressure-consistent method and proves the lower bound. Section 8 treats the penalty threshold. Sections 9–12 contain the extensions: Clough–Tocher refinements, general domains, Navier–Stokes and three dimensions. Section 13 reports the computations.

2. SETTING, METHODS AND ASSUMPTIONS

Domain. Let $R = (-L, L)^2$ with $L > 1$, let D be the open unit disk, $\Omega = R \setminus \overline{D}$ and $\Gamma = \partial D$. Let P_h be a convex polygon with vertices on Γ , $\Gamma_h = \partial P_h$ and $\Omega_h = R \setminus P_h$. Since $P_h \subset D$, we have $\Omega \subset \Omega_h$. The strip $S_h = \Omega_h \setminus \overline{\Omega}$ lies between each chord and its arc.

Mesh. $\mathcal{T}_h$ is a conforming triangulation of Ω_h , and $\mathcal{E}_h^\Gamma$ is its set of edges on Γ_h . We use the parameters

$$h = h_\Omega := \max_{T \in \mathcal{T}_h} \text{diam } T, \quad h_\Gamma := \max_{e \in \mathcal{E}_h^\Gamma} |e|, \quad \rho := \frac{h}{\min_{e \in \mathcal{E}_h^\Gamma} |e|}, \quad \gamma := \frac{\mu h_\Gamma}{h}.$$

Here h is the h in Scott's penalty μ/h , and γ is the effective boundary penalty, so that $\mu/h = \gamma/h_\Gamma$.

Assumption 2.1.

- (M0) Shape regularity, and $c_0 h_\Gamma \leq |e| \leq h_\Gamma$ for all $e \in \mathcal{E}_h^\Gamma$. Hence $\rho/h \leq 1/(c_0 h_\Gamma)$. No quasi-uniformity is assumed; $h_\Gamma \ll h$ is allowed.
- (M1) No vertex of $\mathcal{T}_h$ is singular. A vertex is singular if it is an interior vertex whose incident edges lie on two lines, a boundary vertex with one incident triangle, or a boundary vertex whose incident edges lie on two lines.
- (M1') Every vertex of Γ_h has at least three incident triangles. The angle of Ω_h at a vertex of Γ_h is $\pi + O(h_\Gamma)$, so with two triangles $\Theta(z) = O(h_\Gamma) \rightarrow 0$, contradicting (M2).
- (M2) $\Theta(z) := \max_j |\sin(\theta_j + \theta_{j+1})| \geq \Theta_0 > 0$ at every vertex z , in the notation of Guzmán–Scott [11]. At boundary vertices the wrap-around term is omitted.
- (D) $\tilde{f}$ is smooth and bounded on a fixed neighbourhood of $\overline{\Omega}$, $\tilde{f} = f$ on Ω , and $h \leq h_*$ is small (hence so is $h_\Gamma \leq h$).

Spaces. Fix $k \geq 4$ (Section 9 treats $k \geq 2$ on Clough–Tocher refinements). V_h is the space of continuous piecewise- P_k vector fields, $V_h^R := \{v \in V_h : v|_{\partial R} = 0\}$ and $V_h^0 := \{v \in V_h : v|_{\partial \Omega_h} = 0\}$. Π_h is the space of discontinuous piecewise- P_{k-1} functions and $Z_h := \{v \in V_h^R : \operatorname{div} v = 0\}$. $W_h := \{z \in V_h : z|_{\partial R} = g_I, \operatorname{div} z = 0\}$, where g_I is the P_k Lagrange interpolant of g . Gjerde and Scott use $k = 4$, as do the computations of Section 13 except those of Section 13.8 ($k = 5, 6$, and $k = 2, 3$ on Clough–Tocher refinements).

Forms. n is the outward unit normal of Ω_h on Γ_h (pointing into P_h), and $\langle \cdot, \cdot \rangle$ is the $L^2(\Gamma_h)$ pairing. Let

$$\begin{aligned} a_h(u, v) &= \frac{1}{2}(Du, Dv), & Du &= \nabla u + (\nabla u)^\top, \\ N_h(u, v) &= a_h(u, v) - \langle \partial_n u, v \rangle - \langle u, \partial_n v \rangle + \frac{\mu}{h} \langle u, v \rangle, & \partial_n u &:= (\nabla u)n. \end{aligned}$$

Definition 2.2 (Method GS [8, (28)], as printed). Find $u_h \in V_h$ with $u_h|_{\partial R} = g_I$ and $p_h \in \Pi_h$ such that

$$N_h(u_h, v) - (p_h, \operatorname{div} v) = (\tilde{f}, v) \quad \forall v \in V_h^R, \quad (q, \operatorname{div} u_h) = 0 \quad \forall q \in \Pi_h.$$

Definition 2.3 (Method CNS*). As GS, with momentum row

$$N_h(u_h, v) - (p_h, \operatorname{div} v) + \langle p_h - \bar{p}_\Gamma(p_h), v \cdot n \rangle = (\tilde{f}, v), \quad \bar{p}_\Gamma(q) := |\Gamma_h|^{-1} \int_{\Gamma_h} q.$$

Exact solution and extensions. (u, p) is smooth with $-\Delta u + \nabla p = f$ and $\operatorname{div} u = 0$ in Ω , $u|_\Gamma = 0$ and $u|_{\partial R} = g$; note $-\operatorname{div} D(u) = -\Delta u$. The stream function ψ with $u = \operatorname{curl} \psi = (\partial_y \psi, -\partial_x \psi)$ is single-valued because $u \cdot n = 0$ on Γ . Let $\tilde{\psi}$ be a smooth extension of ψ to a neighbourhood of $\bar{\Omega}$ and $\tilde{u} := \operatorname{curl} \tilde{\psi}$, so $\operatorname{div} \tilde{u} = 0$. Let $\tilde{p}$ be a smooth extension of p . Then $F := \tilde{f} + \Delta \tilde{u} - \nabla \tilde{p}$ vanishes on Ω and is supported in S_h .

Norms and constants. For v with traces on Γ_h ,

$$\|v\|^2 := \|\nabla v\|_{\Omega_h}^2 + \frac{\mu}{h} \|v\|_{\Gamma_h}^2 + \sum_{e \in \mathcal{E}_h^\Gamma} |e| \|\partial_n v\|_{L^2(e)}^2.$$

This is used for discrete fields and for errors $\tilde{u} - z$; since $\tilde{u}$ is smooth, every trace exists. We write $G := \|p - \bar{p}\|_{L^2(\Gamma)}$, with $\bar{p}$ the mean of p over Γ .

C and c depend on (M0)–(M2), k , L and norms of $(u, p, \tilde{\psi})$. They are independent of $h, h_\Gamma, \mu, \gamma, \rho$ unless displayed. C_p is a constant that may also depend on $\|p - \bar{p}\|_{W^{1,\infty}(\Gamma)}$. Where G is tracked explicitly, “ δ small” means $\delta \leq \delta_0(C_p)$.

3. GEOMETRY

Lemma 3.1 (Chords). *Let e be a chord of length ℓ and $x \in e$. Then $\operatorname{dist}(x, \Gamma) \leq \ell^2/8 + O(\ell^4) \leq \ell^2/4$. Let s be the signed arclength from the chord midpoint m_e , x^* the nearest point of Γ , and τ_e the unit tangent of e . Then, up to a fixed orientation sign,*

$$n(x^*) - n_e = -s \tau_e + O(\ell^2).$$

The arclength Jacobian of the radial projection $\Gamma_h \rightarrow \Gamma$ is $1 + O(\ell^2)$.

Proof. The sagitta is $1 - \sqrt{1 - \ell^2/4} = \ell^2/8 + \ell^4/128 + \dots$. The normal at x^* is the radial unit vector. Its angle differs from that of n_e by the central angle to x^* , which is $s + O(s^3)$. $\square$

Lemma 3.2 (Uniform trace, Friedrichs and Korn). *There are $C_{\text{tr}}, C_F, \kappa$, independent of h , such that every $v \in H^1(\Omega_h)^2$ with $v|_{\partial R} = 0$ satisfies*

$$\|v\|_{L^2(\Gamma_h)} \leq C_{\text{tr}} \|\nabla v\|, \quad \|v\| \leq C_F \|\nabla v\|, \quad \|\nabla v\|^2 \leq \kappa a_h(v, v).$$

Proof. (1) Write Γ_h in polar form as $r = \rho_h(\theta)$. On a chord with endpoint angles $\theta_m \pm \alpha$ we have $\rho_h = \cos \alpha / \cos(\theta - \theta_m)$. Hence $\|\rho_h - 1\|_\infty \leq Ch_\Gamma^2$ and $\|\rho'_h\|_\infty \leq Ch_\Gamma$.
(2) Let χ be smooth, with $\chi(1) = 1$ and $\chi \equiv 0$ outside a fixed collar in Ω . Define $\Phi_h(r, \theta) = (r + \chi(r)(\rho_h(\theta) - 1), \theta)$.
(3) Φ_h maps Ω onto Ω_h and Γ onto Γ_h , is the identity near ∂R , and is bi-Lipschitz with $\|D\Phi_h - I\|_\infty \leq Ch_\Gamma$ almost everywhere.
(4) For $w = v \circ \Phi_h$ we have $D(w) = D(v) \circ \Phi_h + E$ with $|E| \leq Ch_\Gamma |\nabla v \circ \Phi_h|$.
(5) On the fixed domain Ω , w vanishes on ∂R , a set of positive measure. So the first Korn inequality, the trace inequality and the Friedrichs inequality hold.
(6) Changing variables back gives $\|\nabla v\| \leq C(\|D(v)\| + h_\Gamma \|\nabla v\|)$; absorb the last term for $h_\Gamma \leq h_*$. The other two inequalities transfer the same way. $\square$

Lemma 3.3 (Extension). $\|\tilde{u}\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^2$, so $\|\tilde{u}\|_{L^2(\Gamma_h)} \leq Ch_\Gamma^2$. For $v \in V_h^R$, $|(F, v)| \leq Ch_\Gamma^2 \|\nabla v\|$.

Proof. The first bound follows from Lemma 3.1 and $\tilde{u}|_\Gamma = 0$. For the second, apply a one-dimensional Poincaré inequality across strips of width at most Ch_Γ^2 , and use $|S_h| \leq Ch_\Gamma^2$, $\|F\|_\infty \leq C$ (by (D)) and Lemma 3.2. $\square$

Lemma 3.4 (Wall structure). On Γ : $\nabla u = \partial_n u \otimes n$, $\partial_n u \cdot n = 0$ and $(\nabla u)^\top n = 0$.

Proof. The tangential derivatives vanish, and $\operatorname{div} u = n \cdot \partial_n u$. $\square$

Lemma 3.5 (Transposed-gradient term). For $v \in V_h^R$,

$$|(\nabla \tilde{u})^\top n, v| \leq C \sum_{e \in \mathcal{E}_h^\Gamma} (|e|^2 \|v\|_{L^1(e)} + |e|^2 \|\partial_\tau v\|_{L^1(e)}) \leq Ch_\Gamma^{3/2} \|\nabla v\|.$$

The rate $h_\Gamma^{3/2}$ is sharp on V_h^R : a first-layer field with odd trace on each chord attains it, and the dual-norm test of Section 13 gives 1.56–1.57. For the method itself, which is tested on Z_h , Theorem 7.12 proves the matching lower bound $ch_\Gamma^{3/2}$ for the H^1 error, by a different route (unconditionally for $k \geq 7$, and under a local condition on the boundary stars for $4 \leq k \leq 6$). The CNS* H^1 rates of Table 4 decrease monotonically toward 3/2. Lemma 3.5 is a sharper dual-norm statement than the upper bounds need: by Remark 12.7, Lemma 5.2 and Theorems 6.4 and 7.2 also follow from the cruder energy-dual bound that holds without the odd-in- s cancellation.

Proof. (1) Taylor-expand about x^* and apply Lemmas 3.1 and 3.4:

$$(\nabla \tilde{u}(x))^\top n_e = n(x^*) (\partial_n u(x^*) \cdot (n_e - n(x^*))) + O(h_\Gamma^2) = s a_e + O(h_\Gamma^2),$$

where $a_e := n(m_e^*) (\partial_n u(m_e^*) \cdot \tau_e)$ is constant on e . The $O(h_\Gamma^2)$ remainder collects $|x - x^*|$, the $O(\ell^2)$ term of Lemma 3.1, and $s \cdot O(s)$.

(2) Since $\int_e s ds = 0$,

$$\left| \int_e s a_e \cdot v \right| = \left| \int_e s a_e \cdot (v - v(m_e)) \right| \leq \frac{|e|^2}{4} \|\partial_\tau v\|_{L^1(e)}.$$

(3) Use $\|\cdot\|_{L^1(e)} \leq |e|^{1/2} \|\cdot\|_{L^2(e)}$ and the inverse trace inequality $|e| \|\partial_\tau v\|_{L^2(e)}^2 \leq C_I^2 \|\nabla v\|_{T_e}^2$.

(4) Summing, $\sum_e |e|^{5/2} \|\partial_\tau v\|_{L^2(e)} \leq Ch_\Gamma^2 (\#\mathcal{E}_h^\Gamma)^{1/2} \|\nabla v\| \leq Ch_\Gamma^{3/2} \|\nabla v\|$. The zeroth-order term is handled the same way, finishing with Lemma 3.2. $\square$

4. DISCRETE TOOLS

Lemma 4.1 (Inverse traces). *For $v \in V_h$,*

$$\sum_{e \in \mathcal{E}_h^\Gamma} |e| \|\partial_n v\|_{L^2(e)}^2 \leq C_I^2 \|\nabla v\|^2, \quad \text{hence} \quad \|\partial_n v\|_{\Gamma_h} \leq C_I (\rho/h)^{1/2} \|\nabla v\|.$$

For $q \in \Pi_h$,

$$\|q\|_{\Gamma_h}^2 \leq C_I^2 \sum_{T \in \mathcal{T}_h^\Gamma} |e_T|^{-1} \|q\|_T^2 \leq C_I^2 (\rho/h) \|q\|^2.$$

Lemma 4.2 (Coercivity and continuity). *Let $\mu_0 := 4\kappa\rho C_I^2$. The condition $\gamma \geq \gamma_0 := 4\kappa C_I^2/c_0$ is sufficient for $\mu \geq \mu_0$. If $\mu \geq \mu_0$, then*

$$N_h(v, v) \geq c_1 \|v\|^2 \quad \forall v \in V_h^R, \quad |N_h(w, v)| \leq M \|w\| \|v\|,$$

with $c_1 = 1/\max(2\kappa(1+C_I^2), 2)$ and $M = 3 + (\kappa C_I^2)^{-1/2}$. Both constants are independent of h , μ and ρ .

Proof. (1) By Lemma 4.1 and Young's inequality with $\varepsilon = 1/(2\kappa)$,

$$2|\langle \partial_n v, v \rangle| \leq 2C_I \|\nabla v\| (\rho/h)^{1/2} \|v\|_{\Gamma_h} \leq \frac{1}{2} a_h(v, v) + 2\kappa\rho C_I^2 h^{-1} \|v\|_{\Gamma_h}^2.$$

(2) Hence $N_h(v, v) \geq \frac{1}{2} a_h(v, v) + (\mu - 2\kappa\rho C_I^2) h^{-1} \|v\|^2$. For $\mu \geq \mu_0$ this is at least $(2\kappa)^{-1} \|\nabla v\|^2 + (\mu/2h) \|v\|^2$.

(3) Adding $\sum_e |e| \|\partial_n v\|^2 \leq C_I^2 \|\nabla v\|^2$ gives the coercivity constant.

(4) For continuity, $a_h(w, v) \leq 2\|\nabla w\| \|\nabla v\|$. Each ∂_n -term is at most $(\rho/\mu)^{1/2} \|w\| \|v\| \leq (4\kappa C_I^2)^{-1/2} \|w\| \|v\|$. $\square$

Lemma 4.3 (Divergence range). *Under (M1), $\operatorname{div} V_h^R = \Pi_h$ and $W_h \neq \emptyset$. The velocities of GS and CNS* are pointwise divergence-free.*

Proof. (1) $\operatorname{div} V_h^0 = \Pi_h \cap L_0^2$ by the inf-sup condition of Lemma 4.5 below (Guzmán–Scott [11], extending Scott–Vogelius [17]). This holds on the doubly connected domain Ω_h ; there is no circularity, since Lemma 4.5 does not use the present lemma.

(2) The edge bubble $b = \beta_e n_e$ on one $e \subset \Gamma_h$ lies in V_h^R and has $\int \operatorname{div} b = \int_{\Gamma_h} b \cdot n = |e|/6 \neq 0$. This adds the constants, so $\operatorname{div} V_h^R = \Pi_h$.

(3) For any $\mathcal{G} \in V_h$ with trace g_I , we have $\operatorname{div} \mathcal{G} \in \Pi_h = \operatorname{div} V_h^R$. Hence $\mathcal{G} - v \in W_h$ for a suitable $v \in V_h^R$.

(4) For the discrete velocity, $u_h - \mathcal{G} \in V_h^R$, and the continuity row forces $\operatorname{div} u_h = 0$. $\square$

Remark 4.4. If a vertex of ∂R is singular (for example a corner with one triangle), then $A_z(\operatorname{div} \mathcal{G}) = 0$ is a condition on g_I that generic data violate, and $W_h = \emptyset$.

Lemma 4.5 (Uniform inf-sup constant). *There is $\beta > 0$, depending only on k , Θ_0 , shape regularity and L , such that for every $q \in \Pi_h \cap L_0^2(\Omega_h)$ there exists $v \in V_h^0$ with $\operatorname{div} v = q$ and $\|\nabla v\| \leq \beta^{-1} \|q\|$.*

Proof. Guzmán and Scott [11, Theorem 1] prove the inf-sup condition for $(V_h^0, \Pi_h \cap L_0^2)$ for $k \geq 4$ under (M1)–(M2). Their proof needs neither quasi-uniformity, nor small h , nor an interior vertex in each triangle. The discrete Fortin operator is built locally from vertex and edge patches, using shape regularity and (M2) only. The domain enters only through the continuous inf-sup constant (via the P_2 – P_0 Bernardi–Raugel pair) and a Poincaré constant. The Poincaré constant is uniform by Lemma 3.2, so it remains to show that the continuous inf-sup constant is uniform over the domains Ω_h .

- Cover Ω_h by the h -dependent pieces $W_j \cap \Omega_h$, where the W_j are fixed angular wedges of half-angle $\alpha < \pi/2$ about the origin. Each $W_j \cap R$ is convex, and $W_j \cap \Omega_h = (W_j \cap R) \setminus P_h$.

- Each $W_j \cap \Omega_h$ is star-shaped with respect to a ball $B(c_j, r_B) \subset W_j \cap R$ with $|c_j| = r_b$, provided every chord line meeting W_j leaves the ball on its outer side. This holds uniformly when $(r_b - r_B) \cos(\alpha + \arcsin(r_B/r_b) + h_\Gamma) > 1$, which can be arranged with fixed α, r_b, r_B because $P_h \subset D$.
- The pieces have diameters, ball radii and pairwise overlap measures bounded above and below independently of h .

Galdi's Lemma III.3.4 [7] then gives a Bogovskii constant, and hence a continuous inf-sup constant, independent of h . The equivalence of the inf-sup condition with a bounded right inverse of div gives the statement. $\square$

Lemma 4.6 (Approximation in W_h). *Let $k \geq 4$. Let $\sigma_k := k + 2$ for even k and $\sigma_k := k + 1$ for odd k . Then*

$$E_h := \inf_{z \in W_h} \|\tilde{u} - z\| \leq C\varepsilon_h, \quad \varepsilon_h := (1 + (\gamma h_\Gamma)^{1/2}) h^k + h^{\sigma_k} (h_\Gamma^{-1/2} + (\mu/h)^{1/2}).$$

The second term comes from the net flux of the interpolated boundary data. It cannot be removed for the method as posed. Let ν be the outward normal of R and $m_R := \int_{\partial R} g_I \cdot \nu$. Every $z \in W_h$ (including the discrete velocity) has $\int_{\Gamma_h} z \cdot n = -m_R$, while $\int_{\Gamma_h} \tilde{u} \cdot n = 0$. Hence $\|\tilde{u} - z\| \geq (\mu/h)^{1/2} |m_R|/|\Gamma_h|^{1/2}$, and $|m_R| \asymp h^{\sigma_k}$ for generic g .

For bounded γ the term is at most Ch^k whenever $h_\Gamma \geq h^{2(\sigma_k - k)}$: that is, $h_\Gamma \geq h^4$ for even k and $h_\Gamma \geq h^2$ for odd k . If g_I is flux-corrected so that $m_R = 0$, or for the test data of Section 13, where $g \cdot \nu$ is a polynomial of degree at most 5 on each side and the flux integrals over ∂R are exact, only the contribution of Γ_h to the net flux remains. By step 2 of the proof below it is $O(h_\Gamma^{\sigma_k})$, so the term reduces to $C(1 + \gamma^{1/2})h_\Gamma^{\sigma_k - 1/2}$, which is below h^k for bounded γ .

Proof. (1) *Interpolation.* Let $w_h := I_h \tilde{u} \in V_h$ be the Lagrange interpolant of $\tilde{u}$, which is smooth on Ω_h . On ∂R , $w_h = g_I$. Boundary triangles have diameter at most Ch_Γ by shape regularity, so

$$\|\nabla(\tilde{u} - w_h)\| \leq Ch^k, \quad \|\tilde{u} - w_h\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^{k+1}, \quad \|\partial_n(\tilde{u} - w_h)\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^k.$$

Hence $\|\tilde{u} - w_h\| \leq C(h^k + (\mu/h)^{1/2} h_\Gamma^{k+1} + h_\Gamma^k) \leq C(1 + (\gamma h_\Gamma)^{1/2}) h^k$. Moreover $\operatorname{div} w_h = \operatorname{div}(w_h - \tilde{u}) \in \Pi_h$, with $\|\operatorname{div} w_h\| \leq Ch^k$.

- (2) *Net flux.* Let ν be the outward normal of R . Since $\tilde{u}|_{\partial R} = g$ and $\operatorname{div} \tilde{u} = 0$, we have $\int_{\Gamma_h} \tilde{u} \cdot n = -\int_{\partial R} g \cdot \nu = 0$. Therefore

$$m := \int_{\Omega_h} \operatorname{div} w_h = \int_{\partial R} (g_I - g) \cdot \nu + \int_{\Gamma_h} (w_h - \tilde{u}) \cdot n, \quad |m| \leq Ch^{\sigma_k}.$$

Here equispaced P_k Lagrange interpolation on an edge integrates like the closed $(k+1)$ -point Newton–Cotes rule. That rule is exact for polynomials of degree $\sigma_k - 1$, so each edge contributes $O(|e|^{\sigma_k+1})$.

- (3) *Removing the mean.* Let $b := \sum_{e \in \mathcal{E}_h^\Gamma} \beta_e n_e$, where β_e is the quadratic edge bubble of the boundary triangle on e . Then $b \in V_h^R$, $\int \operatorname{div} b = \int_{\Gamma_h} b \cdot n = |\Gamma_h|/6$, and with $a := 6m/|\Gamma_h|$,

$$\|ab\| \leq C|a|(h_\Gamma^{-1/2} + (\mu/h)^{1/2}) \leq Ch^{\sigma_k} (h_\Gamma^{-1/2} + (\mu/h)^{1/2}) \leq C\varepsilon_h,$$

using $\#\mathcal{E}_h^\Gamma \leq C/h_\Gamma$. Also $\|\operatorname{div}(ab)\| \leq C|a|h_\Gamma^{-1/2} \leq C\varepsilon_h$.

- (4) *Divergence correction.* The field $q := \operatorname{div}(w_h - ab) \in \Pi_h$ has mean zero and $\|q\| \leq C\varepsilon_h$. Lemma 4.5 gives $v \in V_h^0$ with $\operatorname{div} v = q$ and $\|\nabla v\| \leq \beta^{-1} \|q\| \leq C\varepsilon_h$. Since $v = 0$ on Γ_h , $\|v\|^2 = \|\nabla v\|^2 + \sum_e |e| \|\partial_n v\|_{L^2(e)}^2 \leq (1 + C_I^2) \|\nabla v\|^2$.

(5) Set $z := w_h - ab - v$. Then $z|_{\partial R} = g_I$ and $\operatorname{div} z = 0$, so $z \in W_h$, and

$$\|\tilde{u} - z\| \leq \|\tilde{u} - w_h\| + \|ab\| + \|v\| \leq C\varepsilon_h. \quad \square$$

Remark 4.7. Version 1 of this preprint proved Lemma 4.6 for $k = 4$ by matching Argyris stream-function data on ∂R . The proof above avoids stream functions and holds for every $k \geq 4$.

5. THE ERROR EQUATION

Lemma 5.1. For $v \in V_h^R$,

$$N_h(\tilde{u}, v) - (\tilde{p}, \operatorname{div} v) + \langle \tilde{p}, v \cdot n \rangle = (\tilde{f}, v) + \mathcal{G}(v),$$

where

$$\mathcal{G}(v) := \langle (\nabla \tilde{u})^\top n, v \rangle - \langle \tilde{u}, \partial_n v \rangle + \frac{\mu}{h} \langle \tilde{u}, v \rangle - (F, v).$$

Proof. The argument has four steps.

(1) By symmetry of $D(\tilde{u})$ and since $v = 0$ on ∂R ,

$$a_h(\tilde{u}, v) = (D(\tilde{u}), \nabla v) = -(\operatorname{div} D(\tilde{u}), v) + \langle D(\tilde{u})n, v \rangle.$$

(2) $\operatorname{div} D(\tilde{u}) = \Delta \tilde{u}$.

(3) $-\Delta \tilde{u} = \tilde{f} - F - \nabla \tilde{p}$, and $-(\nabla \tilde{p}, v) = (\tilde{p}, \operatorname{div} v) - \langle \tilde{p}, v \cdot n \rangle$.

(4) $D(\tilde{u})n - \partial_n \tilde{u} = (\nabla \tilde{u})^\top n$. □

Lemma 5.2. For $v \in V_h^R$, $|\mathcal{G}(v)| \leq C(1 + \gamma^{1/2})h_\Gamma^{3/2}\|v\|$. For $v \in V_h^0$, $|\mathcal{G}(v)| \leq Ch_\Gamma^{3/2}\|\nabla v\|$.

Proof. We bound the four terms of $\mathcal{G}(v)$ in turn.

- The transposed term is at most $Ch_\Gamma^{3/2}\|\nabla v\|$ by Lemma 3.5.
- $|\langle \tilde{u}, \partial_n v \rangle| \leq Ch_\Gamma^2 \cdot C_I(\rho/h)^{1/2}\|\nabla v\| \leq Ch_\Gamma^{3/2}\|\nabla v\|$.
- The penalty term is at most $(\mu/h)^{1/2}Ch_\Gamma^2\|v\| = C\gamma^{1/2}h_\Gamma^{3/2}\|v\|$.
- The F term is bounded by Lemma 3.3.

For $v \in V_h^0$, the transposed and penalty terms vanish. □

Corollary 5.3 (GS on Z_h). Let $v \in Z_h$. Then $(\cdot, \operatorname{div} v) = 0$ and $\int_{\Gamma_h} v \cdot n = 0$, so for every constant c ,

$$N_h(\tilde{u} - u_h, v) = \mathcal{R}(v) + \mathcal{G}(v), \quad \mathcal{R}(v) := -\langle \tilde{p} - c, v \cdot n \rangle.$$

6. THE PRINTED METHOD

Throughout this section $\gamma \geq \gamma_0$.

Test field. Let $\varphi_0(\theta) := \int_0^\theta (p - \bar{p})(\theta') d\theta'$, which is periodic, and $\varphi := \chi_1(r)\varphi_0(\theta)$, where $\chi_1 \equiv 1$ near $r = 1$ (so $\chi_1'(1) = 0$) and $\chi_1 \equiv 0$ near ∂R . Then $w := \operatorname{curl} \varphi$ has components $w_r = \partial_\theta \varphi / r$ and $w_\theta = -\partial_r \varphi$, and on Γ

$$w = (p - \bar{p})e_r = -(p - \bar{p})n.$$

Let Π_{k+1} be the C^1 piecewise- P_{k+1} interpolant of Morgan and Scott [13] (for $k = 4$, the Argyris interpolant [1]), which exists because $k + 1 \geq 5$; its constants depend on the angles, hence on (M0) and (M2). Since $\varphi \equiv 0$ near ∂R and $h \leq h_*$, $v_\varphi := \operatorname{curl} \Pi_{k+1} \varphi \in Z_h$, and $\|v_\varphi - w\|_{W^{1,\infty}} \leq Ch^k$.

Theorem 6.1 (Energy norm: exact rate 1/2). The error satisfies the upper bound

$$\|\tilde{u} - u_h\| \leq C \left[(h_\Gamma/\gamma)^{1/2}G + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h \right].$$

If $G > 0$ and $h \leq h_0(G)$, with h_0 independent of $\gamma \geq \gamma_0$, it also satisfies the lower bound

$$\|\tilde{u} - u_h\| \geq (2M)^{-1}(h_\Gamma/\gamma)^{1/2}G - C \left[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h \right].$$

In Scott's variables, $(h_\Gamma/\gamma)^{1/2} = (h/\mu)^{1/2}$.

Proof of the upper bound. For $z \in W_h$, $e_h := z - u_h \in Z_h$. By Corollary 5.3 and Lemma 4.2,

$$c_1 \|e_h\|^2 \leq N_h(z - \tilde{u}, e_h) + \mathcal{R}(e_h) + \mathcal{G}(e_h).$$

Using the Jacobian bound of Lemma 3.1, $|\mathcal{R}(e_h)| \leq (G + Ch_\Gamma^2)(h/\mu)^{1/2} \|e_h\|$. Then apply Lemmas 5.2 and 4.6. $\square$

Proof of the lower bound. (1) We have $v_\varphi \cdot n_e = -(p - \bar{p})(x^*) n(x^*) \cdot n_e + O(h^k)$, $\mathcal{R} = -\langle \tilde{p} - \bar{p}, v \cdot n \rangle$ and $n(x^*) \cdot n_e = 1 + O(s^2)$. Hence

$$\mathcal{R}(v_\varphi) = \langle (\tilde{p} - \bar{p})(p - \bar{p})(x^*), n(x^*) \cdot n_e \rangle + O(h^k) = G^2 + O(h_\Gamma^2 + h^k).$$

$$(2) \|v_\varphi\|^2 \leq (\mu/h)(G + Ch_\Gamma^2 + Ch^k)^2 + C.$$

(3) By continuity, $M\|\tilde{u} - u_h\| \geq (|\mathcal{R}(v_\varphi)| - |\mathcal{G}(v_\varphi)|)/\|v_\varphi\|$, and Lemma 5.2 bounds $\mathcal{G}$.

(4) $|\mathcal{R}|/\|v_\varphi\| \geq (h/\mu)^{1/2} G(1 - C(h_\Gamma^2 + h^k)/G^2 - C(h_\Gamma^2 + h^k)/G - C(h_\Gamma/\gamma_0)^{1/2}/G)$, which is at least $\frac{1}{2}(h/\mu)^{1/2} G$ for $h \leq h_0(G)$. $\square$

The lower bound is carried by the boundary parts of $\|\cdot\|$. By Theorem 6.4, $\|\nabla(\tilde{u} - u_h)\|$ is only $O(h/\mu)$.

Corollary 6.2. *If $G > 0$, $\gamma h_\Gamma \ll G$ and $\varepsilon_h \ll (h_\Gamma/\gamma)^{1/2} G$, then $\|\tilde{u} - u_h\| \asymp (h/\mu)^{1/2}$. The mesh-dependent energy rate is exactly $1/2$.*

Lemma 6.3 (Normal-load cancellation). *Let S be edgewise $W^{1,\infty}$ on Γ_h , continuous at the vertices of Γ_h , with $|S \cdot \tau_e| \leq Ch_\Gamma$ on every $e \in \mathcal{E}_h^\Gamma$. (For instance, S smooth with $S \cdot \tau = 0$ on Γ .) Then for $v \in Z_h$,*

$$|\langle S, \partial_n v \rangle| \leq C(\|v\|_{\Gamma_h} + h_\Gamma^{1/2} \|\nabla v\|) \leq C\|\nabla v\|.$$

Proof. (1) *Tangential part.* By Lemma 4.1 and (M0), $|\langle (S \cdot \tau)\tau, \partial_n v \rangle| \leq Ch_\Gamma \|\partial_n v\|_{\Gamma_h} \leq Ch_\Gamma^{1/2} \|\nabla v\|$.

(2) *Normal part.* On e , v is a polynomial on T_e with $\text{div } v = 0$, so $\partial_n v \cdot n_e = -\partial_\tau(v \cdot \tau_e)$. With $\sigma_e := S \cdot n_e$,

$$\int_e \sigma_e \partial_n v \cdot n_e = \int_e \partial_\tau \sigma_e (v \cdot \tau_e) - [\sigma_e v \cdot \tau_e]_{\partial e}.$$

(3) At a polygon vertex x shared by e and e' , the endpoint terms combine to $v(x) \cdot (\sigma_{e'} \tau_{e'} - \sigma_e \tau_e)$, of size at most $C(|e| + |e'|)|v(x)|$.

(4) $\sum_e |e| \|v\|_{L^\infty(e)} \leq C \sum_e |e|^{1/2} \|v\|_{L^2(e)} \leq C|\Gamma_h|^{1/2} \|v\|_{\Gamma_h}$.

(5) Finish with Lemma 3.2. $\square$

Numerically (Section 13), the dual norm of $v \mapsto \langle S, \partial_n v \rangle$ stays bounded on Z_h for normal S . It grows like $h_\Gamma^{-1/2}$ on V_h^R (no divergence constraint), and also on Z_h for tangential S . Incompressibility is exactly what the proof uses.

Theorem 6.4 (H^1 seminorm: rate 1). *With $C_p := C(1 + \|p - \bar{p}\|_{W^{1,\infty}(\Gamma)})$,*

$$\|\nabla(\tilde{u} - u_h)\| \leq C_p (h_\Gamma/\gamma) + C \left[(1 + \gamma^{1/2}) h_\Gamma^{3/2} + \varepsilon_h \right].$$

Proof. (1) *Split.* For $z \in W_h$, write $z - u_h = \delta_R + \delta_G$, where $\delta_R, \delta_G \in Z_h$ solve

$$N_h(\delta_R, v) = \mathcal{R}(v), \quad N_h(\delta_G, v) = \mathcal{G}(v) - N_h(\tilde{u} - z, v) \quad \forall v \in Z_h.$$

Lax–Milgram on Z_h (Lemma 4.2) gives existence and uniqueness.

(2) *Geometric part.* $\|\nabla \delta_G\| \leq \|\delta_G\| \leq c_1^{-1} (C(1 + \gamma^{1/2}) h_\Gamma^{3/2} + ME_h)$.

- (3) *Ansatz.* Heuristically $\delta_R \approx -(h/\mu)(p - \bar{p})n = (h/\mu)w$ on Γ_h . Set $S_\varphi := v_\varphi \in Z_h$ and $r_h := \delta_R - (h/\mu)S_\varphi \in Z_h$. Then $N_h(r_h, v) = m(v) + \ell(v)$ with

$$\begin{aligned} m(v) &:= \mathcal{R}(v) - \langle S_\varphi, v \rangle = -\langle (\tilde{p} - \bar{p})n + S_\varphi, v \rangle, \\ \ell(v) &:= -(h/\mu)[a_h(S_\varphi, v) - \langle \partial_n S_\varphi, v \rangle - \langle S_\varphi, \partial_n v \rangle]. \end{aligned}$$

- (4) *Bound on m .* On e ,

$$(\tilde{p} - \bar{p})n_e + S_\varphi = (p - \bar{p})(x^*)(n_e - n(x^*)) + O(h_\Gamma^2) + O(h^k) = (p - \bar{p})(m_e^*)s\tau_e + O(h_\Gamma^2) + O(h^k).$$

The leading term is odd in s with a constant coefficient. The argument of Lemma 3.5 handles it, and the $O(h^k)$ remainder contributes $Ch^k\|v\|_{L^1(\Gamma_h)}$. Hence $|m(v)| \leq C_p(h_\Gamma^{3/2} + h^k)\|\nabla v\|$.

- (5) *Bound on ℓ .* $|a_h(S_\varphi, v)| + |\langle \partial_n S_\varphi, v \rangle| \leq C_p\|\nabla v\|$. By Lemma 6.3 applied to $S = w$,

$$|\langle S_\varphi, \partial_n v \rangle| \leq |\langle w, \partial_n v \rangle| + |\langle S_\varphi - w, \partial_n v \rangle| \leq C_p\|\nabla v\| + Ch^k h_\Gamma^{-1/2}\|\nabla v\|.$$

So $|\ell(v)| \leq (h/\mu)(C_p + Ch^k h_\Gamma^{-1/2})\|\nabla v\|$.

- (6) *Close.* Testing with $v = r_h$,

$$c_1\|r_h\|^2 \leq [C_p(h/\mu + h_\Gamma^{3/2} + h^k) + C(h/\mu)h^k h_\Gamma^{-1/2}]\|\nabla r_h\|.$$

The last two terms are at most $C\varepsilon_h$, since $(h/\mu)h^k h_\Gamma^{-1/2} = \gamma^{-1}h^k h_\Gamma^{1/2} \leq Ch^k$. Hence

$$\|\nabla \delta_R\| \leq (h/\mu)\|\nabla S_\varphi\| + \|\nabla r_h\| \leq C_p(h_\Gamma/\gamma) + C(h_\Gamma^{3/2} + \varepsilon_h). \quad \square$$

Remark 6.5 (Why rate 1 and not 1/2). Without incompressibility, Lemma 6.3 would give only $Ch_\Gamma^{-1/2}$. The H^1 error would then be $(h/\mu)h_\Gamma^{-1/2}$: rate 1/2, with element-scale oscillation in the trace. The inconsistency $-pn$ is purely normal, and $\operatorname{div} v = 0$ converts $\partial_n v \cdot n$ into a tangential derivative. That is why GS still converges at rate 1 in H^1 .

Theorem 6.6 (Boundary slip; H^1 lower bound). *Let $\gamma \geq \gamma_0$ and $G > 0$, and put $X := C_p(h_\Gamma/\gamma) + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h$, the right-hand side of Theorem 6.4. Suppose that, for a fixed small δ ,*

$$X \leq \delta G \min(G, h_\Gamma^{1/2}), \quad h_\Gamma/\gamma \leq \delta G, \quad h_\Gamma \leq h_0(G).$$

All three hold for fixed γ and bounded ρ as $h_\Gamma \rightarrow 0$. Then

- (i) $c(h_\Gamma/\gamma)G \leq \|\tilde{u} - u_h\|_{L^2(\Gamma_h)} \leq C(h_\Gamma/\gamma)^{1/2}\|\tilde{u} - u_h\|$;
- (ii) $\|\nabla(\tilde{u} - u_h)\| \geq c(h_\Gamma/\gamma)G - Ch^{k+1/2}$.

Combined with Theorem 6.4, $\|\nabla(\tilde{u} - u_h)\| \asymp h/\mu$ in this regime.

Proof of (i). Let $e_u := \tilde{u} - u_h$ and $v := v_\varphi$. Corollary 5.3 rearranges to

$$\frac{\mu}{h}\langle e_u, v \rangle = \mathcal{R}(v) + \mathcal{G}(v) - a_h(e_u, v) + \langle \partial_n e_u, v \rangle + \langle e_u, \partial_n v \rangle.$$

We bound the terms, using that v is smooth up to $O(h^k)$.

- (1) $\mathcal{R}(v) = G^2 + O(h_\Gamma^2 + h^k)$.
- (2) $|\mathcal{G}(v)| \leq C(h_\Gamma^{3/2} + \gamma h_\Gamma^3 + \gamma h_\Gamma h^k)$. On the chord, $\tilde{u} = \pm d \partial_n u(x^*) + O(d^2)$ is tangential while $v \approx -(p - \bar{p})n$ is normal. Hence $\tilde{u} \cdot v = O(h_\Gamma^4 + h_\Gamma^2 h^k)$, the second term coming from $v_\varphi - w$.
- (3) $|a_h(e_u, v)| \leq C_p\|\nabla e_u\| \leq C_p X \leq C_p \delta G^2$, by Theorem 6.4.
- (4) By Theorem 6.1 and $h/\mu = h_\Gamma/\gamma \leq \delta G$,

$$|\langle e_u, \partial_n v \rangle| \leq C_p\|e_u\|_{\Gamma_h} \leq C_p(h/\mu)^{1/2}\|e_u\| \leq C_p[(h/\mu)G + X] \leq 2C_p\delta G^2.$$

(5) We have

$$|\langle \partial_n e_u, v \rangle| \leq \|\partial_n(\tilde{u} - z)\|_{\Gamma_h} \|v\| + C_I(\rho/h)^{1/2} (\|\nabla(z - \tilde{u})\| + \|\nabla e_u\|) \|v\|_{\Gamma_h}.$$

By Theorem 6.4 and $(\rho/h)^{1/2} \leq (c_0 h_\Gamma)^{-1/2}$, this is at most $C h_\Gamma^{-1/2} XG \leq C\delta G^2$.

For δ small, $|\langle e_u, v \rangle| \geq \frac{1}{2}(h/\mu)G^2$. Dividing by $\|v\|_{\Gamma_h} \leq G + Ch_\Gamma^2$ gives the lower bound. The upper bound is $\|e_u\|_{\Gamma_h} \leq (h/\mu)^{1/2} \|e_u\|$. $\square$

Proof of (ii). Let $E \in H^1(\Omega_h)$ lift $(g - g_I)|_{\partial R}$, supported near ∂R , with $E|_{\Gamma_h} = 0$ and $\|E\|_{H^1} \leq C\|g - g_I\|_{H^{1/2}(\partial R)} \leq Ch^{k+1/2}$. By Lemma 3.2, $\|e_u\|_{\Gamma_h} = \|e_u - E\|_{\Gamma_h} \leq C_{\text{tr}}(\|\nabla e_u\| + \|\nabla E\|)$. $\square$

Corollary 6.7 (Exact leading term). *Let $G > 0$ and $h_\Gamma \rightarrow 0$ with $h/\mu \rightarrow 0$, $\gamma h_\Gamma \rightarrow 0$ and $\varepsilon_h/(h_\Gamma/\gamma)^{1/2} \rightarrow 0$. Then*

$$\tilde{u} - u_h = (h/\mu) v_\varphi + \vartheta_h, \quad \|\vartheta_h\| = o((h/\mu)^{1/2}), \quad \|\vartheta_h\|_{\Gamma_h} = o(h/\mu),$$

and consequently

$$\frac{\|\tilde{u} - u_h\|_{L^2(\Gamma_h)}}{(h/\mu)G} \rightarrow 1, \quad \frac{\|\tilde{u} - u_h\|}{(h/\mu)^{1/2}G} \rightarrow 1.$$

On Γ_h , $v_\varphi \approx -(p - \bar{p})n$. So to leading order the printed method replaces the no-slip condition by the leak condition $(\mu/h) u_h \cdot n \approx p - \bar{p}$: the penalty, not a traction, balances the pressure.

Proof. Write $\tilde{u} - u_h = (\tilde{u} - z) + \delta_R + \delta_G$ in the notation of Theorem 6.4, with $\delta_R = (h/\mu)S_\varphi + r_h$ and $S_\varphi = v_\varphi$.

(1) From step 6 of Theorem 6.4,

$$c_1 \|r_h\|^2 \leq [C_p(h/\mu + h_\Gamma^{3/2}) + C\varepsilon_h] \|r_h\|.$$

So $\|r_h\| \leq C_p(h/\mu + h_\Gamma^{3/2}) + C\varepsilon_h$ and $\|r_h\|_{\Gamma_h} \leq (h/\mu)^{1/2} \|r_h\|$.

(2) $\|\delta_G\| + \|\tilde{u} - z\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$, and the Γ_h norm carries an extra factor $(h/\mu)^{1/2}$.

(3) Relative to $(h/\mu)^{1/2}G$ (energy) and $(h/\mu)G$ (slip), steps 1–2 contribute

$$O((h/\mu)^{1/2}/G + (\gamma^{1/2} + \gamma)h_\Gamma/G + \varepsilon_h(\gamma/h_\Gamma)^{1/2}/G) \rightarrow 0.$$

(4) By Lemma 3.1 (Jacobian and displacement of Γ_h from Γ), $\|v_\varphi\|_{\Gamma_h} = G(1 + O(h_\Gamma^2)) + O(h^k)$ and $\|v_\varphi\|^2 = (\mu/h)\|v_\varphi\|_{\Gamma_h}^2 + O(1)$. $\square$

This gives the sharp answer to Scott's question for the printed method. The estimate $h_\Gamma^{3/2} + h^k$ fails exactly when p is not constant on Γ . The error is then the explicit field $(h/\mu)v_\varphi$, with the universal constant 1 in both the slip norm and the energy norm.

6.1. The H^1 leak constant. Corollary 6.7 identifies the slip and energy constants of GS (both $\rightarrow 1$), but not the H^1 constant $\|\nabla(\tilde{u} - u_h)\| \mu/(hG)$, because the remainder $r_h = \delta_R - (h/\mu)S_\varphi$ of Theorem 6.4 has the same H^1 order as $(h/\mu)S_\varphi$. We identify the limit of $(\mu/h)\delta_R$: it is the Stokes flow in Ω driven by the leak as *Dirichlet* data. The H^1 constant is $K = \|\nabla U\|/G$, a global quantity that depends on the outer box (it tends to $3/\sqrt{7}$ as $L \rightarrow \infty$ for the pressure of test P). For test P and $L = 2.5$ a converged series computation gives $K = 2.7565157$, inside the rigorous bracket $[2.0746, 3.0526]$ of Proposition 6.13.

6.1.1. *The limit problem.*

Definition 6.8 (Leak flow). Let (U, P) solve

$$-\Delta U + \nabla P = 0, \quad \operatorname{div} U = 0 \text{ in } \Omega, \quad U = (p - \bar{p}) e_r = -(p - \bar{p}) n \text{ on } \Gamma, \quad U = 0 \text{ on } \partial R,$$

and put $K := \|\nabla U\|_{L^2(\Omega)}/G$.

The data are compatible, since $\int_{\Gamma} U \cdot n = -\int_{\Gamma} (p - \bar{p}) = 0$. Hence (U, P) exists and is unique (P up to a constant), $U = \operatorname{curl} \Psi$ with a single-valued stream function, and $U|_{\Gamma} = w|_{\Gamma}$ with $w = \operatorname{curl} \varphi$ the test field above. U is smooth up to Γ (smooth boundary and data) and $U \in H^2(\Omega)$ (the corners of R are convex). Since $-\Delta U = -\nabla P$, for every divergence-free V with the same boundary values $\int \nabla U : \nabla(V - U) = -\int \nabla P \cdot (V - U) = 0$, so

$$\|\nabla U\|^2 = \min \left\{ \|\nabla V\|_{L^2(\Omega)}^2 : V \in H^1(\Omega)^2, \operatorname{div} V = 0, V|_{\Gamma} = (p - \bar{p}) e_r, V|_{\partial R} = 0 \right\}. \quad (1)$$

In particular K depends on $p|_{\Gamma}$ and on the outer boundary: it is not a local quantity.

6.1.2. *Convergence.* Extend Ψ smoothly across Γ to a fixed neighbourhood and put $\tilde{U} := \operatorname{curl} \tilde{\Psi}$ on Ω_h , so $\operatorname{div} \tilde{U} = 0$; extend P smoothly to $\tilde{P}$. Then $F_U := -\Delta \tilde{U} + \nabla \tilde{P}$ vanishes on Ω , is bounded, and is supported in the strip S_{φ} .

Lemma 6.9. *There is $z_U \in Z_h$ with $\|\tilde{U} - z_U\| \leq E_U := C(h\|U\|_{H^2(\Omega)} + (1 + (\gamma h_{\Gamma})^{1/2})h^k + (1 + \gamma^{1/2})h_{\Gamma}^{\sigma_k - 1/2})$. For bounded γ , $E_U \leq Ch$, since $\sigma_k - \frac{1}{2} \geq k + \frac{1}{2}$.*

Proof. Repeat the proof of Lemma 4.6 with $\tilde{u}$ replaced by $\tilde{U}$ and $g = 0$. Away from Γ only $\tilde{U} \in H^2$ is available, so the interpolation step gives $\|\nabla(\tilde{U} - I_h \tilde{U})\| \leq Ch\|U\|_{H^2(\Omega)}$. Near Γ , $\tilde{U}$ is smooth and the boundary terms are as before. The net flux $m = \int_{\Gamma_h} (I_h \tilde{U} - \tilde{U}) \cdot n$ is $O(h_{\Gamma}^{\sigma_k})$ because $\int_{\Gamma_h} \tilde{U} \cdot n = \int_{\Omega_h} \operatorname{div} \tilde{U} = 0$. As in step 3 of that proof, the flux bubble then costs $C|m|(h_{\Gamma}^{-1/2} + (\mu/h)^{1/2}) \leq C(1 + \gamma^{1/2})h_{\Gamma}^{\sigma_k - 1/2}$; the Bogovskii correction (Lemma 4.5) is unchanged. $\square$

Theorem 6.10 (The leak flow is the H^1 limit). *Let $\gamma \geq \gamma_0$ and let $\delta_R \in Z_h$ solve $N_h(\delta_R, v) = \mathcal{R}(v)$ for all $v \in Z_h$ (Theorem 6.4, step 1). Then*

$$\left\| \nabla \left(\frac{\mu}{h} \delta_R - \tilde{U} \right) \right\|_{\Omega_h} \leq \left\| \frac{\mu}{h} \delta_R - z_U \right\| + E_U \leq C_U \left[\min(\gamma h_{\Gamma}^{1/2}, (\gamma h_{\Gamma})^{1/2}) + h_{\Gamma}^{1/2} + (h/\mu)^{1/2} + E_U \right],$$

where C_U depends on $(M0)-(M2)$, k , L and on norms of p and U near Γ .

Proof. Put $\epsilon := \delta_R - (h/\mu)z_U \in Z_h$. For $v \in Z_h$,

$$N_h(\epsilon, v) = \mathcal{R}(v) - \frac{h}{\mu} N_h(\tilde{U}, v) + \frac{h}{\mu} N_h(\tilde{U} - z_U, v). \quad (2)$$

(1) *Green's formula for $\tilde{U}$.* As in Lemma 5.1, $\operatorname{div} D(\tilde{U}) = \Delta \tilde{U}$, $\operatorname{div} v = 0$, $v|_{\partial R} = 0$ and $\int_{\Gamma_h} v \cdot n = 0$ give, for any constant c ,

$$N_h(\tilde{U}, v) = (F_U, v) + \langle \sigma_U, v \rangle - \langle \tilde{U}, \partial_n v \rangle + \frac{\mu}{h} \langle \tilde{U}, v \rangle, \quad \sigma_U := (\nabla \tilde{U})^T n - (\tilde{P} - c) n.$$

(2) *The penalty terms cancel to leading order.* With $\mathcal{R}(v) = -\langle (\tilde{p} - \bar{p})n, v \rangle$, (2) becomes

$$N_h(\epsilon, v) = -\langle \zeta, v \rangle - \frac{h}{\mu} \left[(F_U, v) + \langle \sigma_U, v \rangle - \langle \tilde{U}, \partial_n v \rangle \right] + \frac{h}{\mu} N_h(\tilde{U} - z_U, v), \quad \zeta := (\tilde{p} - \bar{p})n + \tilde{U}.$$

(3) *The mismatch ζ .* On $e \in \mathcal{E}_h^\Gamma$, by Lemma 3.1 and $\tilde{U}(x^*) = -(p - \bar{p})(x^*)n(x^*)$,

$$\zeta = (p - \bar{p})(x^*)(n_e - n(x^*)) + O(h_\Gamma^2) = (p - \bar{p})(m_e^*)s\tau_e + O(h_\Gamma^2),$$

exactly the functional m of Theorem 6.4, step 4 (without the h^k term). Hence $|\langle \zeta, v \rangle| \leq Ch_\Gamma^{3/2} \|\nabla v\|$. Alternatively $\|\zeta\|_{\Gamma_h} \leq Ch_\Gamma$ gives $|\langle \zeta, v \rangle| \leq Ch_\Gamma(h/\mu)^{1/2} \|v\|$.

(4) *The $O(h/\mu)$ terms.* $|(F_U, v)| \leq Ch_\Gamma^2 \|\nabla v\|$ (Lemma 3.3); $|\langle \sigma_U, v \rangle| \leq C\|v\|_{\Gamma_h} \leq C(h/\mu)^{1/2} \|v\|$. On Γ_h , $\tilde{U}$ is edgewise smooth, continuous at the vertices, and $\tilde{U} \cdot \tau_e = -(p - \bar{p})(x^*)n(x^*) \cdot \tau_e + O(h_\Gamma^2) = O(h_\Gamma)$. So Lemma 6.3 applies with $S = \tilde{U}$: $|\langle \tilde{U}, \partial_n v \rangle| \leq C(\|v\|_{\Gamma_h} + h_\Gamma^{1/2} \|\nabla v\|) \leq C((h/\mu)^{1/2} + h_\Gamma^{1/2}) \|v\|$. Finally $|N_h(\tilde{U} - z_U, v)| \leq ME_U \|v\|$ (Lemmas 4.2 and 6.9).

(5) *Close.* Test with $v = \epsilon$ and use Lemma 4.2:

$$c_1 \|\epsilon\| \leq C \left[\min(h_\Gamma^{3/2}, h_\Gamma(h/\mu)^{1/2}) + \frac{h}{\mu} (h_\Gamma^2 + (h/\mu)^{1/2} + h_\Gamma^{1/2} + E_U) \right].$$

Divide by h/μ , and use $h_\Gamma^{3/2} \mu/h = \gamma h_\Gamma^{1/2}$ and $h_\Gamma(\mu/h)^{1/2} = (\gamma h_\Gamma)^{1/2}$. The first inequality of the theorem is the triangle inequality with Lemma 6.9. $\square$

Corollary 6.11 (The H^1 constant). (i) *For GS with general data, in the notation of Theorem 6.4,*

$$\left\| \nabla \left(\frac{\mu}{h} (\tilde{u} - u_h) - \tilde{U} \right) \right\|_{\Omega_h} \leq (\text{bound of Theorem 6.10}) + C \frac{\mu}{h} \left[(1 + \gamma^{1/2}) h_\Gamma^{3/2} + \varepsilon_h \right].$$

(ii) *Consequently, if $h_\Gamma \rightarrow 0$ with $h/\mu \rightarrow 0$, $\gamma h_\Gamma \rightarrow 0$, $\gamma(1 + \gamma^{1/2}) h_\Gamma^{1/2} \rightarrow 0$ and $(\mu/h) \varepsilon_h \rightarrow 0$ (all of which hold for fixed γ and bounded ρ), then*

$$\frac{\|\nabla(\tilde{u} - u_h)\|_{\Omega_h}}{(h/\mu)G} \rightarrow K = \frac{\|\nabla U\|_{L^2(\Omega)}}{G}.$$

(iii) *On test P ($u = 0$, $f = \nabla p$) one has $\tilde{u} = 0$, $z = 0$ and $\mathcal{G} \equiv 0$, so $\tilde{u} - u_h = \delta_R$ exactly and Theorem 6.10 applies with no further condition.*

(iv) *The remainder of Theorem 6.4 converges: $(\mu/h)r_h \rightarrow U - w$ in H^1 . The limit is the Stokes correction of the cut-off field w : it vanishes on $\partial\Omega$ and solves $-\Delta(U - w) + \nabla P = \Delta w$. It depends on the cut-off χ_1 , whereas U does not.*

Proof. (i) is $\tilde{u} - u_h = (\tilde{u} - z) + \delta_R + \delta_G$ with steps 2 of Theorem 6.4 and Lemma 4.6. (ii) Multiply out; $\|\nabla \tilde{U}\|_{\Omega_h}^2 - \|\nabla U\|_\Omega^2 = \|\nabla \tilde{U}\|_{S_h}^2 = O(h_\Gamma^2)$. (iii) $F = \tilde{f} - \nabla \tilde{p} = 0$ and all terms of $\mathcal{G}$ contain $\tilde{u}$. (iv) $r_h = \delta_R - (h/\mu)v_\varphi$ and $\|\nabla(v_\varphi - w)\| \leq Ch^k$. $\square$

Remark 6.12 (The Dirichlet-dominated regime). The hypothesis $\gamma h_\Gamma = \mu h_\Gamma^2/h \rightarrow 0$ is the one of Corollary 6.7. It fails for the large- μ runs of Table 6 ($\gamma h_\Gamma \approx 240$ at $N = 96$, $\mu = 10^4$). There the discrete problem is close to its $\mu \rightarrow \infty$ limit, a discrete Dirichlet problem whose data $-(p - \bar{p})n_e$ jumps in direction at every vertex of Γ_h . The computations below show that $(\mu/h)\delta_R \rightarrow U$ in that regime too, with relative H^1 distance d_h decaying roughly like $h_\Gamma^{0.6} - h_\Gamma^{0.8}$, the decay slowing as N grows, but Theorem 6.10 does not cover it. Proving convergence uniformly in $\mu \geq \mu_0$ is open.

6.1.3. Rigorous bracket and exact asymptotics. For $\varrho > 1$ let $\mathcal{A}_\varrho = \{1 < r < \varrho\}$ and let $\mathcal{E}(\varrho)$ be the minimum (1) over $\mathcal{A}_\varrho$ (zero data on $r = \varrho$). Extending by zero shows that the minimum decreases as the domain grows.

Proposition 6.13. *If $\mathcal{A}_{\varrho_1} \subset \Omega \subset \mathcal{A}_{\varrho_2}$, then $\mathcal{E}(\varrho_2) \leq \|\nabla U\|^2 \leq \mathcal{E}(\varrho_1)$. In the annulus the Fourier modes decouple: if $\Psi|_\Gamma = \sum_n (a_n \cos n\theta + b_n \sin n\theta)$, then $\mathcal{E}(\varrho) = \sum_n (a_n^2 + b_n^2) E_n(\varrho)$, where $E_n(\varrho) = \pi \int_1^\varrho [f''^2 + 2n^2((f/r)')^2 + (f'/r - n^2 f/r^2)^2] r dr$, and f is the combination of $r^n, r^{-n}, r^{n+2}, r^{2-n}$ ($r, r^{-1}, r^3, r \log r$ for $n = 1$) with $f(1) = 1$, $f'(1) = 0$, $f(\varrho) = f'(\varrho) = 0$.*

For test P, $p - \bar{p} = -\frac{3}{4}\sin\theta + \frac{1}{4}\sin 3\theta + \frac{1}{2}\cos\theta$ on Γ , so $\Psi|_\Gamma = \frac{3}{4}\cos\theta + \frac{1}{2}\sin\theta - \frac{1}{12}\cos 3\theta$ and $G^2 = 7\pi/8$. With $\varrho_1 = L = 2.5$ and $\varrho_2 = L\sqrt{2}$ (40-digit quadrature):

$$2.0746004 \leq K \leq 3.0525769$$

The same formulas give the dependence on the box. $E_3(\varrho) \rightarrow 45\pi$ (the exterior clamped solution is $f = \frac{3}{2}r^{-1} - \frac{1}{2}r^{-3}$). $E_1(\varrho) \rightarrow \pi$: the $n = 1$ data split into a potential dipole (exterior energy exactly π) and a rigid translation, whose energy vanishes as $\varrho \rightarrow \infty$ (the Stokes paradox). The annulus values $E_1(\varrho) - \pi = 2.47, 0.87, 0.38$ at $\varrho = 10, 10^2, 10^4$ are consistent with the classical $1/\log \varrho$ rate, which we observe numerically here rather than prove. Hence

$$K(L) \rightarrow \sqrt{\frac{(\frac{9}{16} + \frac{1}{4})\pi + \frac{45\pi}{144}}{7\pi/8}} = \frac{3}{\sqrt{7}} \approx 1.134 \quad (L \rightarrow \infty),$$

but only logarithmically: the annulus values are $K = 1.420, 1.242, 1.183$ at $\varrho = 10, 10^2, 10^4$. So the constant 2.7–2.8 of Table 6 belongs to the box $L = 2.5$, not to the method.

6.1.4. *The value for test P. Series solution.* Write $\Psi = \sum_{n \text{ odd} \leq n_{\max}} f_n(r) \cos n\theta$ for data $\cos\theta$ and $\cos 3\theta$ separately (the class of $\cos n\theta$, n odd, is invariant under both reflections of the square; the $\sin\theta$ datum is the rotation of the $\cos\theta$ one by $\pi/2$, and cross terms between the two reflection classes vanish). Imposing $\Psi = 0$ on ∂R , rather than an unknown constant, is justified by symmetry: for data $\cos n\theta$ with n odd, the reflection $x \mapsto -x$ maps Ω to itself and changes the sign of the data, so by uniqueness Ψ is odd under it, and its constant value on ∂R is zero; the $\sin\theta$ datum follows by rotation. The conditions on $r = 1$ are imposed exactly, mode by mode, and $\Psi = \partial_\nu \Psi = 0$ on ∂R by least squares at 16 000 Chebyshev-clustered points. The energy is integrated by Gauss quadrature on eight polar sectors. The value is stable to 8 digits under refinement of both the truncation $n_{\max}$ and the quadrature:

$n_{\max}$	max boundary residual	$\ \nabla U\ $	K
15	$4.4 \cdot 10^{-3}$	4.57012143	2.75644148
31	$6.1 \cdot 10^{-5}$	4.57024239	2.75651444
47	$5.1 \cdot 10^{-6}$	4.57024449	2.75651571
63	$1.1 \cdot 10^{-6}$	4.57024450	2.75651571
79	$1.8 \cdot 10^{-7}$	4.57024450	2.75651571

$$K_P = 2.7565157$$

This lies inside the rigorous bracket.

Computation: `code/h1_limit.py annulus`; `code/h1_limit.py mps`; output `logs/h1_limit_series.log`.

Finite element check. Our solver (SuperLU with refinement, as in Section 13) was run on test P, and the discrete field was compared *pointwise* with the series U at the quadrature points. Both the normalised H^1 error K_h and the relative distance $d_h := \|\nabla((\mu/h)(\tilde{u} - u_h) - \tilde{U})\|/\|\nabla U\|$ are shown below. The $\mu = 100$ and $N = 96$ values of K_h reproduce Table 6 to all printed digits.

N	$\mu = 10^2$		$\mu = 10^3$		$\mu = 10^4$		$\mu = 10^8$	
	K_h	d_h	K_h	d_h	K_h	d_h	K_h	d_h
16	2.4642	0.193	2.7844	0.318	3.0562	0.539	3.3254	0.717
24	2.5629	0.147	2.7913	0.259	2.9309	0.398	3.0452	0.497
32	2.6117	0.124	2.7880	0.222	2.8751	0.323	2.9339	0.385
48	2.6601	0.098	2.7808	0.179	2.8283	0.246	2.8514	0.277
64	2.6843	0.084	2.7759	0.154	2.8082	0.206	2.8208	0.226
96	2.7084	0.067	2.7701	0.124	2.7899	0.163	2.7959	0.175

Computation: `code/h1_limit.py fem`; output `logs/h1_limit_fem.log`.

The measured values 2.71–2.79 bracket the limit 2.7565 from both sides. The approach runs in two directions.

- At $\mu = 100$, K_h increases towards K and d_h decays like $h^{0.6}$ ($0.193 \rightarrow 0.067$ as h drops by a factor 5.3). This is consistent with Theorem 6.10: for fixed μ , the terms $(h/\mu)^{1/2}$ and $h_\Gamma^{1/2}$ are both $\propto h^{1/2}$, and $(\gamma h_\Gamma)^{1/2} \propto h^{1/2}$ too.
- At large μ (Remark 6.12), K_h decreases towards K from above. The excess is the energy of an element-scale oscillation that is nearly orthogonal to U : $K_h^2 \approx K^2(1 + d_h^2)$, for example $2.7565\sqrt{1 + 0.175^2} = 2.798$ against 2.796 at $N = 96$.

At fixed $N = 96$ the μ -dependence of Table 6 ($2.708 \rightarrow 2.770 \rightarrow 2.790$) is the crossover between the two regimes, not a sign of a different limit.

7. THE PRESSURE-CONSISTENT METHOD

Proposition 7.1 (The naive consistent form is singular). *Replace $\langle p_h - \bar{p}_\Gamma(p_h), v \cdot n \rangle$ by $\langle p_h, v \cdot n \rangle$. The resulting linear system is square ($V_h^R \times \Pi_h$ tested by $V_h^R \times \Pi_h$). It has the kernel vector $(0, 1)$ and is therefore singular.*

Proof. For $v \in V_h^R$,

$$-(1, \operatorname{div} v) + \langle 1, v \cdot n \rangle = - \int_{\partial\Omega_h} v \cdot n + \int_{\Gamma_h} v \cdot n = 0.$$

The continuity row vanishes at $u = 0$. $\square$

The system is solvable only for right-hand sides orthogonal to a left null vector, which this argument does not identify. With generic data (Section 13), an iterated-penalty solver and a bordered direct solve both returned discrete velocities that are not divergence-free (discrete divergence about $4.5 \cdot 10^{-3}$). This is consistent with incompatibility.

Let β be the uniform inf-sup constant of Lemma 4.5.

Theorem 7.2 (CNS*: well posed, with Scott's rate). *There is $\gamma_2 \asymp \max(\gamma_0, \beta^{-2})$ such that for $\gamma \geq \gamma_2$ the method CNS* has a unique solution and*

$$\|\tilde{u} - u_h\| \leq C \left[(1 + \gamma^{1/2}) h_\Gamma^{3/2} + \varepsilon_h \right].$$

Consequently, for every fixed $k \geq 4$, for γ in a bounded range $[\gamma_2, \gamma_{\max}]$ and $h_\Gamma \geq h_\Omega^{2(\sigma_k - k)}$ (that is, $h_\Gamma \geq h_\Omega^4$ for even k and $h_\Gamma \geq h_\Omega^2$ for odd k),

$$\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C (h_\Gamma^{3/2} + h_\Omega^k).$$

This condition ensures that the flux term of ε_h is at most Ch^k . Without it, that single extra term is kept, and by the remark after Lemma 4.6 it is then genuinely present unless g_Γ is flux-corrected (which reduces it to a term below h^k). This is the rate Scott conjectured, for general Stokes data. The L^2 part of the H^1 norm uses the lift E from the proof of Theorem 6.6(i).

Proof. (1) *Consistency.* Write $B^*(q, v) := -(q, \operatorname{div} v) + \langle q - \bar{p}_\Gamma(q), v \cdot n \rangle$.

- For a constant c , $B^*(q + c, v) = B^*(q, v) - c \int_{\Gamma_h} v \cdot n$.
- $B^*(\tilde{p}, v)$ equals the pressure form of Lemma 5.1 minus $\bar{p}_\Gamma(\tilde{p}) \int_{\Gamma_h} v \cdot n$.
- Hence $p^* := \tilde{p} - \bar{p}_\Gamma(\tilde{p})$ satisfies $N_h(\tilde{u}, v) + B^*(p^*, v) = (\tilde{f}, v) + \mathcal{G}(v)$ for all $v \in V_h^R$.

With $e_u := \tilde{u} - u_h$ and $e_p := p^* - p_h$ this gives

$$N_h(e_u, v) + B^*(e_p, v) = \mathcal{G}(v) \quad \forall v \in V_h^R. \quad (3)$$

- (2) *Split.* Write $e_p = \eta + \xi$ with $\eta := p^* - \pi_h p^*$ and $\xi \in \Pi_h$. For $v \in V_h^0$, $(\eta, \operatorname{div} v) = 0$ because $\operatorname{div} V_h^0 \subset \Pi_h$.
- (3) *Pressure.* For $v \in V_h^0$, (3) gives

$$(\xi - \bar{\xi}, \operatorname{div} v) = N_h(e_u, v) - \mathcal{G}(v) = a_h(e_u, v) - \langle e_u, \partial_n v \rangle - \mathcal{G}(v).$$

Using $(\rho/\mu)^{1/2} \leq C$ and Lemma 5.2, the right-hand side is at most $C(\|e_u\| + h_\Gamma^{3/2})\|\nabla v\|$. Lemma 4.5 then gives $\beta\|\xi - \bar{\xi}\| \leq C(\|e_u\| + h_\Gamma^{3/2})$.

- (4) *Velocity.* For $z \in W_h$, $e_h := z - u_h \in Z_h$. Test (3) with $v = e_h$. Since $(e_p, \operatorname{div} e_h) = 0$ and $\langle \bar{p}_\Gamma(e_p), e_h \cdot n \rangle = 0$,

$$c_1 \|e_h\|^2 \leq M E_h \|e_h\| + |\mathcal{G}(e_h)| + |\langle \eta, e_h \cdot n \rangle| + |\langle \xi - \bar{\xi}, e_h \cdot n \rangle|,$$

where

$$|\langle \eta, e_h \cdot n \rangle| \leq C h_\Gamma^k (h/\mu)^{1/2} \|e_h\|,$$

$$|\langle \xi - \bar{\xi}, e_h \cdot n \rangle| \leq C_I (\rho/h)^{1/2} \|\xi - \bar{\xi}\| (h/\mu)^{1/2} \|e_h\| \leq C (c_0 \gamma)^{-1/2} \|\xi - \bar{\xi}\| \|e_h\|.$$

- (5) *Absorb.* Insert step 3 and $\|e_u\| \leq E_h + \|e_h\|$. If $\gamma \geq \gamma_2$ with $C(c_0 \gamma_2)^{-1/2} \beta^{-1} \leq c_1/2$, the pressure term is absorbed.
- (6) *Uniqueness.* Apply steps 3–5 to the homogeneous problem ($g_I = 0$, so $z = 0 \in W_h$). This gives $u_h = 0$, and then $p_h = c$ is constant. Now (3) reads $-c \int_{\Gamma_h} v \cdot n = 0$ for all v , and the edge bubble of Lemma 4.3 gives $c = 0$. The system is square, so uniqueness gives existence. $\square$

7.1. A matching lower bound: the rate $h_\Gamma^{3/2}$ is sharp. Theorem 7.2 gives $\|\nabla(\tilde{u} - u_h)\| \leq \|\tilde{u} - u_h\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$ for CNS*, and Theorem 6.1 gives the same bound for GS when $G = 0$. Here we prove the reverse inequality in the H^1 seminorm whenever the wall shear is not identically zero.

Status of the hypothesis. The proof needs a local condition (M3) on the boundary stars (Assumption 7.3). For $k \geq 7$ it holds on every mesh satisfying (M0)–(M2) (Lemma 7.4), so the lower bound is unconditional there. For $4 \leq k \leq 6$ we have *not* proved (M3). It is a finite-dimensional, scale-invariant condition that we verified numerically for $k = 4$ at every boundary vertex of the meshes of Section 13 (Table 1); since the space it involves grows with k , this verification also covers $k = 5, 6$ on those meshes (Remark 7.5). For $4 \leq k \leq 6$ on other meshes, the lower bound is conditional on (M3). The proof does *not* go through the transposed-gradient term of Lemma 3.5; see Section 7.1.5 for why that route fails for CNS*.

7.1.1. Notation. On Γ let n be the unit normal pointing into D and τ the unit tangent, oriented so that $\tau(x^*) \cdot \tau_e > 0$ for $x \in e$ (τ_e is n_e rotated in the same sense). By Lemma 3.4, $\partial_n u = W\tau$ on Γ , where

$$W := \partial_n u \cdot \tau \quad (\text{the wall shear}).$$

For $e \in \mathcal{E}_h^\Gamma$, T_e is the triangle with edge e , s is arclength on e from its midpoint m_e , and $W_e := W(m_e^*)$. No triangle has two edges on Γ_h : two such edges would share a vertex z of Γ_h , and the angle of Ω_h at z is $\pi + O(h_\Gamma) > \pi$. Define

$$Y_h := Z_h \cap V_h^0, \quad Y_h^1 := \left\{ v \in Y_h : \int_e \partial_n v = 0 \quad \forall e \in \mathcal{E}_h^\Gamma \right\}.$$

7.1.2. The local assumption. Write $\mathcal{M}(v) := -\frac{1}{2} \sum_{e \in \mathcal{E}_h^\Gamma} \int_e s^2 \partial_n v \cdot \tau_e ds$. For a vertex z of Γ_h , let e_z be the edge of Γ_h that follows z counterclockwise; $z \mapsto e_z$ is a bijection between the vertices and the edges of Γ_h .

Assumption 7.3 (Boundary test fields). **(M3)** There are constants $\kappa_0 > 0$ and K_{ov}, C_s such that every vertex z of Γ_h carries a field $v_z \in Y_h^1$ with the following properties:

- (i) $\|\nabla v_z\| = 1$;
- (ii) v_z is supported in a patch ω_z of triangles within distance $C_s h_\Gamma$ of z , and every triangle lies in at most K_{ov} patches;
- (iii) $\mathcal{M}(v_z) \geq \kappa_0 |e_z|^2$, with e_z as above.

Condition (iii) is invariant under scaling, since $\|\nabla v\|$ is scale invariant in two dimensions. Only the sign of v_z is chosen in (iii): $\mathcal{M}(-v) = -\mathcal{M}(v)$.

Lemma 7.4 ((M3) for $k \geq 7$). *If $k \geq 7$, (M3) holds with $\omega_z = T_{e_z}$. Here $K_{\text{ov}} = 1$ and κ_0 depends only on (M0).*

Proof. Let $T = T_{e_z}$, and let $\lambda_1, \lambda_2, \lambda_3$ be its barycentric coordinates with $\lambda_3 = 0$ on $e := e_z$. Put $b := \lambda_1 \lambda_2 \lambda_3$ and $\sigma := 2s/|e| = \lambda_1 - \lambda_2$ on e (up to sign), and set

$$\phi := b^2((\lambda_1 - \lambda_2)^2 - \frac{1}{7}) \in P_8, \quad v := \text{curl } \phi \in P_7^2.$$

- ϕ vanishes to second order on ∂T , so $v = 0$ on ∂T . Extended by zero, v is continuous, lies in $V_h^0 \subset V_h^R$ (because $k \geq 7$), and satisfies $\text{div } v = 0$. Hence $v \in Y_h$.
- On e , $\partial_n v \cdot \tau_e = \pm 2|\nabla \lambda_3|^2(\lambda_1 \lambda_2)^2(\sigma^2 - \frac{1}{7}) = \pm \frac{1}{8}|\nabla \lambda_3|^2(1 - \sigma^2)^2(\sigma^2 - \frac{1}{7})$.
- Since $\int_{-1}^1 (1 - \sigma^2)^2 d\sigma = \frac{16}{15}$ and $\int_{-1}^1 \sigma^2(1 - \sigma^2)^2 d\sigma = \frac{16}{105}$, this has mean zero. With $\partial_n v \cdot n_e = 0$ (Lemma 7.6 below), $\int_e \partial_n v = 0$, so $v \in Y_h^1$.
- Its second moment is $\int_{-1}^1 \sigma^2(1 - \sigma^2)^2(\sigma^2 - \frac{1}{7}) d\sigma = \frac{16}{315} - \frac{16}{735} = \frac{64}{2205} \neq 0$.

The other two edges of T are interior edges and do not enter $\mathcal{M}$. Hence $\mathcal{M}(v) \neq 0$. The ratio $|\mathcal{M}(v)|/(|e|^2 \|\nabla v\|)$ is invariant under similarity. It is a continuous, nonvanishing function of the shape of T , and the shapes allowed by (M0) form a compact set, so the ratio is bounded below by some $\kappa_0 > 0$. Normalise and choose the sign. $\square$

Remark 7.5 ($k = 4$). For $k = 4$ no field in Y_h^1 can be supported in a single triangle, or in the two triangles of one first-layer quadrilateral; for the production meshes the latter space is $\{0\}$ (Section 7.1.6). The smallest natural patch is the boundary star ω_z (the triangles at z).

Write $v = \text{curl } \phi$ with ϕ a C^1 piecewise quintic; since $v = 0$ on $\partial\omega_z$ and on Γ_h , we may take $\phi = 0$ and $\nabla \phi = 0$ on $\partial\omega_z$ and on the two edges e, e' of Γ_h at z . On such an edge e , $\partial_n v \cdot \tau_e = \pm \partial_{n_e n_e} \phi|_{T_e}$ is a cubic with zero mean ($v \in Y_h^1$). At the far endpoint of e the gradient of $\phi|_{T_e}$ vanishes along two independent lines, so its Hessian vanishes there. If also $\partial_{n_e n_e} \phi|_{T_e}(z) = 0$, the cubic vanishes at both ends of e and has zero mean, so it is odd about the midpoint and $\int_e s^2 \partial_n v \cdot \tau_e = 0$. Hence (M3) requires $\partial_{n_e n_e} \phi|_{T_e}(z) \neq 0$ for at least one of the two edges e at z , the value being taken from the triangle T_e . This is possible only because C^1 quintics can have Hessians that jump across the interior edges at z : an Argyris (C^2 -vertex) stream function has a single Hessian at z , which vanishes on both boundary directions and hence is zero. Such jumps exist at a boundary vertex with at least three triangles, which is (M1').

We have not proved (M3) for $k = 4$ on general meshes. It is a finite-dimensional, scale-invariant condition on each boundary star, and we verified it numerically:

- for every boundary vertex of every production mesh ($N = 8, 12, 16, 20, 24, 32, 48, 64, 96, 128, 192, 256$) and also for $N = 512$ and 1024 (Table 1): $\dim Y^1(\omega_z) = 1$, $\min_z \kappa_z \geq 0.00288$ in every case, and $\min_z \kappa_z$ converges to about 0.0029 as N grows;
- for randomly generated three- and four-triangle boundary stars: $\dim Y^1(\omega_z) = 2m - 5$ for m triangles, and $\kappa_z > 0$ in every sample.

(M3) is monotone in the degree: a field in Y_h^1 for degree k is continuous piecewise $P_{k'}$ for every $k' \geq k$, and $\mathcal{M}$ and $\|\nabla \cdot\|$ do not depend on k . So (M3) for $k = 4$ on a mesh implies (M3) for $k = 5, 6$ on the same mesh, and the numerical verification above covers $4 \leq k \leq 6$ on the meshes of Section 13. It is a numerical verification in floating point, not a proof.

7.1.3. *Three lemmas.*

Lemma 7.6. *If $v \in Y_h$ and $e \in \mathcal{E}_h^\Gamma$, then $\partial_n v \cdot n_e = 0$ on e .*

Proof. $v|_{T_e}$ is a polynomial that vanishes on e , so $\partial_\tau v = 0$ on e . Then $0 = \operatorname{div} v = \partial_\tau(v \cdot \tau_e) + \partial_n(v \cdot n_e)$ on e . $\square$

Lemma 7.7 (The error identity on Y_h). *Let u_h be the velocity of CNS* ($\gamma \geq \gamma_2$), or of GS ($\gamma \geq \gamma_0$, any G), and let $e_u := \tilde{u} - u_h$. Then for all $v \in Y_h$,*

$$a_h(e_u, v) - \langle e_u, \partial_n v \rangle = -\langle \tilde{u}, \partial_n v \rangle - (F, v).$$

Proof. For CNS* use (3): $B^*(e_p, v) = -(e_p, \operatorname{div} v) + \langle e_p - \bar{p}_\Gamma(e_p), v \cdot n \rangle = 0$, because $\operatorname{div} v = 0$ and $v = 0$ on Γ_h . For GS use Corollary 5.3: $\mathcal{R}(v) = -\langle \bar{p} - c, v \cdot n \rangle = 0$. In both cases $N_h(e_u, v) = \mathcal{G}(v)$. Since $v = 0$ on Γ_h , $N_h(e_u, v) = a_h(e_u, v) - \langle e_u, \partial_n v \rangle$, and in $\mathcal{G}(v)$ the transposed and penalty terms vanish, leaving $\mathcal{G}(v) = -\langle \tilde{u}, \partial_n v \rangle - (F, v)$. $\square$

Neither the pressure error nor the penalty enters this identity. It is simply the discrete momentum equation $a_h(u_h, v) - \langle u_h, \partial_n v \rangle = (f, v)$ on Y_h , which $\tilde{u}$ violates by $-\langle \tilde{u}, \partial_n v \rangle$ (Nitsche's symmetric term applied to the nonzero trace of $\tilde{u}$ on the chords) and by (F, v) .

Lemma 7.8 (Trace control on Y_h^1). *Let C_{PT} be the constant in $\|w - \bar{w}_T\|_{L^2(e)} \leq C_{PT}|e|^{1/2}\|\nabla w\|_T$ for $e \subset \partial T$, where $\bar{w}_T$ is the mean of w over T ; it depends only on shape regularity. Then for $w \in H^1(\Omega_h)^2$ and $v \in Y_h^1$,*

$$|\langle w, \partial_n v \rangle| \leq C_{PT} C_I \|\nabla w\| \|\nabla v\|.$$

Proof. On each e , $\int_e \partial_n v = 0$, so $\int_e w \cdot \partial_n v = \int_e (w - \bar{w}_{T_e}) \cdot \partial_n v$. Hence

$$|\langle w, \partial_n v \rangle| \leq \sum_e C_{PT} \|\nabla w\|_{T_e} |e|^{1/2} \|\partial_n v\|_{L^2(e)} \leq C_{PT} C_I \sum_e \|\nabla w\|_{T_e} \|\nabla v\|_{T_e},$$

using the local form of Lemma 4.1. Finish with Cauchy–Schwarz; the T_e are distinct. $\square$

Corollary 7.9 (Abstract lower bound). *With $\Lambda_0 := 2 + C_{PT} C_I$ and $e_u = \tilde{u} - u_h$ as in Lemma 7.7,*

$$\Lambda_0 \|\nabla e_u\| \geq \sup_{0 \neq v \in Y_h^1} \frac{|\langle \tilde{u}, \partial_n v \rangle + (F, v)|}{\|\nabla v\|}.$$

Proof. $|a_h(e_u, v)| \leq 2 \|\nabla e_u\| \|\nabla v\|$; apply Lemmas 7.7 and 7.8. $\square$

Lemma 7.10 (Leading term). *For $v \in Y_h^1$,*

$$\left| \langle \tilde{u}, \partial_n v \rangle - \ell_W(v) \right| \leq C h_\Gamma^{5/2} \|\nabla v\|, \quad \ell_W(v) := -\frac{1}{2} \sum_{e \in \mathcal{E}_h^\Gamma} W_e \int_e s^2 \partial_n v \cdot \tau_e ds,$$

and $|(F, v)| \leq C h_\Gamma^2 \|\nabla v\|$.

Proof. (1) *Geometry.* Let $x \in e$ and let $x^* = x/|x|$ be its nearest point on Γ . Then $x = x^* + d n(x^*)$ with $d = 1 - |x|$. Put $\ell = |e|$. Since $|x|^2 = 1 - \ell^2/4 + s^2$,

$$d = \frac{1}{2} \left(\frac{\ell^2}{4} - s^2 \right) + O(\ell^4).$$

(2) *Taylor expansion.* Using $\tilde{u}(x^*) = 0$ and Lemma 3.4,

$$\tilde{u}(x) = d W(x^*) \tau(x^*) + O(d^2).$$

Moreover $\tau(x^*) \cdot \tau_e = 1 + O(s^2)$ and $W(x^*) = W_e + O(|s|)$. Hence

$$\tilde{u} \cdot \tau_e = \frac{1}{2} \left(\frac{\ell^2}{4} - s^2 \right) W_e + O(h_\Gamma^3) \quad \text{on } e.$$

- (3) *Pairing on one edge.* By Lemma 7.6, $\tilde{u} \cdot \partial_n v = (\tilde{u} \cdot \tau_e)(\partial_n v \cdot \tau_e)$ on e . The constant part $\frac{\ell^2}{8} W_e$ integrates to zero against $\partial_n v \cdot \tau_e$, whose mean vanishes because $v \in Y_h^1$. Therefore

$$\int_e \tilde{u} \cdot \partial_n v = -\frac{1}{2} W_e \int_e s^2 \partial_n v \cdot \tau_e + O(h_\Gamma^3) \|\partial_n v\|_{L^1(e)}.$$

- (4) *Summation.* $\sum_e \|\partial_n v\|_{L^1(e)} \leq \sum_e |e|^{1/2} \|\partial_n v\|_{L^2(e)} \leq C_I (\#\mathcal{E}_h^\Gamma)^{1/2} \|\nabla v\| \leq C h_\Gamma^{-1/2} \|\nabla v\|$.
 (5) *The F term.* This is Lemma 3.3. $\square$

Lemma 7.11 (Assembling the test field). *Assume (M3), $W \not\equiv 0$, and $h_\Gamma \leq h_1$, where h_1 depends on $\|W\|_{W^{1,\infty}(\Gamma)}/\|W\|_{L^2(\Gamma)}$, on (M0), on C_I , and on the constants κ_0 , C_s , K_{ov} of (M3). Then $v := \sum_z W(z) v_z \in Y_h^1$ satisfies*

$$\ell_W(v) \geq \frac{1}{2} \kappa_0 c_0^{5/2} K_{\text{ov}}^{-1/2} \|W\|_{L^2(\Gamma)} h_\Gamma^{3/2} \|\nabla v\|.$$

Proof. (1) *Norm.* By bounded overlap, $\|\nabla v\|^2 \leq K_{\text{ov}} \sum_z W(z)^2$.

- (2) *Splitting ℓ_W .* Write $m_{z,e} := -\frac{1}{2} \int_e s^2 \partial_n v_z \cdot \tau_e$, so that $\sum_e m_{z,e} = \mathcal{M}(v_z)$. Then $|m_{z,e}| \leq \frac{h_\Gamma^2}{8} C_I$, and $m_{z,e} \neq 0$ only for the boundedly many edges e in ω_z . For those, $|W_e - W(z)| \leq C h_\Gamma \|W\|_{W^{1,\infty}}$. Hence

$$\ell_W(v) = \sum_z W(z) \sum_e W_e m_{z,e} \geq \sum_z W(z)^2 \mathcal{M}(v_z) - C h_\Gamma^3 \|W\|_{W^{1,\infty}} \sum_z |W(z)|.$$

- (3) *Lower bound.* By (M3)(iii) and (M0), $\mathcal{M}(v_z) \geq \kappa_0 c_0^2 h_\Gamma^2$. Moreover $\sum_z |W(z)| \leq \#\mathcal{E}_h^\Gamma \|W\|_\infty \leq C h_\Gamma^{-1} \|W\|_\infty$.
 (4) *Riemann sum.* Let $\Sigma_W := \sum_z W(z)^2 |e_z|$. Since $z \mapsto e_z$ is a bijection onto the edges, this is a Riemann sum for $\int_\Gamma W^2$, and $|\Sigma_W - \|W\|_{L^2(\Gamma)}^2| \leq C h_\Gamma \|W\|_{W^{1,\infty}}^2$; and $\sum_z W(z)^2 \in [\Sigma_W/h_\Gamma, \Sigma_W/(c_0 h_\Gamma)]$.
 (5) Combining,

$$\ell_W(v) \geq \kappa_0 c_0^2 h_\Gamma \Sigma_W - C h_\Gamma^2 \|W\|_{W^{1,\infty}}^2, \quad \|\nabla v\| \leq (K_{\text{ov}} \Sigma_W / (c_0 h_\Gamma))^{1/2}.$$

For $h_\Gamma \leq h_1$ we have $\Sigma_W \geq \frac{3}{4} \|W\|_{L^2(\Gamma)}^2$, and the negative term is at most a quarter of the positive one. $\square$

7.1.4. The theorem.

Theorem 7.12 (H^1 lower bound). *Assume (M0)–(M2) and (D), and assume (M3); for $k \geq 7$ the latter holds automatically (Lemma 7.4), while for $4 \leq k \leq 6$ it is a hypothesis. Let $W = \partial_n u \cdot \tau \not\equiv 0$ on Γ . Let u_h be the velocity of CNS* with $\gamma \geq \gamma_2$, or of GS with $\gamma \geq \gamma_0$ and any G . Then, for $h_\Gamma \leq h_1$,*

$$\|\nabla(\tilde{u} - u_h)\|_{\Omega_h} \geq c_{\text{LB}} \kappa_0 \|W\|_{L^2(\Gamma)} h_\Gamma^{3/2} - C h_\Gamma^2, \quad c_{\text{LB}} := \frac{c_0^{5/2}}{2 \Lambda_0 K_{\text{ov}}^{1/2}}.$$

Here c_{LB} depends on c_0 , on K_{ov} from (M3), and on C_{PT} and C_I (hence on (M0)–(M2)); h_1 is as in Lemma 7.11; and C depends on $\|\tilde{u}\|_{W^{2,\infty}}$, $\|W\|_{W^{1,\infty}}$ and $\|F\|_\infty$. Neither depends on γ , μ , ρ , h_Ω , the pressure, or ε_h .

Proof. Take v from Lemma 7.11 in Corollary 7.9, and use Lemma 7.10:

$$\Lambda_0 \|\nabla e_u\| \geq \frac{\ell_W(v)}{\|\nabla v\|} - C h_\Gamma^{5/2} - C h_\Gamma^2.$$

Lemma 7.11 bounds the first term from below. $\square$

Corollary 7.13 (Sharpness of Theorem 7.2 and of Theorem 6.1 at $G = 0$). *Assume (M0)–(M3), $W \not\equiv 0$, $\gamma \in [\gamma_2, \gamma_{\max}]$ and $\varepsilon_h \leq Ch_\Gamma^{3/2}$ (for instance $h_\Omega^k \leq h_\Gamma^{3/2}$ and the flux condition of Theorem 7.2). Then for CNS*, and for GS when p is constant on Γ ,*

$$ch_\Gamma^{3/2} \leq \|\nabla(\tilde{u} - u_h)\| \leq \|\tilde{u} - u_h\| \leq Ch_\Gamma^{3/2}.$$

So the H^1 rate is exactly 3/2. For GS with $G > 0$, Theorems 6.6 and 7.12 together give

$$\|\nabla(\tilde{u} - u_h)\| \geq c \max((h_\Gamma/\gamma)G, h_\Gamma^{3/2})$$

in the regime of Theorem 6.6.

- Remark 7.14.** (1) *Only the chordal bump matters.* The lower bound is carried by the part of $\tilde{u}|_{\Gamma_h}$ that oscillates on each chord, $-\frac{1}{2}s^2W_e$ minus its mean. It has amplitude h_Γ^2 and wavelength h_Γ , hence $H^{1/2}(\Gamma_h)$ -size $h_\Gamma^{3/2}$. No discrete velocity that is Nitsche-consistent on Y_h can reproduce it. The mean part $\frac{\ell^2}{12}W_e$ is smooth along Γ_h and contributes only $O(h_\Gamma^2)$.
- (2) *The terms that do not carry it.* The transposed-gradient term and the penalty term of $\mathcal{G}$ vanish identically on Y_h , as does every pressure term. The lower bound therefore holds for CNS* even though its pressure error is itself $O(h_\Gamma^{3/2})$.
- (3) *A method-independent statement.* Corollary 7.9 holds for *any* discrete velocity with $a_h(u_h, v) - \langle u_h, \partial_n v \rangle = (\tilde{f}, v)$ for all $v \in Y_h$. Every Nitsche variant with this volume form and zero boundary data, tested on Z_h , has this property.

7.1.5. *The route through the transposed-gradient term.* The natural first attempt tests with a field whose trace is odd on each chord, and uses $\mathcal{G}(v) \approx \langle s a_e, v \rangle$. We record what that route gives.

- (1) *The relevant trace is normal, not tangential.* Since $a_e = n(m_e^*)W_e$, we have $\langle s a_e, v \rangle = \sum_e W_e \int_e s v \cdot n_e + O(h_\Gamma^2)\|v\|_{L^1}$. An odd *tangential* trace pairs to zero at leading order.
- (2) *Incompressibility allows an odd normal trace.* For $v = \text{curl } \phi$, $v \cdot n_e = -\partial_\tau \phi$, so $\int_e v \cdot n_e = 0$ just says that ϕ takes equal values at the two ends of e . The identity $\partial_n v \cdot n_e = -\partial_\tau(v \cdot \tau_e)$ constrains the normal *derivative*, not the trace. So there is no obstruction there.
- (3) *The penalty term reinforces, at relative order γh_Γ .* On e ,

$$\tilde{u} \cdot n_e = dW(x^*)\tau(x^*) \cdot n_e + O(h_\Gamma^4) \quad \text{and} \quad (\nabla \tilde{u})^\top n_e \cdot n_e = W(x^*)\tau(x^*) \cdot n_e + O(h_\Gamma^2),$$

where $\tau(x^*) \cdot n_e = \pm s + O(s^3)$. So on the normal trace the penalty term multiplies the transposed term pointwise by $1 + (\mu/h)d(s)$, with $0 \leq (\mu/h)d \leq \gamma h_\Gamma/8$. It is lower order for fixed γ , of the same sign, and becomes leading once $\gamma \gtrsim h_\Gamma^{-1}$. On the tangential trace the penalty pairs $\frac{1}{2}(\frac{\ell^2}{4} - s^2)W_e$ with $v \cdot \tau_e$. This is not lower order unless $v \cdot \tau_e$ is odd or zero; choosing $\partial_n \phi = 0$ on Γ_h removes it.

- (4) *The Nitsche symmetric term is not lower order.* $-\langle \tilde{u}, \partial_n v \rangle$ pairs the chordal bump with $\partial_n v \cdot \tau_e$, which for a first-layer field of amplitude A is of size A/h_Γ . It has the same order $h_\Gamma A$ as the transposed term. This is the term Theorem 7.12 uses.
- (5) *Where the route fails: the pressure.* A test field with $v \cdot n \neq 0$ lies outside Y_h . For CNS* the error equation then contains $\langle \xi - \bar{\xi}, v \cdot n \rangle$, which by Theorem 7.2 (steps 3–4) is only bounded by

$$C(c_0\gamma)^{-1/2}\beta^{-1}(\|e_u\| + h_\Gamma^{3/2})\|v\|.$$

That is the same order as the term one is trying to bound below, so it can cancel the lower bound unless γ is large compared with constants we do not control.

- (6) *What the route gives when it works.* For GS with $G = 0$, where $|\mathcal{R}(v)| \leq Ch_\Gamma^2\|v\|_{L^1(\Gamma_h)}$ is lower order, the route gives only $\|\tilde{u} - u_h\| \gtrsim h_\Gamma^{3/2}/(1 + \gamma^{1/2})$ in the *energy* norm. That lower bound may be carried entirely by the slip part $(\mu/h)^{1/2}\|e_u\|_{\Gamma_h}$. Numerically this part

dominates: for GS on test A at $N = 64$, $\mu = 100$, it is 0.0795 against $\|\nabla e_u\| = 0.0182$. So it says nothing about H^1 .

7.1.6. *Numerical checks.* All computations use $k = 4$ and the meshes of Section 13. The script is `code/lower_bound_tests.py`.

TABLE 1. (M3) on the production meshes: star constants $\kappa_z := \mathcal{M}(v_z)/|e_z|^2$, with $\|\nabla v_z\| = 1$ and v_z supported in the three triangles at z . All production levels $N = 8, \dots, 256$ were checked; selected rows are shown.

N	$\dim Y^1(\omega_z)$	$\min_z \kappa_z$	mean	$\max_z \kappa_z$
8	1	0.00288	0.0055	0.0080
16	1	0.00555	0.0109	0.0179
64	1	0.00366	0.0145	0.0252
256	1	0.00303	0.0149	0.0260
1024	1	0.00288	0.0149	0.0261

Computation: `code/lower_bound_tests.py stars 8,12,...,1024`; output `logs/lower_bound_stars.log`.

The minimum is attained in the diagonal directions of the square, where the first-layer triangles are most elongated.

TABLE 2. The assembled field $v = \sum_z W(z)v_z$ of Lemma 7.11: $\langle \tilde{u}, \partial_n v \rangle / (\|\nabla v\| h_\Gamma^{3/2})$, with the leading term $\ell_W(v)$ in parentheses.

N	test A	rate	test B ₁	rate
32	0.0606 (0.0605)	–	0.0184 (0.0181)	–
64	0.0651 (0.0651)	1.39	0.0204 (0.0203)	1.35
128	0.0665 (0.0665)	1.47	0.0209 (0.0209)	1.46
256	0.0669 (0.0669)	1.49	0.0211 (0.0211)	1.49
512	0.0670 (0.0670)	1.50	0.0212 (0.0212)	1.50

Computation: `code/lower_bound_tests.py field 16,...,512`; output `logs/lower_bound_field.log`.

On the full spaces, for test A with $\mu = 100$ and $N = 16 \rightarrow 64$ (`logs/lower_bound_global.log`):

- The exact dual norm of $v \mapsto \langle \tilde{u}, \partial_n v \rangle$ on Y_h^1 has rates 1.12, 1.27, 1.35, 1.40; its ratio to $h_\Gamma^{3/2}$ increases from 0.086 to 0.115.
- It is between 14% and 24% of the actual CNS* error $\|\nabla(\tilde{u} - u_h)\|$.
- On Z_h , the energy-dual norms of $\mathcal{G}$ and of its transposed part decay at rates $1.70 \rightarrow 1.54$ and $1.61 \rightarrow 1.51$. So Lemma 3.5 is sharp on Z_h as well as on V_h^R .
- The $\|\nabla \cdot\|$ -dual norm of $\mathcal{G}$ on Z_h decays only like h_Γ^1 . It is dominated by the penalty part, which is of size γh_Γ in that norm. The $h_\Gamma^{3/2}$ upper bound genuinely needs the mesh-dependent norm.

8. THE PENALTY THRESHOLD SCALES WITH ρ

Proposition 8.1.

(a) Sufficiency. $\mu \geq \mu_0 = 4\kappa C_I^2 \rho$ implies coercivity (Lemma 4.2).

(b) Necessity, with an exact certificate. *Let z be a vertex of a straight part of the Nitsche boundary, and let ℓ_z be the length of its boundary edges. Suppose its star, dilated by the factor $1/\ell_z$, is congruent to one of the reference stars $\mathcal{S}_3$ or $\mathcal{S}'_4$ below. If $\mu/(h/\ell_z) < 9$, then N_h is indefinite on Z_h .*

The same holds, by continuity, at a vertex of Γ_h with interior angle $\pi + O(h_\Gamma)$ whose star is a small perturbation of $\mathcal{S}_3$ or $\mathcal{S}'_4$. This uses that both reference stars are non-singular: every vertex, including the rim vertices, has non-collinear incident edges. The dimension of the divergence-free subspace therefore equals the generic count and is locally constant. This fails at the singular $m = 2$ configuration, which $(M1')$ excludes.

So, for meshes that contain such a boundary star at a vertex with $\rho \asymp h/\ell_z$, the condition $\mu_0 \asymp \rho$ is both necessary and sufficient. Theorem 8.2 and Corollary 8.4 below replace the star hypothesis by a condition on a single boundary triangle. The threshold is local: what is required is $\mu/h \gtrsim c/\ell_z$. The ρ -scaling is the price of Scott's normalisation of the penalty by the global h .

Proof of (b). (1) In two dimensions, $a(v, v)$ and $\langle \partial_n v, v \rangle$ are invariant under dilation, and $\|v\|_{L^2(\Gamma_h)}^2$ scales linearly.

(2) A witness v on the reference star, extended by zero, lies in Z_h : it is continuous, vanishes on the star's outer chords and on ∂R , and is divergence-free.

(3) It satisfies $N_h(v, v) = a(v, v) - 2\langle \partial_n v, v \rangle + (\mu\ell_z/h)\|v\|_{\text{ref}}^2$.

(4) By the certificate below, $2B - A - 9C > 0$ on each reference star, so $N_h(v, v) < 0$ when $\mu\ell_z/h < 9$. $\square$

The exact certificate. The reference stars are fans at $O = (0, 0)$ with rational rim points:

- $\mathcal{S}_3$: rim points $(1, 0), (1, 1), (-1, 1), (-1, 0)$;
- $\mathcal{S}'_4$: rim points $(1, 0), (1, 1), (0, \frac{6}{5}), (-1, 1), (-1, 0)$.

The apex of $\mathcal{S}'_4$ is moved off the line $y = 1$. With apex $(0, 1)$ the rim vertex is singular, the kernel dimension jumps from 16 to 17, and the perturbation argument does not apply.

The Nitsche boundary is $\{y = 0\}$, and the rim chords carry zero Dirichlet data. The velocity is represented on each triangle by P_4 polynomials in global coordinates with rational coefficients. The computation is done in exact rational arithmetic (SymPy):

- Continuity across the spokes, vanishing on the rim, and pointwise $\text{div } v = 0$ are imposed as polynomial identities. This gives 80 and 110 linear constraints and kernel dimensions 14 and 16, the generic counts.
- The quadratic forms $A = \sum_T \int \frac{1}{2} Dv : Dv$, $B = \int (\partial_n v) \cdot v$ and $C = \int |v|^2$ are integrated exactly.
- A rational witness in the kernel satisfies $2B - A - 9C > 0$ *exactly* on each star. Its exact Rayleigh quotients $(2B - A)/C$ are 9.97854... ($\mathcal{S}_3$) and 9.26747... ($\mathcal{S}'_4$); these are lower bounds for the largest Rayleigh quotient λ^* of each star.

Computation: `code/lamstar_exact.py`, `code/lamstar_exact_v2.py`; output `logs/lamstar_exact_*.log`.

The witnesses are continuous piecewise- P_4 and divergence-free, so they lie in Z_h for every $k \geq 4$. The necessity statement therefore holds for all $k \geq 4$; for $k > 4$ the true local threshold is at least as large.

In floating point, the largest Rayleigh quotient λ^* is about 15 on uniform fans with $m = 3, 4$ and ranges over 14.8–38.5 on random 4-fans (`code/lamstar.py`). As a global check, the measured threshold $\gamma^* = \mu^* h_\Gamma / h$ stays in 18.1–20.1 as h/h_Γ varies over 3.3–44 on two mesh levels. Under refinement at fixed $h/h_\Gamma \approx 3.3$ it lies in 14.7–20.8, with increments that roughly halve per doubling (Table 7). Section 8.1 removes the reference-star hypothesis and explains the gap between the certified 9 and the measured 18–21.

8.1. Penalty necessity beyond two reference stars. Proposition 8.1(b) certifies indefiniteness of N_h on Z_h for $\mu\ell_z/h < 9$ only at vertices whose star is congruent to (or a perturbation of) $\mathcal{S}_3$ or $\mathcal{S}'_4$. We replace it by a statement about *every* boundary triangle, with an explicit closed-form constant and an exact certificate over a two-parameter family of shapes. We then explain the gap between the local constant and the global $\gamma^* \approx 18$ –21 of Table 7.

8.1.1. A witness supported in one boundary triangle. Let $T \in \mathcal{T}_h$ have an edge $e \subset \Gamma_h$, and let T not meet ∂R . Rotate, translate and dilate by $1/|e|$ so that $e = [(0, 0), (1, 0)]$, the fluid lies above and the apex is (a, b) , $b > 0$. Let $\lambda_0, \lambda_1, \lambda_2$ be the barycentric coordinates of the apex and of the base vertices $(0, 0)$, $(1, 0)$. For $c \in \mathbb{R}^3$ put

$$\psi_c := \lambda_1^2 \lambda_2^2 (c_0 \lambda_0 + c_1 \lambda_1 + c_2 \lambda_2) \in P_5, \quad v_c := \text{curl } \psi_c \in P_4^2.$$

Since ψ_c vanishes to second order on the two interior edges $\{\lambda_1 = 0\}$ and $\{\lambda_2 = 0\}$, v_c vanishes there. Extended by zero, v_c is continuous, zero on ∂R and pointwise divergence-free. So $v_c \in Z_h$ for every $k \geq 4$ and every mesh containing T . No condition on the vertices (singular or not), on the other triangles, or on the straightness of Γ_h beyond the edge e itself is needed.

Let $A_T = \int_T \frac{1}{2} Dv : Dv$, $B_T = \int_e (\partial_n v) \cdot v$ and $C_T = \int_e |v|^2$, with $n = (0, -1)$, as quadratic forms in c . They are computed exactly (SymPy) and, writing $\varpi := 4a^2 + 4b^2$, are

$$C_T = \frac{1}{1260 b^2} \begin{pmatrix} 2 & a-6 & -(a+5) \\ a-6 & 2(\varpi-3a+9) & \varpi-4a+15 \\ -(a+5) & \varpi-4a+15 & 2(\varpi-5a+10) \end{pmatrix},$$

$$B_T = \frac{1}{840 b^3} \begin{pmatrix} 12 & -(\varpi-6a+27) & -(\varpi-2a+25) \\ -(\varpi-6a+27) & 6(\varpi-3a+9) & 3(\varpi-4a+15) \\ -(\varpi-2a+25) & 3(\varpi-4a+15) & 6(\varpi-5a+10) \end{pmatrix},$$

and $A_T = (\alpha_{ij})/b^3$ with

$$\begin{aligned} 630 \alpha_{00} &= 3a^4 - 6a^3 + 6a^2b^2 + 12a^2 - 6ab^2 - 9a + 3b^4 + 4b^2 + 9, \\ 420 \alpha_{01} &= 3a^4 + 6a^2b^2 - 5a^2 + 6a + 3b^4 - 7b^2 - 9, \\ 420 \alpha_{02} &= 3a^4 - 12a^3 + 6a^2b^2 + 13a^2 - 12ab^2 - 8a + 3b^4 - b^2 - 5, \\ 70 \alpha_{11} &= 3a^4 - 3a^3 + 6a^2b^2 + 5a^2 - 3ab^2 - 3a + 3b^4 + 7b^2 + 3, \\ 140 \alpha_{12} &= a^4 - 2a^3 + 2a^2b^2 + 5a^2 - 2ab^2 - 4a + b^4 + 7b^2 + 5, \\ 70 \alpha_{22} &= 3a^4 - 9a^3 + 6a^2b^2 + 14a^2 - 9ab^2 - 10a + 3b^4 + 10b^2 + 5. \end{aligned}$$

$\det C_T = (3a^4 - 6a^3 + 7a^2b^2 + 3a^2 - 7ab^2 + 4b^4 + 5b^2)/(83\,349\,000 b^6) > 0$. (The forms were checked against an independent quadrature/finite-difference evaluation, relative difference $4.5 \cdot 10^{-9}$, and λ_T below against our penalty-threshold code on single-triangle patches of the Section 13 meshes: 10.48912 vs. 10.48912.)

Theorem 8.2 (Single-triangle necessity). *Let λ_T be the largest eigenvalue of the pencil $(2B_T - A_T, C_T)$; it depends only on the shape of T . If $\mu|e|/h < \lambda_T$, then N_h is indefinite on Z_h for every $k \geq 4$. Consequently every mesh satisfies the necessary condition*

$$\mu \geq h \max_{e \in \mathcal{E}_h^\Gamma} \frac{\lambda_{T_e}}{|e|}.$$

Proof. In two dimensions A and B are dilation invariant and C scales like $|e|$. So for the eigenvector c , $N_h(v_c, v_c) = A_T - 2B_T + (\mu|e|/h)C_T = (\mu|e|/h - \lambda_T) C_T(c) < 0$. $\square$

Proposition 8.3 (Exact certificates over shape families). *For $\varphi \in (0, \pi)$ let Q_φ be the set of apex positions (a, b) for which the apex angle is at least φ and $b \geq s \max(a, 1 - a)$, with $s = 0.087 < \tan 5^\circ$. This contains every triangle with apex angle $\geq \varphi$ and base angles $\geq 5^\circ$. Then*

$$\lambda_T > 9.3 \text{ on } Q_{60^\circ}, \quad \lambda_T > 4 \text{ on } Q_{45^\circ}, \quad \lambda_T > 1 \text{ on } Q_{35^\circ}.$$

The certificate proves only the three strict lower bounds. As numerical observations, not certified (`logs/lamstar_family_levels.log`): the first bound is nearly sharp, since a scan gives the minimum 9.31381 over Q_{60° , at the equilateral triangle; along the level sets of the apex angle the scanned minima are 6.07 (50°), 4.47 (45°), 2.83 (40°), 1.12 (35°) and -0.73 (30°), each at the isosceles shape.

Proof (computer-assisted, exact). Q_φ is the part of the disk through $(0, 0)$, $(1, 0)$ with inscribed angle φ that lies above the two lines. Cover a rational box containing Q_φ by dyadic sub-boxes. A box is discarded only if it provably misses Q_φ , using outward-rounded rational enclosures of the disk. On every other box, take a rational vector q (the eigenvector at the box centre, rounded to denominators ≤ 1000) and verify

$$\Phi_q(a, b) := 2520 b^3 q^\top (2B_T - A_T - cC_T)q > 0$$

on the whole box. Φ_q is a polynomial with rational coefficients (the denominators clear exactly), and the lower bound is evaluated monomial-wise in exact rational interval arithmetic. The boxes are split until this succeeds.

- $c = 9$ and $c = 9.3$ on Q_{60° : 323 and 716 boxes, depth ≤ 11 .
- $c = 4$ on Q_{45° : 259 boxes.
- $c = 1$ on Q_{35° : 297 boxes.

As negative controls, the same procedure fails for $c = 9.4$ on Q_{60° and for $c = 4.5$ on Q_{45° , as it must. $\square$

Computation: `code/lamstar_family.py bubble`; `code/lamstar_family.py certify`; output `logs/lamstar_family_bubble.log`, `logs/lamstar_family_certify.log`.

Corollary 8.4 (ρ -scaling is necessary on a broad class). *Let e_* be a shortest edge of Γ_h and suppose its triangle T_{e_*} has apex angle $\geq 35^\circ$ and base angles $\geq 5^\circ$, and does not meet ∂R (automatic for small h , since $\text{diam } T_{e_*} \leq Ch_\Gamma < L - 1$). Then coercivity of N_h on Z_h requires $\mu > \rho$. If the apex angle is $\geq 45^\circ$ (resp. $\geq 60^\circ$) it requires $\mu > 4\rho$ (resp. $\mu > 9.3\rho$). Together with Lemma 4.2, $\mu_0 \asymp \rho$ is necessary and sufficient on every such mesh, with no star-congruence or perturbation hypothesis.*

Remark 8.5 (What the single triangle cannot do). $\lambda_T < 0$ for tall triangles, for example every right triangle with the right angle at a base vertex and apex angle $\leq 30^\circ$ (numerically $\lambda_T \approx -0.2500$ at apex 30° , decreasing to -15.7 at 10°). A uniform statement under (M0)–(M2) alone would need witnesses on several triangles. In the Section 13 meshes such tall boundary triangles occur near the corner directions; at $N = 64$ the smallest value is $\lambda_T = -3.8$, with scaled apex about $(-0.08, 2.38)$ (`logs/lamstar_family_levels.log`). There the vertex stars and first-layer windows still have positive λ^* . Proving $\lambda_{\text{star}}^* \geq c(\Theta_0) > 0$ for every nonsingular boundary star remains open. The float evidence (random 4-fans, $\lambda^* \in [14.8, 38.5]$) suggests it holds.

8.1.2. *Why the certified constant is below the global threshold.* The global threshold is $\gamma^* = h_\Gamma \sup_{Z_h} (2B - A)/C$, where A is taken over Ω_h , B, C over Γ_h , and $h_\Gamma = \max |e|$. Every patch witness gives a lower bound. For the Section 13 meshes we computed the patch thresholds with the penalty code of Section 13 (bisection on the inertia of $A_\mu + rB^\top M^{-1}B$, fields vanishing outside the patch). We maximised over all boundary triangles and single-vertex stars. *winW* is the first layer restricted to W columns centred at $\theta = 0$, and the column J is the patch made of the first J

full boundary layers, computed for $J \leq 2$ here (the log also has $J = 3, 4$, which change the values by at most 0.002). The *global* column is Table 7(a); for $N \leq 16$ the full mesh is $J = N/4 \leq 4$ layers and the log reproduces it.

N	triangle	star	win2	win4	win8	win16	$J = 1$	$J = 2$	global
8	6.13	13.71	13.91	14.65	–	–	14.68	14.74	14.74
16	8.93	16.58	16.89	17.91	17.94	–	17.95	18.09	18.09
32	10.08	17.48	17.82	19.54	19.69	19.69	19.69	19.85	19.85
64	10.49	17.70	18.03	20.29	20.68	20.69	20.69	20.80	20.80

Computation: `code/lamstar_family.py mesh`; output `logs/lamstar_family_mesh.log`.

The limit geometry. The meshes of Section 13 do not have equal chords. The square points are equispaced and projected radially, so $\max |e| / \min |e|$ grows from 1.43 ($N = 16$) to 1.97 ($N = 256$). The longest chords are at the side midpoints $\theta = 0, \pi/2, \dots$. There the first layer, scaled by the chord, tends to columns $[i, i+1] \times [0, \frac{3}{4}]$ (layer height $(L-1)/2 = \frac{3}{4}$) split on the forward diagonal. The single-vertex star of that geometry is $\mathcal{S}_{3/4}$, with rim $(1, 0), (1, \frac{3}{4}), (0, \frac{3}{4}), (-1, 0)$. The general exact code (continuity, Dirichlet and divergence constraints over $\mathbb{Q}$, exact barycentric moments) gives:

	star	$W = 2$	$W = 4$	$W = 8$	$W = 16$	$W = 32$
1 layer	17.7064	17.9759	20.4872	21.1726	21.3358	21.3758
2 layers	–	18.1738	20.6781	21.2432	21.3675	21.3974
3 layers	–	18.1916	20.6811	21.2462	21.3708	–

Exact certificates in the limit geometry: $2B - A - 17C > 0$ on $\mathcal{S}_{3/4}$ (dimension 14, witness quotient 17.70640; also confirmed by `code/lamstar_exact.py`), and $2B - A - 20C > 0$ on the $W = 4$ window (dimension 41, witness quotient 20.48725).

Computation: `code/lamstar_family.py limit`; `lamstar_exact.run` on $\mathcal{S}_{3/4}$; output `logs/lamstar_family_limit.log`, `logs/lamstar_family_limitstar.log`.

The perturbation argument of Proposition 8.1(b) transfers these strict inequalities to the actual meshes near $\theta = 0$ for all sufficiently large N (non-explicit N_0). All patch vertices keep at least three distinct edge lines, or are Dirichlet corners where v and ∇v vanish, so the kernel dimension is locally constant. This gives $\gamma^*(N) > 20$ for $N \geq N_0$. The finite- N values in the first table agree: win4 is 20.29 at $N = 64$.

The gap between the certified 9 and the measured 18–21 is therefore accounted for, in order of size, by:

- (1) *Shape.* The reference stars $\mathcal{S}_3, \mathcal{S}'_4$ have layer height 1 per unit chord. The meshes have height $\frac{3}{4}$ at the longest chords, which is where γ^* is decided because γ is normalised by $\max |e|$. λ^* grows fast as the layer flattens: the exact witness quotients, which are lower bounds for λ^* , are 9.98 for $\mathcal{S}_3$ against 17.71 for $\mathcal{S}_{3/4}$.
- (2) *Collective boundary modes.* The optimal witness spreads over 4–8 boundary edges and adds about 3.7 ($17.7 \rightarrow 21.4$). Windows wider than about 16 edges add nothing.
- (3) *Depth.* In the limit geometry the second layer adds 0.20 ($W = 2$) and 0.19 ($W = 4$), decreasing to 0.02 at $W = 32$; the third layer adds 0.018 at $W = 2$ and at most 0.004 for $W \geq 4$. On the actual meshes the second layer adds 0.11–0.16 (compare $J = 1$ and $J = 2$). γ^* is a boundary-layer quantity (compare $J = 2$ with the global value).
- (4) *Finite N .* Away from the limit geometry (curvature, the varying chord and height along the layer) $\gamma^*(N)$ is lower. Richardson extrapolation of the window sequences (differences shrink by a factor of about 4 per doubling of W) gives $\gamma_\infty^* \approx 21.39$ (one layer) and 21.41 (two layers), so we expect $\gamma^*(N) \rightarrow \approx 21.4$ on this mesh family.

9. CLOUGH–TOCHER REFINEMENTS: EVERY $k \geq 2$

Sections 2–8 assume continuous P_k velocities with $k \geq 4$ on a mesh satisfying (M0)–(M2). Here we show that the results of Sections 3–8, through Theorem 7.2 and Proposition 8.1, hold for all $k \geq 2$ on Clough–Tocher (barycentric, Alfeld) refinements, and that (M1), (M1') and (M2) can be dropped (the precise scope is Theorem 9.8 and Remark 9.12). Only four ingredients change: the inf-sup constant, the divergence range, the flux exponent in Lemma 4.6, and the C^1 test field v_φ . The penalty-necessity constants of Proposition 8.1 also change, and we recertify them exactly.

9.1. Setting. Let $\mathcal{T}_h^M$ be a conforming triangulation of Ω_h (the *macro mesh*). Its Clough–Tocher refinement $\mathcal{T}_h$ splits every $T \in \mathcal{T}_h^M$ at its barycentre g_T into three triangles. Macro edges are not split, so the edges of $\mathcal{T}_h$ on Γ_h and on ∂R are exactly the macro edges there, and $\mathcal{E}_h^\Gamma$ is unchanged. For $k \geq 2$, V_h is the space of continuous piecewise- P_k fields on $\mathcal{T}_h$ and Π_h the space of discontinuous piecewise- P_{k-1} functions on $\mathcal{T}_h$. V_h^R , V_h^0 , Z_h , W_h , the forms a_h and N_h , the methods GS and CNS* and the norm $\|\cdot\|$ are defined by the formulas of Section 2, applied to these spaces. We put $h := \max_{T \in \mathcal{T}_h^M} \text{diam } T$ and keep h_Γ , ρ , γ as before.

Assumption 9.1.

(CT0) $\mathcal{T}_h^M$ is shape regular: $\text{diam } T \leq \sigma r_T$ for every $T \in \mathcal{T}_h^M$, where r_T is the inradius. Moreover $c_0 h_\Gamma \leq |e| \leq h_\Gamma$ for all $e \in \mathcal{E}_h^\Gamma$.

Assumption (D) of Section 2 is kept. No analogue of (M1), (M1') or (M2) is assumed.

In this section C and c depend on σ , c_0 , k , L and norms of $(u, p, \tilde{\psi})$, and on nothing else.

Lemma 9.2 (Structure of the refined mesh). *Assume (CT0) and $h \leq h_*$.*

- (a) $\mathcal{T}_h$ is shape regular with a constant depending only on σ , and $\text{diam } K \geq c \text{diam } T$ for $K \subset T$.
- (b) Each $e \in \mathcal{E}_h^\Gamma$ is an edge of exactly one $K_e \in \mathcal{T}_h$, namely $K_e = \text{conv}(e \cup \{g_{T_e}\})$, where $T_e \in \mathcal{T}_h^M$ is the macro triangle on e . Its height over e is one third of that of T_e , and $\text{diam } K_e \leq C|e|$. No macro triangle has two edges on Γ_h . Every vertex of Γ_h lies in at least two macro triangles, hence in at least four triangles of $\mathcal{T}_h$.
- (c) $\mathcal{T}_h$ has no singular vertex in the sense of (M1). It satisfies (M2) with a constant $\Theta_0 = \Theta_0(\sigma) > 0$.

In particular, $\mathcal{T}_h$ satisfies (M0)–(M2) of Assumption 2.1, with constants depending only on σ and c_0 .

Proof. (a) For $K \subset T$ we have $|K| = |T|/3$ and $\text{diam } K \leq \text{diam } T$. Hence $\text{diam } K^2/|K| \leq 3 \text{diam } T^2/|T| \leq C(\sigma)$, which bounds the angles of K from below. Also $\text{diam } K \geq$ the length of the macro edge of K , which is $\geq c \text{diam } T$.

- (b) The barycentre lies at one third of each height, so K_e is the only refined triangle with an edge on e , and $\text{diam } K_e \leq \text{diam } T_e \leq C|e|$.

The interior angle of Ω_h at a vertex z of Γ_h is $\pi + O(h_\Gamma)$. Every macro angle is at most $\pi - 2\theta_{\min}(\sigma)$. So for $h \leq h_*$ at least two macro triangles meet at z . A macro triangle with two edges on Γ_h would have both edges at a common vertex z of Γ_h , and its angle at z would be that interior angle, which is impossible.

- (c) Consider a macro vertex z with incident macro triangles $T_1, \dots, T_m$. The refined edges at z are the macro edges and the medians zg_{T_i} , and each median lies strictly inside the angle of T_i at z .

- If z is interior, $m \geq 3$, so z has $2m \geq 6$ refined edges. At most four rays from z lie on two lines, so z is not singular.
- If z is a boundary vertex, it has $2m + 1 \geq 3$ refined edges. For $m \geq 2$ there are at least 5 > 4 of them. For $m = 1$ the three rays are two macro edges, at an angle $< \pi$, and a median strictly between them, so no two are opposite.

- A barycentre has three edges, and no two are opposite since g_T is interior.
- Every vertex lies in at least two refined triangles.

For (M2), at a macro vertex the two refined angles α'_i, α''_i inside T_i are consecutive (never the wrap-around pair), and they add up to the macro angle $\alpha_i \in [\theta_{\min}, \pi - 2\theta_{\min}]$. Hence $\Theta(z) \geq \sin \theta_{\min}$.

At a barycentre, the angles ϕ_1, ϕ_2, ϕ_3 satisfy $\sum \phi_j = 2\pi$ and, by (a), $\phi_j \leq \pi - 2\theta'_{\min}(\sigma)$. Moreover $|\sin(\phi_j + \phi_{j+1})| = \sin \phi_{j+2}$ and $\max_j \phi_j \geq 2\pi/3$. Hence $\Theta(g_T) \geq \min(\sin \frac{2\pi}{3}, \sin 2\theta'_{\min})$. $\square$

By Lemma 9.2, for $k \geq 4$ the paper applies verbatim to $\mathcal{T}_h$. The new content is $k \in \{2, 3\}$, together with the fact that no angle or singular-vertex hypothesis on the macro mesh is needed for any k . The proofs below cover all $k \geq 2$ uniformly, except the C^1 test field of Lemma 9.7.

9.2. Divergence: local and global. For a triangle T with Clough–Tocher split T^{ct} , let $V_0(T)$ be the space of continuous piecewise- P_k fields on T^{ct} vanishing on ∂T , and $Q_0(T)$ the space of discontinuous piecewise- P_{k-1} functions on T^{ct} with mean zero on T .

Lemma 9.3 (Local surjectivity). *Let $k \geq 2$. Then $\text{div } V_0(T) = Q_0(T)$. For every $q \in Q_0(T)$ there is $v \in V_0(T)$ with $\text{div } v = q$ and $\|\nabla v\|_T \leq C_{\text{loc}} \|q\|_T$, where C_{loc} depends only on k and the shape regularity of T .*

Proof. (1) *Reference element.* Let $\hat{T}$ be the reference triangle, with its Clough–Tocher split. We show that $\text{div} : V_0(\hat{T}) \rightarrow Q_0(\hat{T})$ is onto.

- $k = 2$. Here $\dim V_0(\hat{T}) = 8 = \dim Q_0(\hat{T})$, so it suffices that div is injective. A divergence-free $v \in V_0(\hat{T})$ is $\text{curl } \psi$ with $\psi \in C^1(\hat{T})$ piecewise cubic on the split and $\psi = \nabla \psi = 0$ on $\partial \hat{T}$ (the boundary is connected). All twelve Hsieh–Clough–Tocher degrees of freedom of ψ [4, 3] then vanish, so $\psi = 0$.
- $k \geq 4$. The split is a three-triangle mesh of $\hat{T}$ with no singular vertex (Lemma 9.2(c) applied to one macro triangle), so the Scott–Vogelius theorem [17, 11] applies on the domain $\hat{T}$.
- $k = 3$ (and, as a check, $k = 2, 4, 5, 6$). An exact rational rank computation gives $\text{rank}(\text{div}|_{V_0(\hat{T})}) = \dim Q_0(\hat{T})$. The dimensions are $\dim V_0 = 8, 20, 38, 62, 92$, $\dim Q_0 = 8, 17, 29, 44, 62$ and $\dim \ker \text{div} = 0, 3, 9, 18, 30$.
- The statement for all $k \geq d = 2$ is also the local ingredient of Guzmán–Neilan [10]; for $k = 2$ it is Arnold–Qin [2].

A surjective linear map between finite-dimensional spaces has a bounded right inverse. Let $\hat{C}_k$ be its norm.

- (2) *Piola transfer.* Let $F(\hat{x}) = B\hat{x} + b$ map $\hat{T}$ onto T . Affine maps preserve barycentres, so F maps $\hat{T}^{\text{ct}}$ onto T^{ct} . For $q \in Q_0(T)$, put $\hat{q} := (\det B) q \circ F \in Q_0(\hat{T})$. Take $\hat{v}$ with $\widehat{\text{div}} \hat{v} = \hat{q}$ and $\|\hat{\nabla} \hat{v}\| \leq \hat{C}_k \|\hat{q}\|$, and set $v := (\det B)^{-1} B \hat{v} \circ F^{-1}$. Then $v \in V_0(T)$, $\text{div } v = q$ and $\|\nabla v\|_T \leq \hat{C}_k \|B\| \|B^{-1}\| \|q\|_T$. $\square$

Computation: `code/lamstar_exact_ct.py` (Part 1); output `logs/lamstar_exact_ct.log`.

Lemma 9.4 (Uniform inf-sup constant, Clough–Tocher). *Let $k \geq 2$ and assume (CT0). There is $\beta > 0$, depending only on k, σ and L , such that for every $q \in \Pi_h \cap L_0^2(\Omega_h)$ there exists $v \in V_h^0$ with $\text{div } v = q$ and $\|\nabla v\| \leq \beta^{-1} \|q\|$.*

Proof. (1) *Continuous level.* By the second half of the proof of Lemma 4.5 (the Galdi wedge cover, which involves only Ω_h and not the mesh), there is $\beta_c > 0$ independent of h with the following property: for each $f \in L_0^2(\Omega_h)$ there is $w \in H_0^1(\Omega_h)^2$ with $\text{div } w = f$ and $\|\nabla w\| \leq \beta_c^{-1} \|f\|$.

- (2) *Macro Fortin step.* Let Π^{SZ} be the Scott–Zhang interpolant [18] onto continuous piecewise- P_1 fields on $\mathcal{T}_h^M$ that vanish on $\partial\Omega_h$. For an interior macro edge E , let $b_E = \lambda_a \lambda_b$ be its quadratic macro bubble. Set

$$w_h := \Pi^{\text{SZ}} w + \sum_E c_E b_E, \quad c_E := \frac{6}{|E|} \int_E (w - \Pi^{\text{SZ}} w) \in \mathbb{R}^2.$$

Then w_h is continuous piecewise- P_2 on $\mathcal{T}_h^M$, hence in V_h^0 for $k \geq 2$, and $\int_E w_h = \int_E w$ on every macro edge (both vanish on $\partial\Omega_h$). By the divergence theorem, $\int_T \operatorname{div} w_h = \int_T \operatorname{div} w = \int_T q$ for every $T \in \mathcal{T}_h^M$. The trace inequality and the local estimates of Π^{SZ} give $|c_E| \leq C \|\nabla w\|_{\omega_E}$. Since $\|\nabla b_E\| \leq C$ in two dimensions, $\|\nabla w_h\| \leq C \|\nabla w\| \leq C \beta_c^{-1} \|q\|$.

- (3) *Local step.* On each $T \in \mathcal{T}_h^M$, $r := q - \operatorname{div} w_h$ is piecewise P_{k-1} on T^{ct} (since $\operatorname{div} w_h|_T \in P_1$) and has mean zero on T . Lemma 9.3 gives $v_T \in V_0(T)$ with $\operatorname{div} v_T = r|_T$ and $\|\nabla v_T\|_T \leq C_{\text{loc}} \|r\|_T$. Extended by zero, $v_T \in V_h^0$.
- (4) Put $v := w_h + \sum_T v_T \in V_h^0$. Then $\operatorname{div} v = q$ and $\|\nabla v\| \leq \|\nabla w_h\| + C_{\text{loc}} (\|q\| + \sqrt{2} \|\nabla w_h\|) \leq \beta^{-1} \|q\|$. $\square$

The domain enters only through β_c , in step 1. The transfer of the Galdi argument is therefore immediate: unlike Lemma 4.5, neither a Poincaré constant nor an angle condition is involved. For $k \geq 4$ one may instead apply Lemma 4.5 to $\mathcal{T}_h$, using Lemma 9.2(c).

Lemma 9.5 (Divergence range). *Let $k \geq 2$ and assume (CT0). Then $\operatorname{div} V_h^R = \Pi_h$, and $W_h \neq \emptyset$ for every boundary datum g . The velocities of GS and CNS* are pointwise divergence-free.*

Proof. Follow the proof of Lemma 4.3, using Lemma 9.4 for $\operatorname{div} V_h^0 = \Pi_h \cap L_0^2$, and the quadratic edge bubble $\beta_e n_e$ of the refined triangle K_e for the constants ($\int_e \beta_e = |e|/6$). The corner obstruction in the remark after Lemma 4.3 cannot occur, because by Lemma 9.2(c) no vertex of ∂R is singular in $\mathcal{T}_h$. $\square$

9.3. Approximation in W_h and the test field.

Lemma 9.6 (Approximation in W_h , $k \geq 2$). *Let $k \geq 2$, assume (CT0), and let $\sigma_k := k + 2$ for even k and $\sigma_k := k + 1$ for odd k . In particular $\sigma_2 = \sigma_3 = 4$. Then*

$$E_h = \inf_{z \in W_h} \|\tilde{u} - z\| \leq C \varepsilon_h, \quad \varepsilon_h = (1 + (\gamma h_\Gamma)^{1/2}) h^k + h^{\sigma_k} (h_\Gamma^{-1/2} + (\mu/h)^{1/2}).$$

The lower bound $\|\tilde{u} - z\| \geq (\mu/h)^{1/2} |m_R|/|\Gamma_h|^{1/2}$ for every $z \in W_h$, stated after Lemma 4.6, holds unchanged. The flux term is at most Ch^k for bounded γ whenever $h_\Gamma \geq h^{2(\sigma_k-k)}$: that is, $h_\Gamma \geq h^4$ for $k = 2$ and $h_\Gamma \geq h^2$ for $k = 3$. If $g \cdot \nu$ is a polynomial of degree at most $\sigma_k - 1$ on each side of R , the contribution of ∂R vanishes and the flux term reduces to a term below h^k (as in the remark after Lemma 4.6).

Proof. Steps 1–5 of the proof of Lemma 4.6 go through word for word, with three substitutions.

- (1) The Lagrange interpolant is taken on $\mathcal{T}_h$. Refined triangles meeting Γ_h have diameter at most Ch_Γ by Lemma 9.2(a,b), so the three interpolation bounds of step 1 hold for every $k \geq 1$.
- (2) In step 2, the trace of $I_h \tilde{u}$ on a boundary edge is its equispaced P_k Lagrange interpolant, and the edge is not split. Integrating it is the closed $(k+1)$ -point Newton–Cotes rule, which is exact on P_{σ_k-1} . For $k = 2$ this is Simpson’s rule and for $k = 3$ Simpson’s 3/8 rule, both exact on cubics. Each edge therefore contributes $O(|e|^{\sigma_k+1})$, and $|m| \leq Ch^{\sigma_k}$.
- (3) Step 3 uses the bubbles β_e of the refined triangles K_e , and step 4 uses Lemma 9.4 in place of Lemma 4.5. $\square$

Lemma 9.7 (The C^1 test field). *Let $k \geq 2$, assume (CT0) and $h \leq h_*$, and let φ and $w = \text{curl } \varphi$ be as in Section 6. There is $v_\varphi \in Z_h$ with*

$$\|v_\varphi - w\|_{L^\infty(\Omega_h)} \leq Ch^k, \quad \max_{K \in \mathcal{T}_h} \|\nabla(v_\varphi - w)\|_{L^\infty(K)} \leq Ch. \quad (4)$$

Proof. • $k \in \{2, 3\}$. Let $\Pi^{\text{HCT}}\varphi$ be the Hsieh–Clough–Tocher interpolant on $\mathcal{T}_h^M$ [4]. It is C^1 on Ω_h and piecewise cubic on $\mathcal{T}_h$. Its degrees of freedom are the values and gradients at the macro vertices and the normal derivatives at the macro-edge midpoints, and it reproduces P_3 . The HCT interpolation estimate [3, Section 6.1] gives

$$|\varphi - \Pi^{\text{HCT}}\varphi|_{W^{j,\infty}(T)} \leq Ch_T^{4-j} |\varphi|_{W^{4,\infty}(T)}, \quad j \leq 2,$$

with C depending on σ . Put $v_\varphi := \text{curl } \Pi^{\text{HCT}}\varphi$. It is continuous and piecewise $P_2 \subset P_k$ on $\mathcal{T}_h$, and divergence-free. Since the interpolant is local and $\varphi \equiv 0$ near ∂R , $v_\varphi = 0$ near ∂R . So $v_\varphi \in Z_h$, with L^∞ error $Ch^3 \leq Ch^k$ and gradient error Ch^2 .

- $k \geq 4$. By Lemma 9.2, $\mathcal{T}_h$ satisfies (M0)–(M2), and the construction of Section 6 (the Morgan–Scott interpolant of degree $k+1$ [13] on $\mathcal{T}_h$) gives the errors Ch^{k+1} and Ch^k . $\square$

Section 6 writes $\|v_\varphi - w\|_{W^{1,\infty}} \leq Ch^k$, but its proofs use only the two bounds in (4). The $O(h^k)$ terms in step 1 of the lower bound of Theorem 6.1, in steps 4–5 of Theorem 6.4, in steps 1–2 of Theorem 6.6(i) and in step 4 of Corollary 6.7 are all pointwise values of $v_\varphi - w$ on Γ_h . Everywhere else only boundedness of ∇v_φ is used: in $\|v_\varphi\|$, $a_h(S_h, \cdot)$, $\langle \partial_n S_h, \cdot \rangle$ and $\langle e, \partial_n v \rangle$.

9.4. The theorem.

Theorem 9.8 (Clough–Tocher version of the paper). *Let $k \geq 2$ be fixed, and let $\mathcal{T}_h$ be the Clough–Tocher refinement of a macro mesh satisfying (CT0). Let V_h, Π_h be as in this section, and assume (D).*

- (i) *The following hold as stated, with constants $C, c, C_I, \kappa, c_1, M, \beta, \gamma_0, \gamma_2, h_0, \delta_0$ depending on σ, c_0, k, L (and on the data as in Section 2) instead of on (M0)–(M2), k, L :*

- Lemmas 3.1–3.5, 4.1, 4.2, 5.1, 5.2 and 6.3, and Corollary 5.3;
- Lemma 4.3 in the form of Lemma 9.5, Lemma 4.5 in the form of Lemma 9.4, and Lemma 4.6 in the form of Lemma 9.6;
- Theorems 6.1, 6.4, 6.6 and 7.2, Corollaries 6.2 and 6.7, Proposition 7.1 and Proposition 8.1(a).

In Lemma 4.1 the triangle T_e is replaced by K_e ; no triangle has two edges on Γ_h (Lemma 9.2(b)).

- (ii) *In particular, for $\gamma \in [\gamma_2, \gamma_{\max}]$ the method CNS^* satisfies*

$$\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h^k) \quad \text{whenever } h_\Gamma \geq h^{2(\sigma_k - k)}.$$

This is $h_\Gamma^{3/2} + h^2$ for $k = 2$ with $h_\Gamma \geq h^4$, and $h_\Gamma^{3/2} + h^3$ for $k = 3$ with $h_\Gamma \geq h^2$. The printed method GS has the leading error $(h/\mu)v_\varphi$ of Corollary 6.7, with constant 1 in the slip and energy norms.

Proof. We go through the paper section by section and record every place where the mesh, the degree or the element enters.

- (1) *Section 3.* Lemmas 3.1–3.4 concern only Γ, Γ_h and Ω_h , which are unchanged. Lemma 3.5 uses, besides these, only the inverse trace inequality $|e| \|\partial_\tau v\|_{L^2(e)}^2 \leq C_I^2 \|\nabla v\|_{K_e}^2$ on the refined triangle K_e . This holds by Lemma 9.2(a,b) and a scaling argument.
- (2) *Section 4.* Lemma 4.1 holds on K_e for the same reason, and the K_e are distinct for distinct e . Lemma 4.2 uses only Lemmas 3.2 and 4.1, so its formulas for μ_0, γ_0, c_1, M are unchanged, with the new C_I . For the remaining three lemmas, see Lemmas 9.5, 9.4 and 9.6.

- (3) *Section 5.* Lemma 5.1 is an identity for smooth $\tilde{u}$ and any $v \in V_h^R$. Lemma 5.2 uses Lemmas 3.3, 3.5 and 4.1. Corollary 5.3 uses $\operatorname{div} v = 0$ pointwise for $v \in Z_h$.
- (4) *Section 6.* Take v_φ from Lemma 9.7; by the discussion after it, (4) is all that Theorems 6.1, 6.4, 6.6 and Corollaries 6.2, 6.7 need. Lemma 6.3 uses only three facts: on e , $v \in Z_h$ is a polynomial on K_e with $\operatorname{div} v = 0$ pointwise, so $\partial_n v \cdot n_e = -\partial_\tau(v \cdot \tau_e)$ on e ; v is continuous at the vertices of Γ_h ; and Lemmas 4.1 and 3.2 hold. All three are available.
- The remaining steps (the split $\delta_R + \delta_G$, Lax–Milgram on Z_h , the ansatz r_h , and the lower-bound arguments) are abstract: they use only Lemma 4.2, Lemma 4.6 through $E_h \leq C\varepsilon_h$, and Lemma 3.2. The lift E in the proof of Theorem 6.6(ii) uses $\|g - g_I\|_{H^{1/2}(\partial R)} \leq Ch^{k+1/2}$, which holds for every $k \geq 1$.
- (5) *Section 7.* Proposition 7.1 is algebraic. In Theorem 7.2, step 2 uses $\operatorname{div} V_h^0 \subset \Pi_h$ with π_h the $L^2(\Omega_h)$ projection onto Π_h (on $\mathcal{T}_h$). Step 3 uses Lemma 9.4. Step 4 uses $\|\eta\|_{L^\infty(K_e)} \leq C(\operatorname{diam} K_e)^k \leq Ch_\Gamma^k$. Step 6 uses the bubble $\beta_e n_e$ on K_e .
- (6) *Section 8(a)* is Lemma 4.2.

The rate in (ii) is Theorem 7.2 combined with Lemma 9.6. $\square$

Remark 9.9. For $k = 2$ the bulk term h^2 dominates $h_\Gamma^{3/2}$ exactly when $h_\Gamma \leq h^{4/3}$. So for quasi-uniform meshes the CNS* error is $O(h^{3/2})$ for every $k \geq 2$, and the degree matters only through ε_h . The constant-pressure benchmark of [8] is the case $G = 0$ of Theorem 6.1. It was analysed on Clough–Tocher splits with $k \geq 2$ in [5]; the proof here does not use [5].

9.5. The penalty threshold on Clough–Tocher stars. The necessity half of Proposition 8.1 depends on the element, and we recertify it. A *Clough–Tocher star* at a boundary vertex z is the union of the macro triangles containing z , each split at its barycentre. We use the reference stars of Section 8, now split, together with two more. All are fans at $O = (0, 0)$ with Nitsche boundary $\{y = 0\}$:

- $\mathcal{S}_2$: rim points $(1, 0), (0, 1), (-1, 0)$;
- $\mathcal{S}_3$: rim points $(1, 0), (1, 1), (-1, 1), (-1, 0)$;
- $\mathcal{S}_4$: rim points $(1, 0), (1, 1), (0, 1), (-1, 1), (-1, 0)$;
- $\mathcal{S}'_4$: rim points $(1, 0), (1, 1), (0, \frac{6}{5}), (-1, 1), (-1, 0)$.

Without the split, $\mathcal{S}_2$ and $\mathcal{S}_4$ contain singular vertices, and Section 8 had to exclude them. After the split every vertex is non-singular (Lemma 9.2(c)).

Proposition 9.10 (Penalty necessity, Clough–Tocher). *Let z be a vertex of a straight part of the Nitsche boundary, and let ℓ_z be the length of its boundary edges. Suppose its Clough–Tocher star, dilated by the factor $1/\ell_z$, is congruent to the split star $\mathcal{S}$. Let $k \geq 2$, and let $\lambda_k(\mathcal{S})$ be the certified value in Table 3, taking $\lambda_k := \lambda_4$ for $k \geq 5$. If $\mu\ell_z/h < \lambda_k(\mathcal{S})$, then N_h is indefinite on Z_h .*

The same holds, with the same constants and by continuity, at every vertex of Γ_h with interior angle $\pi + O(h_\Gamma)$ whose split star is a small perturbation of a split reference star. No exclusion like $(M1')$ is needed; in particular the case of two macro triangles ($\mathcal{S}_2$), which is typical at vertices of Γ_h , is covered.

Computation: `code/lamstar_exact_ct.py` (Part 2); output `logs/lamstar_exact_ct.log`.

Proof. (1) The witnesses. Steps 1–4 of the proof of Proposition 8.1(b) apply verbatim. The witness is a continuous piecewise- P_k field on the split star, pointwise divergence-free, vanishing on the rim chords, which are macro edges. Extended by zero it therefore lies in Z_h .

- (2) Monotonicity in k . A witness for degree k is also continuous piecewise- $P_{k'}$ for $k' \geq k$, so λ^* is nondecreasing in k ; this justifies $\lambda_k := \lambda_4$ for $k \geq 5$. The witnesses of Section 8 (unsplit P_4 fans) are also piecewise P_4 on the split fans. For $k \geq 4$ they certify only 9, which Table 3 supersedes.

		$\mathcal{S}_2$	$\mathcal{S}_3$	$\mathcal{S}_4$	$\mathcal{S}'_4$
$k = 2$	kernel dimension	6	7	8	8
	certified λ_2	5.71	7.54	7.54	7.52
	exact quotient	5.71715	7.54884	7.54467	7.52180
$k = 3$	kernel dimension	18	24	30	30
	certified λ_3	24.15	24.81	24.80	24.80
	exact quotient	24.15473	24.81227	24.80848	24.80696
$k = 4$	kernel dimension	36	50	64	64
	certified λ_4	48.38	48.78	48.78	48.77
	exact quotient	48.38946	48.78255	48.78181	48.77803

TABLE 3. Clough–Tocher split stars. *Kernel dimension*: the exact dimension of the divergence-free space of continuous piecewise- P_k fields on the split star vanishing on the rim. It equals $\dim V_{\mathcal{S}} - \dim \Pi_{\mathcal{S}}$ in every case, with $\text{rank div} = \dim \Pi_{\mathcal{S}}$. *Certified λ_k* : the rational λ_0 for which $2B - A - \lambda_0 C > 0$ is verified exactly on a rational witness. *Exact quotient*: the exact value of $(2B - A)/C$ at that witness. (Rounded to five decimals; the certified row is the rigorous statement.) It is a lower bound for the largest Rayleigh quotient $\lambda^*(\mathcal{S})$, and it agrees with a floating-point computation of $\lambda^*(\mathcal{S})$ to about 10^{-5} relative.

- (3) The certificate. The computation follows Section 8, with two changes. First, the per-element polynomials live on the split triangles, whose vertices (including barycentres) are rational. Second, the maximisation is reduced exactly. The kernel $\ker \text{div}$ and the splitting of it into $\ker C$ (fields with zero trace on the Nitsche boundary) and a complement are computed in exact rational arithmetic. On $\ker C$, $B = 0$ and $A > 0$, so the Schur complement of $2B - A$ onto the complement is formed exactly. Only the top eigenvector of this small exact pencil is computed in floating point, then rationalised.

A floating-point splitting misclassifies directions with tiny but non-zero trace. For $k = 3$ it overestimated λ^* (a spurious 25.11), which the exact check then rejected. We report only exactly verified values.

- (4) Continuity. For every Clough–Tocher star, not only the reference ones, $\text{div} : V_{\mathcal{S}} \rightarrow \Pi_{\mathcal{S}}$ is onto. Here $V_{\mathcal{S}}$ is the space of continuous piecewise- P_k fields vanishing on the rim, and $\Pi_{\mathcal{S}}$ the space of discontinuous piecewise- P_{k-1} functions. Indeed, the proof of Lemma 9.4 on the fixed Lipschitz polygon $\mathcal{S}$ (step 1 now being the continuous inf-sup condition on $\mathcal{S}$) gives $\text{div } V_0(\mathcal{S}) = \Pi_{\mathcal{S}} \cap L_0^2$, and the β_e bubble on a Nitsche edge adds the constants.

Hence $\dim \ker \text{div} = \dim V_{\mathcal{S}} - \dim \Pi_{\mathcal{S}}$ does not depend on the vertex positions, for fixed combinatorics. The divergence-free subspace is the kernel of a constraint matrix whose entries are polynomial in the vertex coordinates and whose rank is constant, so it depends continuously on them. The forms A, B, C (with edgewise normals) do as well, so the strict inequality $2B - A - \lambda_0 C > 0$ persists under small perturbations. This includes the rotation by $O(h_{\Gamma})$ of one Nitsche edge at a vertex of Γ_h and unequal edge lengths ℓ, ℓ' with ℓ'/ℓ near 1. $\square$

Remark 9.11 (Size of the threshold). The local thresholds on split stars are much larger than on unsplit ones: the exact witness quotients, which are lower bounds for λ^* , are 48.8 versus 9.98 on $\mathcal{S}_3$ for $k = 4$. They grow with k (7.5, 24.8, 48.8 for $k = 2, 3, 4$). This is consistent with the inverse-trace constant of the thinner boundary triangle K_e , whose height is one third of that of T_e , and with its growth in k . Sufficiency (Lemma 4.2) and necessity (Proposition 9.10) still both

scale with ρ , so the conclusion $\mu_0 \asymp \rho$ of Section 8 holds for Clough–Tocher meshes whose shortest boundary edge carries such a star. The threshold $\gamma^* \approx 18$ –21 measured on unsplit $k = 4$ meshes does not transfer. On the Clough–Tocher refinements of the same meshes we measured $\gamma^* \approx 11.5$ ($k = 2$), 34.6 ($k = 3$) and 65.5 ($k = 4$) at the finest levels (Table 9), each above the corresponding local certificate of Table 3.

Remark 9.12 (Scope). Theorem 9.8 covers Sections 3–7 up to and including Theorem 7.2, and Proposition 8.1. We have not re-checked the identification of the H^1 constant (Section 6.1), the lower bound (Section 7.1), the single-triangle witness of Section 8.1, or Sections 10–12 on Clough–Tocher meshes, and make no claim about them there.

10. GENERAL SMOOTH DOMAINS

Sections 2–8 treat the exterior of the unit disk in a square. Here we extend them to a bounded connected domain whose no-slip boundary Γ consists of finitely many closed C^3 curves, convex or not, each approximated by an inscribed polygon, while the rest of the boundary, $\Sigma \neq \emptyset$, is polygonal and carries strong Dirichlet data. The main new points are:

- a curvature version of the chord lemma;
- a normal-coordinate map Φ_h that works when $\Omega \setminus \Omega_h$ and $\Omega_h \setminus \Omega$ are both nonempty;
- a chart-based star-shaped cover that gives a uniform Bogovskiĭ constant;
- with several curved components, the correct pressure constant is the *global* boundary mean. This forces a new test field (a divergence-free field without a global stream function) and a new discrete-curl interpolant.

The pressure-consistent method needs only one global correction, and its discrete velocity does not depend on how that constant is normalised; a per-component correction is inconsistent. The geometric lemmas use only $\Gamma \in C^3$. All results also assume the regularity (R1) of the exact solution ($u \in C^{k+1}(\bar{\Omega})$, $p \in C^k(\bar{\Omega})$), which for generic f requires $\Gamma \in C^{k+1,\alpha}$. The results that use the leak test field (the lower bound of Theorem 6.1, Theorems 6.4 and 6.6, Corollary 6.7) are proved under the stronger assumption $\Gamma \in C^{k+3}$, $p|_{\Gamma} \in C^{k+1}$ (assumption (R2) below). The scope is Sections 3–8 up to and including Theorem 7.2 and Proposition 8.1; see Section 10.8. No computations have been run for general domains.

10.1. Setting and assumptions.

10.1.1. Domain.

Assumption 10.1 (Domain). $\Omega \subset \mathbb{R}^2$ is a bounded connected open set with $\partial\Omega = \Gamma \sqcup \Sigma$, where

- $\Gamma = \Gamma_1 \sqcup \cdots \sqcup \Gamma_m$, $m \geq 1$, and each Γ_i is a whole connected component of $\partial\Omega$ and a closed embedded curve of class C^3 ;
- $\Sigma \neq \emptyset$ is a finite disjoint union of closed simple polygons, each a whole component of $\partial\Omega$.

Which components are “inner” or “outer” is not restricted: the outer boundary may be curved (with Σ an inner polygonal hole) or polygonal. Components mixing curved and straight pieces are *not* covered (the junctions are corners of Γ).

For the domain of Section 2, $m = 1$, $\Gamma_1 = \partial D$, $\Sigma = \partial R$.

For each i fix an arclength parametrisation $\gamma_i : \mathbb{R}/|\Gamma_i|\mathbb{Z} \rightarrow \Gamma_i$, let $N_i(s)$ be the unit normal pointing into Ω , and orient γ_i so that (T_i, N_i) , $T_i := \gamma_i'$, is positively oriented. The signed curvature $\varkappa$ is defined by $T_i' = \varkappa N_i$ (so $N_i' = -\varkappa T_i$). Thus $\varkappa > 0$ where Ω is locally convex near Γ and $\varkappa < 0$ where it is locally concave; for the exterior of the unit disk $\varkappa \equiv -1$. The curvature is written $\varkappa$ because κ is the Korn constant of Lemma 3.2. Since $\Gamma \in C^3$, $\varkappa \in C^1(\Gamma)$.

On Γ we write $n := -N$ (the outward normal of Ω); on Γ_h , n is the outward normal of Ω_h , as in Section 2.

Fact 10.2 (Tubular neighbourhood; standard). *There is $r > 0$ with $r\|\varkappa\|_\infty \leq \frac{1}{2}$ such that, with $\mathcal{T}_r := \{x : \text{dist}(x, \Gamma) < r\}$:*

- (i) $\Psi(s, d) := \gamma_i(s) + dN_i(s)$ is a C^2 diffeomorphism from $\bigsqcup_i (\mathbb{R}/|\Gamma_i|\mathbb{Z}) \times (-r, r)$ onto $\mathcal{T}_r$, and d is the signed distance to Γ (positive in Ω), so $\mathcal{T}_r \cap \Omega = \{d > 0\}$;
- (ii) $d \in C^3(\mathcal{T}_r)$, the projection $\pi(x) = x - d(x)\nabla d(x)$ and the arclength coordinate $\sigma := s \circ \Psi^{-1}$ are C^2 , and

$$\nabla d = N \circ \pi, \quad \nabla \sigma = \frac{T \circ \pi}{J}, \quad J(x) := 1 - \varkappa(\pi(x))d(x) \in [\frac{1}{2}, \frac{3}{2}], \quad dx = J dd ds;$$

- (iii) $\text{dist}(\mathcal{T}_r, \Sigma) > 0$, and $\text{dist}(\Gamma_i, \Gamma_j) > 2r$ for $i \neq j$.

In the orthogonal coordinates (s, d) (scale factors J and 1), a field $F N \circ \pi + G T \circ \pi$ has

$$\text{div}(F N \circ \pi + G T \circ \pi) = \frac{1}{J}(\partial_d(JF) + \partial_s G). \quad (5)$$

Fact 10.3 (Uniform local graphs; standard). *There is $a_0 \in (0, r/4]$ with $4\|\varkappa\|_\infty a_0 \leq \frac{1}{5}$ such that for every $y = \gamma_i(s_0) \in \Gamma$, in the Cartesian frame $(y; T_i(s_0), N_i(s_0))$ with coordinates (ξ, η) and with $Q'_y := (-2a_0, 2a_0)^2$,*

$$\Gamma \cap Q'_y = \{(\xi, f_y(\xi)) : |\xi| < 2a_0\}, \quad \Omega \cap Q'_y = \{\eta > f_y(\xi)\},$$

where $f_y \in C^3$, $f_y(0) = f'_y(0) = 0$, $f''_y = \varkappa(1 + f_y'^2)^{3/2}$, $|f'_y(\xi)| \leq 2\|\varkappa\|_\infty|\xi| \leq \frac{1}{5}$, $|f_y(\xi)| \leq \|\varkappa\|_\infty \xi^2$ and $\|f_y'''\|_\infty \leq C_\Gamma$ (depending on $\|\varkappa\|_{C^1}$). Moreover $Q'_y \subset \mathcal{T}_r$.

Both facts concern the fixed curve only (tubular neighbourhood theorem, regularity of the distance function for C^3 curves, implicit function theorem and compactness); no h enters.

10.1.2. Inscribed polygons and mesh.

Assumption 10.4 (Inscribed polygons).

- (G1) For each i , $\Gamma_{h,i}$ is the closed polygon whose vertices $z_{i,0}, \dots, z_{i,N_i-1} \in \Gamma_i$ are in cyclic order along γ_i , consecutive vertices being joined by chords. For a chord e , Γ_e denotes the arc of Γ_i between its endpoints containing no other vertex, and $|\Gamma_e| \leq 2|e|$. $\Gamma_h := \bigcup_i \Gamma_{h,i}$.
- (G2) $h_\Gamma \leq h_*$, where $h_* \leq a_0/4$ is small in terms of Γ , Σ and the data (it is decreased finitely often below).

(G1) only excludes a chord that “wraps around” the long way; it is automatic if, e.g., the vertices are images of a quasi-uniform partition of the parameter.

The domain Ω_h is defined in Lemma 10.8. The mesh assumptions (M0)–(M2), (M1'), (D) of Assumption 2.1 are kept verbatim with ∂R replaced by Σ ; $\mathcal{T}_h$ is a conforming triangulation of Ω_h and $\mathcal{E}_h^\Gamma$ its edges on Γ_h . The spaces $V_h, V_h^R, V_h^0, \Pi_h, Z_h, W_h$, the forms a_h, N_h , the methods GS and CNS*, and the norm $\|\cdot\|$ are those of Section 2, with ∂R replaced by Σ . In CNS*,

$$\bar{p}_\Gamma(q) := |\Gamma_h|^{-1} \int_{\Gamma_h} q,$$

the mean over *all* of Γ_h (see Section 10.6 for why this is the right choice).

10.1.3. Exact solution, regularity and the pressure constant.

Assumption 10.5 (Regularity).

- (R1) $u \in C^{k+1}(\bar{\Omega})$, $p \in C^k(\bar{\Omega})$ solve $-\Delta u + \nabla p = f$, $\text{div } u = 0$ in Ω , $u|_\Gamma = 0$, $u|_\Sigma = g$; $\tilde{f}$ as in (D).
- (R2) (Only for results that use the test field of Section 10.5: the lower bound of Theorem 10.21, Theorems 10.22 and 10.23, Corollary 10.24.) $\Gamma \in C^{k+3}$ and $p|_\Gamma \in C^{k+1}(\Gamma)$.

Since $u|_\Gamma = 0$ and $\text{div } u = 0$, necessarily $\int_\Sigma g \cdot \nu = 0$ (ν the outward normal of Ω).

Definition 10.6 (Pressure constant). $\bar{p} := |\Gamma|^{-1} \int_{\Gamma} p \, ds$ (the mean over all curved components), $q := p - \bar{p}$ on Γ , and

$$G := \|p - \bar{p}\|_{L^2(\Gamma)}, \quad \bar{q}_i := |\Gamma_i|^{-1} \int_{\Gamma_i} q \quad (\text{so } \sum_i |\Gamma_i| \bar{q}_i = 0).$$

$$C_p := C(1 + \|p - \bar{p}\|_{W^{1,\infty}(\Gamma)}).$$

Proposition 10.25 shows that the per-component quantity $G_{\text{per}}^2 := \sum_i \|p - \bar{p}_i\|_{L^2(\Gamma_i)}^2$ ($\bar{p}_i$ the mean over Γ_i) is *not* the right one when $m \geq 2$. Note $G^2 = G_{\text{per}}^2 + \sum_i |\Gamma_i| \bar{q}_i^2$.

Constants. C, c depend on (M0)–(M2), k , the geometry ($\|\kappa\|_{C^1}$, r , a_0 , $|\Gamma|$, Σ , the fixed cover of Lemma 10.14), and norms of $(u, p, \tilde{u}, \tilde{p})$; they are independent of $h, h_{\Gamma}, \mu, \gamma, \rho$.

10.2. Geometry.

Lemma 10.7 (Chords; replaces Lemma 3.1). *Let $e \in \mathcal{E}_h^{\Gamma}$ be a chord of Γ_i of length ℓ , with midpoint m_e and unit tangent τ_e oriented like T_i along Γ_e . For $x \in e$ let $t \in [-\ell/2, \ell/2]$ be its coordinate along e from m_e , $x^* := \pi(x)$, and $\kappa_e := \kappa(m_e^*)$. Let N_e be the unit normal of e with $N_e \cdot N(x^*) > 0$ and $n_e := -N_e$. Then:*

- (a) (sagitta) $d(x) = \frac{1}{2} \kappa(x^*) (\ell^2/4 - t^2) + O(\ell^3)$ and $|d(x)| \leq \|\kappa\|_{\infty} \ell^2/8 + C\ell^4$. If $\kappa \geq 0$ on Γ_e then $e \subset \bar{\Omega}$; if $\kappa \leq 0$ on Γ_e then $e \cap \Omega = \emptyset$.
- (b) (normal rotation) $n(x^*) - n_e = \kappa_e t \tau_e + O(\ell^2)$. In particular $n(x^*) \cdot n_e = 1 + O(\ell^2)$ and $|n(x^*) \cdot \tau_e| + |N(x^*) \cdot \tau_e| \leq C\ell$.
- (c) (Jacobian) $\pi|_e : e \rightarrow \Gamma_e$ is a C^2 bijection with $\frac{d(\sigma \circ \pi)}{dt} = \frac{T(x^*) \cdot \tau_e}{J(x)} = 1 + O(\ell^2)$; hence $|\Gamma_e| = \ell(1 + O(\ell^2))$.
- (d) (turning) If e, e' share a vertex, $|\tau_e - \tau_{e'}| + |n_e - n_{e'}| \leq 2\|\kappa\|_{\infty}(|\Gamma_e| + |\Gamma_{e'}|)$.

The $O(\cdot)$ constants depend only on $\|\kappa\|_{C^1}$. For $\kappa \equiv -1$ this is Lemma 3.1 (there s plays the role of t).

Proof. By (G1), $|\Gamma_e| \leq 2h_{\Gamma} \leq a_0/2$, so Γ_e lies in Q'_z for its first endpoint z and is a graph there. By the mean value theorem there is $y_0 \in \Gamma_e$ with $T(y_0) \parallel e$. Work in the frame of Fact 10.3 at y_0 , with $f := f_{y_0}$ (the arc is still a graph there since $|\Gamma_e| \leq a_0/2$). The chord is the horizontal segment $\{(\xi, c) : \xi_a \leq \xi \leq \xi_b\}$ with $f(\xi_a) = f(\xi_b) = c$, $\xi_b - \xi_a = \ell$, $0 \in [\xi_a, \xi_b]$, and $\tau_e = e_1$, $N_e = e_2$. Put $\xi_m := (\xi_a + \xi_b)/2$, so $t = \xi - \xi_m$.

(a) Linear interpolation error: $c - f(\xi) = \frac{1}{2} f''(\zeta)(\xi - \xi_a)(\xi_b - \xi)$ for some $\zeta \in (\xi_a, \xi_b)$, and $(\xi - \xi_a)(\xi_b - \xi) = \ell^2/4 - t^2$. Since $|f'| \leq 2\|\kappa\|_{\infty} \ell$ on $[\xi_a, \xi_b]$, $f'' = \kappa(1 + f'^2)^{3/2} = \kappa(1 + O(\ell^2))$, and κ at the graph point over ζ differs from $\kappa(x^*)$ by $\|\kappa'\|_{\infty} \cdot O(\ell)$. Next, d is C^2 with $d = 0$ on the graph and $\nabla d = N$, so $d(\xi, c) = (c - f(\xi))N_2 + O((c - f)^2)$ with $N_2 = (1 + f'^2)^{-1/2} = 1 + O(\ell^2)$. Hence $d(x) = \frac{1}{2} \kappa(x^*) (\ell^2/4 - t^2) + O(\ell^3)$, and $|d(x)| \leq \frac{1}{2} \|\kappa\|_{\infty} (1 + C\ell^2) \ell^2/4 + C\ell^4$. The sign statements follow because f'' has the sign of κ and $\Omega \cap Q'_z = \{\eta > f\}$.

(b) $N(x^*) = (1 + f'(\xi^*)^2)^{-1/2}(-f'(\xi^*), 1)$, so $N(x^*) - N_e = -f'(\xi^*)e_1 + O(\ell^2)$. The projection moves the abscissa by $|d|/|f'| = O(\ell^3)$, so $f'(\xi^*) = f'(\xi) + O(\ell^3) = f'(\xi_m) + f''(\cdot)t + O(\ell^3)$. The midpoint rule for divided differences gives $f'(\xi_m) = \frac{f(\xi_b) - f(\xi_a)}{\ell} + O(\ell^2 \|f'''\|_{\infty}) = O(\ell^2)$, and $f''(\cdot) = \kappa(x^*) + O(\ell) = \kappa_e + O(\ell)$. Thus $N(x^*) - N_e = -\kappa_e t \tau_e + O(\ell^2)$; multiply by -1 .

(c) By Fact 10.2(ii), $\frac{d}{dt} \sigma(\pi(x)) = \nabla \sigma \cdot \tau_e = T(x^*) \cdot \tau_e / J(x)$. Here $|T(x^*) - T(y_0)| \leq \|\kappa\|_{\infty} |\Gamma_e|$, so $T(x^*) \cdot \tau_e = 1 + O(\ell^2)$, and $J = 1 + O(\ell^2)$ by (a). The derivative is positive, and the endpoints are fixed by π , so $\pi|_e$ is a bijection onto Γ_e .

(d) $\tau_e = T(y_0)$ and $\tau_{e'} = T(y'_0)$ with $y_0 \in \Gamma_e$, $y'_0 \in \Gamma_{e'}$, and $|T(y) - T(y')| \leq \|\kappa\|_{\infty} \cdot (\text{arc distance})$; n_e is τ_e rotated by a fixed angle. $\square$

Lemma 10.8 (Γ_h is a normal graph; definition of Ω_h). For $h_{\Gamma} \leq h_*$:

(a) $\Gamma_{h,i} = \{\gamma_i(s) + \eta_h(s)N_i(s)\}$ for a Lipschitz, piecewise C^2 function η_h with

$$\|\eta_h\|_\infty \leq \|\varkappa\|_\infty h_\Gamma^2/8 + Ch_\Gamma^4, \quad \|\eta'_h\|_\infty \leq Ch_\Gamma.$$

In particular the $\Gamma_{h,i}$ are simple closed polygons, pairwise disjoint and disjoint from Σ , and the vertex angle of each is $\pi + O(h_\Gamma)$ (this justifies (M1')).

(b) Define

$$\Omega_h := (\Omega \setminus \mathcal{T}_{r/2}) \cup \{x \in \mathcal{T}_{r/2} : d(x) > \eta_h(\sigma(x))\}.$$

Then Ω_h is a bounded polygonal open set with $\partial\Omega_h = \Sigma \sqcup \Gamma_h$, it lies on the Ω -side of each chord (so the outward normal of Ω_h on e is n_e of Lemma 10.7), and

$$\Omega_h \setminus \Omega = \{\eta_h(\sigma) < d \leq 0\}, \quad \Omega \setminus \Omega_h = \{0 < d \leq \eta_h(\sigma)\}, \quad |\Omega \triangle \Omega_h| \leq Ch_\Gamma^2.$$

Both differences are nonempty as soon as $\varkappa$ takes both signs on Γ (for h_Γ small).

Proof. (a) By Lemma 10.7(c), π maps each chord bijectively onto its arc; the arcs Γ_e of consecutive chords tile Γ_i (G1). So $\pi|_{\Gamma_{h,i}} : \Gamma_{h,i} \rightarrow \Gamma_i$ is a bijection and $\eta_h(s) := d((\pi|_{\Gamma_{h,i}})^{-1}(\gamma_i(s)))$ represents $\Gamma_{h,i}$ as stated. The sup bound is Lemma 10.7(a). On a chord, $\frac{d\eta_h}{ds} = \nabla d \cdot \tau_e \frac{dt}{ds} = N(x^*) \cdot \tau_e (1 + O(\ell^2)) = O(\ell)$ by Lemma 10.7(b),(c). Since Ψ is a diffeomorphism and $|\eta_h| < r/2$, $\Gamma_{h,i}$ is embedded; disjointness follows from Fact 10.2(iii). The vertex angle is π plus the turning angle, which is $O(h_\Gamma)$ by Lemma 10.7(d).

(b) Openness and $\partial\Omega_h = \Sigma \sqcup \Gamma_h$ are immediate from the definition and Fact 10.2. The descriptions of the differences follow from $\Omega \cap \mathcal{T}_r = \{d > 0\}$; their measure is $\int_\Gamma \int |\eta_h| J dd ds \leq Ch_\Gamma^2$. If $\varkappa(y) < 0$ at some y , then for small h_Γ the chord e with $y \in \Gamma_e$ has $\varkappa < 0$ on Γ_e , and Lemma 10.7(a) gives $d < 0$ in its interior; similarly for $\varkappa > 0$. $\square$

Lemma 10.9 (Uniform trace, Friedrichs, Korn; replaces Lemma 3.2). *There are $C_{\text{tr}}, C_F, \kappa, C_P$, independent of h , such that every $v \in H^1(\Omega_h)^2$ with $v|_\Sigma = 0$ satisfies*

$$\|v\|_{L^2(\Gamma_h)} \leq C_{\text{tr}} \|\nabla v\|, \quad \|v\| \leq C_F \|\nabla v\|, \quad \|\nabla v\|^2 \leq \kappa a_h(v, v),$$

and every $v \in H^1(\Omega_h)$ satisfies $\inf_{c \in \mathbb{R}} \|v - c\|_{\Omega_h} \leq C_P \|\nabla v\|_{\Omega_h}$. Moreover Ω_h is connected.

Proof. (1) Let $\chi \in C^\infty(\mathbb{R})$, $0 \leq \chi \leq 1$, $\chi = 1$ on $(-\infty, r/8]$, $\chi = 0$ on $[r/4, \infty)$, $|\chi'| \leq 16/r$. Define Φ_h on $\bar{\Omega}$ by

$$\Phi_h(\Psi(s, d)) := \Psi(s, d + \chi(d)\eta_h(s)) \quad (0 \leq d < r), \quad \Phi_h := \text{id} \text{ on } \bar{\Omega} \setminus \mathcal{T}_r.$$

(2) For fixed s , $d \mapsto d + \chi(d)\eta_h(s)$ has derivative $1 + \chi'\eta_h \geq 1 - Ch_\Gamma^2/r \geq \frac{1}{2}$, equals $\eta_h(s)$ at $d = 0$ and d for $d \geq r/4$. So Φ_h is a homeomorphism of $\bar{\Omega}$ onto $\bar{\Omega}_h$ mapping Γ_i onto $\Gamma_{h,i}$, with $\Phi_h = \text{id}$ outside $\mathcal{T}_{r/4}$, in particular near Σ . Hence Ω_h is connected.

(3) In (s, d) coordinates $\hat{\Phi}_h(s, d) = (s, d + \chi\eta_h)$ has $D\hat{\Phi}_h = I + \begin{pmatrix} 0 & 0 \\ \chi\eta'_h & \chi'\eta_h \end{pmatrix} = I + O(h_\Gamma)$ a.e. Since $D\Psi \in C^1$, $D\Psi(s, d')D\Psi(s, d)^{-1} = I + O(|d' - d|) = I + O(h_\Gamma^2)$. Therefore $D\Phi_h = D\Psi \circ \hat{\Phi}_h \cdot D\hat{\Phi}_h \cdot (D\Psi)^{-1}$ satisfies $\|D\Phi_h - I\|_{L^\infty} + \|D\Phi_h^{-1} - I\|_{L^\infty} \leq Ch_\Gamma$, and $\Phi_h|_\Gamma$ has arclength Jacobian $1 + O(h_\Gamma^2)$ (it is the inverse of $\pi|_{\Gamma_h}$, Lemma 10.7(c)).

(4) The rest is Lemma 3.2, steps 4–6, verbatim: for $w = v \circ \Phi_h$, $D(w) = D(v) \circ \Phi_h + E$ with $|E| \leq Ch_\Gamma |\nabla v \circ \Phi_h|$; on the fixed connected Lipschitz domain Ω , w vanishes on Σ , which has positive length, so Korn's first inequality, the trace inequality on Γ and the Friedrichs inequality hold; transfer back and absorb $Ch_\Gamma \|\nabla v\|$ for $h_\Gamma \leq h_*$. The Poincaré inequality transfers the same way from Ω . $\square$

Here $\Sigma \neq \emptyset$ is used (Korn and Friedrichs with zero trace on Σ). The map Φ_h works for arbitrary signs of $\varkappa$: it moves points along normals in both directions.

Lemma 10.10 (Extension; replaces Lemma 3.3 and the extension paragraph of Section 2). *There are $\tilde{u} \in C^{k+1}$ and $\tilde{p} \in C^k$ on the open set $\mathcal{O} := \Omega \cup \mathcal{T}_r$ with $\tilde{u} = u$, $\tilde{p} = p$ on Ω and $\operatorname{div} \tilde{u} = 0$ on $\mathcal{O}$. With $F := \tilde{f} + \Delta \tilde{u} - \nabla \tilde{p}$ on Ω_h (so $F = 0$ on $\Omega_h \cap \Omega$ and $\operatorname{supp} F \subset \Omega_h \setminus \Omega$):*

$$\|\tilde{u}\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^2, \quad |(F, v)_{\Omega_h}| \leq Ch_\Gamma^2 \|\nabla v\| \quad \forall v \in V_h^R.$$

Proof. Construction. Let $U_i := \Psi(\Gamma_i \times (-r, r))$ and $U_i^+ := U_i \cap \Omega$, an annulus. The flux of u through $\Psi(\Gamma_i \times \{d\})$ is independent of $d \in (0, r)$ since $\operatorname{div} u = 0$, and tends to $\int_{\Gamma_i} u \cdot N = 0$. So $u = \operatorname{curl} \psi_i$ on U_i^+ with ψ_i single-valued and $\psi_i \in C^{k+2}(\overline{U_i^+})$. (A global stream function need not exist: when Σ has several components, g may have nonzero flux through each of them. The single-valued ψ of Section 2 is replaced by these local ones.) By Whitney's extension theorem ($\overline{U_i^+}$ has C^1 boundary, so C^{k+2} functions on it are Whitney jets), extend ψ_i to $\tilde{\psi}_i \in C^{k+2}(U_i)$, and set $\tilde{u} := \operatorname{curl} \tilde{\psi}_i$ on $U_i \setminus \Omega$ and $\tilde{u} := u$ on Ω . On U_i , $\tilde{u} = \operatorname{curl} \tilde{\psi}_i$ on both sides, so $\tilde{u} \in C^{k+1}(\mathcal{O})$ and $\operatorname{div} \tilde{u} = 0$. $\tilde{p}$ is a Whitney extension of p .

Bounds. $\tilde{u} = 0$ on Γ and $|x - \pi(x)| = |d(x)| \leq Ch_\Gamma^2$ on Γ_h (Lemma 10.8) give the first bound. For the second, F is bounded on $\Omega_h \setminus \Omega = \{\eta_h(s) < d \leq 0\}$. For $x = \Psi(s, d)$ there, with $y_h(s) := \Psi(s, \eta_h(s)) \in \Gamma_h$,

$$|v(x)| \leq |v(y_h(s))| + \int_{\eta_h(s)}^0 |\nabla v(\Psi(s, t))| dt.$$

Integrate with $dx = J dd ds$ over the strip of width $|\eta_h(s)| \leq Ch_\Gamma^2$:

$$\int_{\Omega_h \setminus \Omega} |v| \leq Ch_\Gamma^2 \left(\int_{\Gamma} |v(y_h(s))| ds + \int_{\Omega_h \setminus \Omega} |\nabla v| \right) \leq Ch_\Gamma^2 (\|v\|_{L^1(\Gamma_h)} + h_\Gamma \|\nabla v\|),$$

using the Jacobian of Lemma 10.7(c) and $|\Omega_h \setminus \Omega|^{1/2} \leq Ch_\Gamma$. Finish with $\|v\|_{L^1(\Gamma_h)} \leq |\Gamma_h|^{1/2} \|v\|_{L^2(\Gamma_h)}$ and Lemma 10.9. $\square$

Remark 10.11 (The part $\Omega \setminus \Omega_h$). Where Γ is locally convex ($\varkappa > 0$) the chords cut off $\Omega \setminus \Omega_h$. Nothing needs to be extended there: all fields live on Ω_h and all integrations by parts are on Ω_h . The errors are measured on Ω_h as before, so $\Omega \setminus \Omega_h$ (area $O(h_\Gamma^2)$) is simply not approximated. On $\Omega_h \setminus \Omega$ the error $\tilde{u} - u_h$ depends on the extension; two C^2 extensions differ there by $O(h_\Gamma^2)$ in $W^{1,\infty}$ (their difference and its gradient vanish on Γ), hence by $O(h_\Gamma^3)$ in $H^1(\Omega_h)$, which is below every rate in this section.

Lemma 3.4 (wall structure) holds verbatim on any C^1 curve, with $n = -N$.

Lemma 10.12 (Transposed-gradient term; replaces Lemma 3.5). *The statement of Lemma 3.5 holds, with*

$$a_e := -\varkappa_e n(m_e^*) (\partial_n u(m_e^*) \cdot \tau_e).$$

Proof. Only step 1 changes. For $x \in e$, $\tilde{u} \in C^2$ gives $(\nabla \tilde{u}(x))^\top n_e = (\nabla u(x^*))^\top n_e + O(h_\Gamma^2)$. By Lemma 3.4, $(\nabla u)^\top n(x^*) = 0$ and $(\nabla u)^\top b = n(\partial_n u \cdot b)$ at x^* . With Lemma 10.7(b),

$$(\nabla u(x^*))^\top n_e = n(x^*) (\partial_n u(x^*) \cdot (n_e - n(x^*))) = -\varkappa_e t n(x^*) (\partial_n u(x^*) \cdot \tau_e) + O(\ell^2) = t a_e + O(h_\Gamma^2),$$

since x^* and m_e^* are $O(\ell)$ apart. The coefficient a_e is constant on e , and $\int_e t dt = 0$. Steps 2–4 are unchanged; the sum over edges uses $\#\mathcal{E}_h^\Gamma \leq |\Gamma|/(c_0 h_\Gamma)$, since $|\Gamma_h| \leq |\Gamma|$. $\square$

For $\varkappa \equiv -1$ this is the a_e of Lemma 3.5. On straight parts of Γ ($\varkappa = 0$) the chords lie on Γ , and the leading term vanishes.

10.3. Discrete tools. Lemmas 4.1 and 4.2 are mesh-only and hold verbatim, with κ from Lemma 10.9. Lemma 4.3 and Remark 4.4 hold verbatim with $\partial R \rightarrow \Sigma$. The only domain property used in Lemma 4.3, step 1, is that Ω_h is connected (so that $\text{div } V_h^0$ is the mean-zero part of Π_h), and this is Lemma 10.9. The edge bubble of step 2 may sit on any edge of any $\Gamma_{h,i}$.

10.3.1. *Uniform continuous inf-sup constant.*

Lemma 10.13 (Decomposition of the divergence equation). *Let $U = \bigcup_{j=1}^N U_j$ be a bounded open set, where each U_j is star-shaped with respect to every point of a ball B_j of radius $\geq r_*$ with $\overline{B_j} \subset U_j$, $\text{diam } U_j \leq d_*$, and, for $j < N$, $|A_j| \geq m_*$ where $A_j := U_j \cap D_j$, $D_j := \bigcup_{l>j} U_l$. Then for every $f \in L_0^2(U)$ there is $v \in H_0^1(U)^2$ with $\text{div } v = f$ and $\|\nabla v\| \leq C_B \|f\|$, where C_B depends only on N, r_*, d_*, m_* and $|U|$.*

Proof. Let $E_j := U_j \setminus D_j$ (so $E_N = U_N$ and the E_j partition U). Put $g_1 := f$ and, for $j = 1, \dots, N-1$,

$$f_j := g_j \mathbf{1}_{E_j} - \frac{\int_{E_j} g_j}{|A_j|} \mathbf{1}_{A_j}, \quad g_{j+1} := g_j - f_j, \quad f_N := g_N.$$

By induction, $\text{supp } g_j \subset E_j \cup D_j$ and $\int g_j = 0$. Hence $\text{supp } f_j \subset U_j$, $\int f_j = 0$ and $\text{supp } g_{j+1} \subset D_j$. Also $\|f_j\| \leq (1 + |U|^{1/2} m_*^{-1/2}) \|g_j\|$ and $\|g_{j+1}\| \leq \|g_j\| + \|f_j\|$. So $\|f_j\| \leq C(N, m_*, |U|) \|f\|$, and $f = \sum_j f_j$. By Bogovskii's theorem on star-shaped domains [7, Thm. III.3.1] there is $v_j \in H_0^1(U_j)^2$ with $\text{div } v_j = f_j$ and $\|\nabla v_j\| \leq c_0(d_*/r_*)^2(1 + d_*/r_*) \|f_j\|$. Take $v := \sum_j v_j$, each extended by zero. $\square$

Lemma 10.14 (A uniform star-shaped cover of Ω_h). *There are $N, r_*, d_*, m_* > 0$, depending only on Ω , and h_* , such that for $h_\Gamma \leq h_*$, Ω_h admits a cover as in Lemma 10.13.*

Proof. Fix $a := a_0$ (Fact 10.3), so $\|\varkappa\|_\infty a \leq 1/20$.

Chart pieces. Choose $y_1, \dots, y_{N_1} \in \Gamma$ ($N_1 \leq 4|\Gamma|/a + m$) such that the arcs of length $a/2$ centred at the y_j cover Γ . In the frame of Fact 10.3 at y_j , let $Q_j := (-a, a)^2 \subset Q'_j := (-2a, 2a)^2$, $f := f_{y_j}$, and $U_j^h := \Omega_h \cap Q_j$.

- (1) $\Gamma_h \cap Q_j$ is the graph of $f_h := I_1 f$, the piecewise-linear interpolant of f at the abscissae of the vertices of Γ_h in Q'_j . A chord meeting Q_j has length $\leq h_\Gamma < a/4$, so both endpoints lie in $Q'_j \cap \Gamma$, i.e. on the graph of f . Chords of other components, and Σ , do not meet $Q'_j \subset \mathcal{T}_r$ (Fact 10.2(iii)). Moreover $|f'_h| \leq \sup_{|\xi| < 2a} |f'| \leq \frac{1}{5} =: L$ (slopes of $I_1 f$ are values of f'), and $|f_h| \leq \|\varkappa\|_\infty (a + h_\Gamma)^2 \leq a/5 =: b_0$.
- (2) $U_j^h = \{(\xi, \eta) \in Q_j : \eta > f_h(\xi)\}$. $Q_j \setminus \Gamma_h$ has exactly two components, above and below the graph. $\Omega_h \cap Q_j$ is relatively open and closed in $Q_j \setminus \Gamma_h$ (because $\partial\Omega_h \cap Q_j = \Gamma_h \cap Q_j$), so it is a union of components. The point $(0, a/2)$ lies in Ω at distance $> a/4$ from Γ , hence in Ω_h (as $\Omega \setminus \Omega_h \subset \{d \leq Ch_\Gamma^2\}$). The point $(0, -a/2)$ has $d = -a/2 < \eta_h$, so it is not in Ω_h .
- (3) U_j^h is star-shaped with respect to every point of $B_j := B((0, \frac{3}{4}a), \frac{a}{8})$. Let $y \in B_j$, $x \in U_j^h$ and $z_t := (1-t)y + tx$. The constraints $|\xi| < a$, $\eta < a$ are convex. Since $g := f_h$ is L -Lipschitz, $g(\xi_{z_t}) \leq g(\xi_x) + L(1-t)|\xi_x - \xi_y| \leq g(\xi_x) + L(1-t)\frac{9}{8}a$. Hence

$$\eta_{z_t} - g(\xi_{z_t}) \geq t(\eta_x - g(\xi_x)) + (1-t)(\frac{5}{8}a - b_0 - \frac{9}{8}La) = t(\eta_x - g(\xi_x)) + (1-t)\frac{a}{5} > 0.$$

Also $\overline{B_j} \subset U_j^h$, since $\frac{5}{8}a > b_0$ and $\frac{7}{8}a < a$. Finally $\text{diam } U_j^h \leq 2\sqrt{2}a$.

Interior pieces. Let $\Omega_* := \{x \in \Omega : \text{dist}(x, \Gamma) > a/4\}$. It is independent of h , and $\Omega_* \subset \Omega_h$ for $h_\Gamma \leq h_*$. It is connected: the map $\Psi(s, d) \mapsto \Psi(s, \frac{a}{4} + (1 - \frac{a}{4r})d)$ on $\{0 < d < r\}$, extended by the identity, is a homeomorphism $\Omega \rightarrow \Omega_*$. It is a bounded Lipschitz domain (its boundary is Σ together with the C^2 parallel curves $\{d = a/4\}$). By [7, Lemma II.1.3] it is a finite union

$\Omega_* = \bigcup_{l=1}^{N_2} V_l$ of domains, each star-shaped with respect to every point of a ball. Since Ω_* is connected, the intersection graph of $\{V_l\}$ is connected. List the V_l in the reverse of a breadth-first order from a root; then each V_l except the last meets a later one in a nonempty open set. All of this is h -independent.

Covering. If $x \in \Omega_h$ and $\text{dist}(x, \Gamma) \geq a/2$, then $x \in \Omega_*$. If $\text{dist}(x, \Gamma) < a/2$, write $x = \pi(x) + dN$ with $\pi(x)$ within arclength $a/4$ of some y_j . In the frame at y_j , $|\xi_x| \leq \frac{a}{4} + \frac{a}{2} \cdot \frac{1}{5} < a$ and $|\eta_x| \leq \frac{a}{5} + \frac{a}{2} < a$, so $x \in U_j^h$. Hence $\Omega_h = \bigcup_j U_j^h \cup \bigcup_l V_l$.

Overlaps. Order the chart pieces first, then the V_l . Then $U_j^h \cap D_j \supset U_j^h \cap \Omega_* \supset y_j + \{|\xi| < \frac{a}{2}, \frac{a}{2} < \eta < a\}$. These points are above f_h , lie in Ω , and are at distance $\geq (\frac{a}{2} - b_0)/\sqrt{1+L^2} > \frac{a}{4}$ from the graph part of Γ , and at distance $> \frac{a}{4}$ from $\Gamma \setminus Q_j'$. So $|A_j| \geq a^2/2$. The overlaps among the V_l are fixed positive numbers. Take $N = N_1 + N_2$, $r_* = \min(a/8, \text{radii of the } V_l)$, $d_* = \max(2\sqrt{2}a, \text{diam } V_l)$ and m_* the minimum overlap. $\square$

Lemma 10.15 (Uniform inf-sup constant; replaces Lemma 4.5). *The statement of Lemma 4.5 holds on Ω_h , with β depending only on k , Θ_0 , shape regularity and Ω .*

Proof. As in Lemma 4.5, the Guzmán–Scott argument [11] depends on the domain only through the continuous inf-sup (Bogovskiĭ) constant and a Poincaré constant. The latter is uniform by Lemma 10.9, the former by Lemmas 10.13 and 10.14. The wedge cover and the hypothesis $P_h \subset D$ used in Lemma 4.5 are no longer needed. $\square$

10.3.2. *Approximation.* Lemma 4.6 and the remark after it hold verbatim with $\partial R \rightarrow \Sigma$, ν the outward normal of Ω on Σ , and $m_R := \int_\Sigma g_I \cdot \nu$. The only domain facts used are:

- $\tilde{u} \in W^{k+1,\infty}(\Omega_h)$ (Lemma 10.10);
- boundary triangles have diameter $\leq Ch_\Gamma$;
- $\int_{\Gamma_h} \tilde{u} \cdot n = \int_{\partial\Omega_h} \tilde{u} \cdot n - \int_\Sigma g \cdot \nu = \int_{\Omega_h} \text{div } \tilde{u} - 0 = 0$;
- the edge-bubble field $b = \sum_{e \in \mathcal{E}_h^\Gamma} \beta_e n_e$ over all components;
- Lemma 10.15.

10.4. **Error equation and the flux structure.** Lemma 5.1 holds verbatim. Its proof integrates by parts on the polygon Ω_h , where $\tilde{u} \in C^2(\overline{\Omega_h})$, $\tilde{p} \in C^1(\overline{\Omega_h})$ and $v = 0$ on Σ . Nothing requires $\Omega \subset \Omega_h$. Lemma 5.2 holds verbatim by Lemmas 10.10 and 10.12.

For $v \in V_h^R$ put $\Phi_i(v) := \int_{\Gamma_{h,i}} v \cdot n$.

Lemma 10.16 (Fluxes).

- For $v \in Z_h$, $\sum_{i=1}^m \Phi_i(v) = 0$.
- If $m \geq 2$ and $h_\Gamma, h \leq h_*$ (with (R2) available for the construction, i.e. $\Gamma \in C^{k+3}$), then $v \mapsto (\Phi_i(v))_i$ maps Z_h onto $\{x \in \mathbb{R}^m : \sum_i x_i = 0\}$. In particular individual fluxes of Z_h functions do not vanish.

Proof. (a) $0 = \int_{\Omega_h} \text{div } v = \int_{\Gamma_h} v \cdot n$, since $v = 0$ on Σ . (b) Given $\alpha \in \mathbb{R}^m$ with $\sum_i |\Gamma_i| \alpha_i = 0$, Lemma 10.18 below, applied with $q := \alpha_i$ on Γ_i (which is C^∞ on Γ , the components being disjoint, and has $\int_\Gamma q = 0$), and Lemma 10.19 give $v_\alpha \in Z_h$ with $v_\alpha = \alpha_i N + O(h_\Gamma^2 + h^k)$ on $\Gamma_{h,i}$. By Lemma 10.7(b) and (c), $\Phi_i(v_\alpha) = -|\Gamma_i| \alpha_i + O(h_\Gamma^2 + h^k) |\alpha|$. The map $\alpha \mapsto (\Phi_i(v_\alpha))_i$ is linear from an $(m-1)$ -dimensional space into the hyperplane, and it is a small perturbation of the isomorphism $\alpha \mapsto (-|\Gamma_i| \alpha_i)_i$. So it is onto for small h . $\square$

Corollary 10.17 (GS on Z_h ; replaces Corollary 5.3). *For $v \in Z_h$ and every constant $c \in \mathbb{R}$,*

$$N_h(\tilde{u} - u_h, v) = \mathcal{R}(v) + \mathcal{G}(v), \quad \mathcal{R}(v) := -\langle \tilde{p} - c, v \cdot n \rangle_{\Gamma_h}.$$

The freedom is one global constant. For piecewise constants c_i on $\Gamma_{h,i}$,

$$-\langle \tilde{p} - \sum_i c_i \mathbf{1}_{\Gamma_{h,i}}, v \cdot n \rangle = \mathcal{R}(v) + \sum_i (c_i - c) \Phi_i(v),$$

and the last sum is not zero on Z_h when $m \geq 2$ and the c_i are not all equal (Lemma 10.16(b)).

Proof. As in Corollary 5.3, using only Lemma 10.16(a). $\square$

10.5. The test field and the printed method.

10.5.1. *The test field.* The field $\varphi = \chi_1(r) \int_0^\theta (p - \bar{p})$ of Section 6 generalises, per component, to $\chi(d) \int_0^s (q - \bar{q}_i)$. This is single-valued only with the *per-component* mean. Since the correct constant is global (Corollary 10.17), w must carry the net fluxes $-|\Gamma_i| \bar{q}_i$ through the Γ_i . Such a w has no global stream function, and the construction below replaces φ .

Lemma 10.18 (Test field). *Assume (R2), and let $q \in C^{k+1}(\Gamma)$ with $\int_\Gamma q = 0$. There is $w \in C^{k+1}(\mathcal{O}')^2$ with:*

- (a) $w = 0$ on $\Omega \cap \mathcal{N}_\Sigma$ for a neighbourhood $\mathcal{N}_\Sigma$ of Σ ; extended by 0 to $\mathcal{N}_\Sigma \setminus \Omega$, w is defined on $\mathcal{O}' := \mathcal{O} \cup \mathcal{N}_\Sigma$ (an open neighbourhood of $\bar{\Omega}$) and $\operatorname{div} w = 0$ there;
- (b) $w = q N = -q n$ on Γ ;
- (c) $\|w\|_{W^{1,\infty}(\mathcal{T}_{r/16})} \leq C \|q\|_{W^{1,\infty}(\Gamma)}$, $\|\nabla w\|_{L^2(\Omega)} \leq C \|q\|_{W^{1,\infty}(\Gamma)}$, and $\|w\|_{W^{k+1,\infty}} \leq C \|q\|_{C^{k+1}(\Gamma)}$.

If $m = 1$, or more generally if all $\bar{q}_i = 0$, then $w = -\operatorname{curl}(\chi(d) Q_i(\sigma))$ near Γ_i , which is the construction of Section 6.

Proof. Let χ be as in Lemma 10.9 (so $\chi = 1$ on $(-r, r/8]$, $\chi = 0$ on $[r/4, r)$), and $Q_i(s) := \int_0^s (q - \bar{q}_i)$, which is $|\Gamma_i|$ -periodic. On $\Psi(\Gamma_i \times (-r, r))$ set

$$w_0 := \frac{\chi(d) q(\sigma)}{J} N \circ \pi - \chi'(d) Q_i(\sigma) T \circ \pi,$$

and $w_0 := 0$ elsewhere in $\mathcal{O}$. By (5),

$$\operatorname{div} w_0 = \frac{1}{J} (\chi' q - \chi'(q - \bar{q}_i)) = \frac{\chi'(d) \bar{q}_i}{J} =: g_0 \quad \text{on } U_i.$$

Here g_0 is supported in the bands $B_i := \Psi(\Gamma_i \times [r/8, r/4]) \subset \Omega$, and

$$\int g_0 = \sum_i \bar{q}_i |\Gamma_i| \int_0^r \chi' = - \sum_i |\Gamma_i| \bar{q}_i = - \int_\Gamma q = 0.$$

Write $g_0 = \sum_{i < m} \bar{q}_i g^{(i)}$ with $g^{(i)} := \chi'(d_i)/J - (|\Gamma_i|/|\Gamma_m|) \chi'(d_m)/J$, where d_j is the signed distance to Γ_j . These are $m-1$ fixed, h -independent, compactly supported mean-zero functions in C^{k+1} (since $\Gamma \in C^{k+3}$ gives $J \in C^{k+1}$).

Let $\Omega_{16} := \{x \in \Omega : \operatorname{dist}(x, \Gamma) > r/16\}$; it is connected, by the argument for Ω_* in Lemma 10.14. Let $K \Subset \Omega_{16}$ be a connected finite union of open balls with closures in Ω_{16} containing $\bigcup_i B_i$. It exists because $\bigcup B_i$ is compact and Ω_{16} is path-connected: join the balls covering the bands by finite chains of balls along paths. A tree argument on the intersection graph of the balls writes each $g^{(i)}$ as a finite sum of mean-zero functions in C_c^{k+1} of single balls: use a smooth partition of unity subordinate to the balls, and transfer masses along tree edges with fixed bumps in $C_c^\infty(B \cap B')$. Bogovskii's operator on each ball maps $W_0^{s,p}$ to $W_0^{s+1,p}$ for all $s \geq 0$, $1 < p < \infty$ [7, Thm. III.3.3]. This gives $w^{(i)} \in W_0^{k+2,p}(K) \subset C_c^{k+1}(K)$ ($p > 2$) with $\operatorname{div} w^{(i)} = g^{(i)}$.

Set $w := w_0 - \sum_{i < m} \bar{q}_i w^{(i)}$. Then:

- (a) holds, since w_0 vanishes outside the collars and $K \Subset \Omega$.
- (b): at $d = 0$, $\chi = 1$, $\chi' = 0$ and $J = 1$, and $w^{(i)} = 0$ near Γ .

- (c): on $\mathcal{T}_{r/16}$ only the first term of w_0 is present and $w^{(i)} = 0$ there, with $W^{1,\infty}$ norm $\leq C\|q\|_{W^{1,\infty}}$. On the bands $|Q_i| \leq |\Gamma_i| \|q\|_\infty$, and $|\bar{q}_i| \leq \|q\|_\infty$. The regularity count is: $N \circ \pi, T \circ \pi, \sigma \in C^{k+2}, J \in C^{k+1}, q \circ \sigma \in C^{k+1}$. $\square$

Lemma 10.19 (Discrete curl of a divergence-free field). *Let $\mathcal{O}'$ be an open neighbourhood of $\bar{\Omega}$, and let $w \in W^{k+1,\infty}(\mathcal{O}')^2$ be divergence-free and vanish on a neighbourhood of Σ . For $T \in \mathcal{T}_h$ let $B_T \subset \mathcal{O}'$ be an open disk of radius $2h$ containing $\bar{T}$. It exists for $h \leq h_*$, since $\bar{\Omega}_h \subset \bar{\Omega} \cup \{|d| \leq Ch_\Gamma^2\}$ lies in a fixed compact subset of $\mathcal{O}'$. Let φ_T be a stream function of w on B_T ($w = \text{curl } \varphi_T$), and define*

$$v_w|_T := \text{curl}(\Pi_{k+1}^T \varphi_T),$$

where Π_{k+1}^T is the element restriction of the C^1 piecewise- P_{k+1} nodal interpolant of [13] (Argyris for $k = 4$), whose degrees of freedom on T are functionals of the jet of the function on $\bar{T}$. Then v_w is well defined, $v_w \in Z_h$, and

$$\|v_w - w\|_{W^{1,\infty}(T)} \leq Ch_T^k |w|_{W^{k+1,\infty}(B_T)}.$$

If $w = \text{curl } \varphi$ globally, then $v_w = \text{curl } \Pi_{k+1} \varphi$ is the v_φ of Section 6.

Proof. Well defined. Two stream functions on B_T differ by a constant, and Π_{k+1}^T is linear and reproduces constants.

Continuity. Let T, T' share an edge e , and let $B \subset \mathcal{O}'$ be a disk containing $T \cup T'$ with stream function φ_B . On the connected set $B \cap B_T$, $\varphi_T - \varphi_B$ is constant, so $\text{curl } \Pi_{k+1}^T \varphi_T = \text{curl } \Pi_{k+1}^T \varphi_B$ on T , and likewise on T' . The global interpolant of φ_B on $T \cup T'$ is C^1 across e . So the two curls agree on e .

$v_w \in Z_h$. v_w is a curl elementwise, so $\text{div } v_w = 0$. If $\bar{T}$ meets Σ , then T lies in the neighbourhood where $w = 0$ (for h small), φ_T is constant and $v_w|_T = 0$. So $v_w = 0$ on Σ .

Error. $\|\text{curl}(\Pi_{k+1}^T \varphi_T - \varphi_T)\|_{W^{1,\infty}(T)} \leq Ch_T^k |\varphi_T|_{W^{k+2,\infty}(T)} = Ch_T^k |w|_{W^{k+1,\infty}(T)}$ by the local interpolation estimate for Π_{k+1} (constants depending on (M0), (M2) as in Section 6). $\square$

Definition 10.20 (General test field). Assume (R2). With $q = p - \bar{p}$ (Definition 10.6), let w be the field of Lemma 10.18 and $v_\varphi := v_w \in Z_h$ from Lemma 10.19. Then $\|v_\varphi - w\|_{W^{1,\infty}(\Omega_h)} \leq Ch^k$, and on Γ , $w = (p - \bar{p})N = -(p - \bar{p})n$, exactly as in Section 6.

The following pointwise facts on a chord e replace the uses of “ $n(x^*) = e_r$ ” in Section 6. They follow from Lemma 10.7, $|x - x^*| \leq Ch_\Gamma^2$, and $w \in W^{1,\infty}$:

$$\begin{aligned} v_\varphi \cdot n_e &= -q(x^*) + O(h_\Gamma^2 + h^k), \\ v_\varphi \cdot \tau_e &= O(\|q\|_\infty \|\varkappa\|_\infty h_\Gamma) + O(h_\Gamma^2 + h^k), \\ (\tilde{p} - \bar{p})(x) n_e + v_\varphi(x) &= -q(m_e^*) \varkappa_e t \tau_e + O(h_\Gamma^2 + h^k). \end{aligned} \tag{6}$$

For the third: $(\tilde{p} - \bar{p})(x) = q(x^*) + O(h_\Gamma^2)$, $v_\varphi(x) = q(x^*)N(x^*) + O(h_\Gamma^2 + h^k) = -q(x^*)n(x^*) + \dots$, then Lemma 10.7(b) and $q(x^*)\varkappa(x^*) = q(m_e^*)\varkappa_e + O(\ell)$. Also, by Lemma 10.7(c),

$$\|\tilde{p} - \bar{p}\|_{L^2(\Gamma_h)} \leq G(1 + Ch_\Gamma^2) + Ch_\Gamma^2, \quad \|v_\varphi\|_{L^2(\Gamma_h)} = G(1 + O(h_\Gamma^2)) + O(h^k). \tag{7}$$

10.5.2. The theorems.

Theorem 10.21 (Theorem 6.1 on general domains). *Under Assumptions 10.1, 10.4, (M0)–(M2), (D) and (R1), the upper bound of Theorem 6.1 holds with G of Definition 10.6. If in addition (R2) holds, the lower bound holds as stated. Corollary 6.2 follows as before.*

Proof. Upper bound. Verbatim, with $c = \bar{p}$ in Corollary 10.17 and (7): $|\mathcal{R}(e_h)| \leq (G + Ch_\Gamma^2)(h/\mu)^{1/2} \|e_h\|$.

Lower bound. Use v_φ of Definition 10.20. Step 1 becomes: by (6) and Lemma 10.7(c),

$$\mathcal{R}(v_\varphi) = \langle \tilde{p} - \bar{p}, q \circ \pi \rangle_{\Gamma_h} + O(h_\Gamma^2 + h^k) = \int_\Gamma q^2 + O(h_\Gamma^2 + h^k) = G^2 + O(h_\Gamma^2 + h^k).$$

Step 2 uses (7) and $\|\nabla v_\varphi\| + (\sum_e |e| \|\partial_n v_\varphi\|_{L^2(e)}^2)^{1/2} \leq C$ (Lemma 10.18(c)). Steps 3–4 are verbatim. $\square$

Theorem 10.22 (Theorem 6.4 on general domains). *Under the assumptions of Theorem 10.21 including (R2), with $C_p = C(1 + \|p - \bar{p}\|_{W^{1,\infty}(\Gamma)})$ ($\bar{p}$ global),*

$$\|\nabla(\tilde{u} - u_h)\| \leq C_p(h_\Gamma/\gamma) + C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h].$$

Proof. Steps 1, 2 and 6 are verbatim. Step 3: $S_\varphi := v_\varphi$ of Definition 10.20.

Step 4: by the third line of (6), on each e ,

$$(\tilde{p} - \bar{p})n_e + S_\varphi = b_e t + O(h_\Gamma^2 + h^k), \quad b_e := -q(m_e^*)\varkappa_e \tau_e \text{ constant on } e, \quad |b_e| \leq \|q\|_\infty \|\varkappa\|_\infty.$$

The argument of Lemma 10.12 gives $|m(v)| \leq C_p(h_\Gamma^{3/2} + h^k)\|\nabla v\|$.

Step 5: $|a_h(S_\varphi, v)| + |\langle \partial_n S_\varphi, v \rangle| \leq C_p \|\nabla v\|$ by Lemma 10.18(c) and Lemma 10.9. For $\langle S_\varphi, \partial_n v \rangle$ apply Lemma 6.3 to $S = w|_{\Gamma_h}$. Its hypotheses hold:

- w is continuous, and edgewise $W^{1,\infty}$ with $\|\nabla w\|_{L^\infty(\mathcal{T}_{r/16})} \leq C\|q\|_{W^{1,\infty}}$;
- $|w \cdot \tau_e| \leq C\|q\|_{W^{1,\infty}} h_\Gamma$ by the second line of (6) (applied to w);
- in step 3 of Lemma 6.3, $|\sigma_{e'}\tau_{e'} - \sigma_e\tau_e| \leq |w(x)|(|n_{e'} - n_e| + |\tau_{e'} - \tau_e|) \leq C\|q\|_\infty \|\varkappa\|_\infty(|e| + |e'|)$ by Lemma 10.7(d);
- the other steps of Lemma 6.3 are local and use only $\operatorname{div} v = 0$ and (M0).

The remainder $\langle S_\varphi - w, \partial_n v \rangle$ is bounded as in Theorem 6.4. $\square$

Theorem 10.23 (Theorem 6.6 on general domains). *Under the assumptions of Theorem 10.22, Theorem 6.6 holds verbatim with the global G .*

Proof. In (i), item 1 is Theorem 10.21, step 1. For item 2, on e we have $\tilde{u}(x) = \tilde{u}(x^*) + d(x)\partial_N \tilde{u}(x^*) + O(d^2) = -d(x)\partial_n u(x^*) + O(h_\Gamma^4)$. This is tangential up to $O(h_\Gamma^4)$, by $\partial_n u \cdot n = 0$ (Lemma 3.4), while $v_\varphi(x) = q(x^*)N(x^*) + O(h_\Gamma^2 + h^k)$. Hence $\tilde{u} \cdot v_\varphi = O(h_\Gamma^4 + h_\Gamma^2 h^k)$ as in Theorem 6.6. Items 3–5 are verbatim. In (ii), the lift E of $(g - g_I)|_\Sigma$ is supported near Σ , where $\Omega_h = \Omega$. $\square$

Corollary 10.24 (Corollary 6.7 on general domains). *Under the assumptions of Theorem 10.22, Corollary 6.7 holds verbatim, with v_φ of Definition 10.20 and the global G . To leading order the printed method imposes the leak condition $(\mu/h)u_h \cdot n \approx p - \bar{p}$ with $\bar{p}$ the mean of p over all curved components. In particular, with n pointing out of Ω_h , the discrete flux through $\Gamma_{h,i}$ is $\Phi_i(u_h) \approx (h/\mu)|\Gamma_i|(\bar{p}_i - \bar{p})$ (Lemma 10.16).*

Proof. Step 4 of the proof of Corollary 6.7 is (7). The rest is verbatim. $\square$

Proposition 10.25 (The constant is global). *Let $m \geq 2$. No estimate $\|\tilde{u} - u_h\| \leq C[(h/\mu)^{1/2}G_{\text{per}} + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$ can hold. Take $u = 0$, $g = 0$, $\tilde{p}$ any smooth function with $\tilde{p}|_{\Gamma_i} \equiv c_i$ (not all equal), and $\tilde{f} := \nabla \tilde{p}$, so that $f = \nabla p$. Then $G_{\text{per}} = 0$, $F = 0$, $\mathcal{G} \equiv 0$ and $E_h = 0$, but for $h \leq h_0(G)$ (in particular for fixed γ and bounded ρ as $h \rightarrow 0$)*

$$\|u_h\| \geq (2M)^{-1}(h/\mu)^{1/2}G, \quad G^2 = \sum_i |\Gamma_i|(c_i - \bar{c})^2 > 0, \quad \bar{c} = \frac{\sum_i |\Gamma_i|c_i}{|\Gamma|}.$$

Proof. Theorem 10.21, lower bound, with $\tilde{u} = 0$ (so $E_h = 0$: take $z = 0 \in W_h$) and $\mathcal{G} \equiv 0$. Here q is piecewise constant, so (R2) reduces to $\Gamma \in C^{k+3}$. $\square$

10.6. The pressure-consistent method. Proposition 7.1 holds verbatim: for $v \in V_h^R$, $-(1, \operatorname{div} v) + \langle 1, v \cdot n \rangle_{\Gamma_h} = -\int_{\Sigma} v \cdot n = 0$.

More generally, for a linear functional ℓ on $\Pi_h + \mathbb{R}$ with $\ell(1) = 1$, let $\operatorname{CNS}_\ell^*$ be CNS^* with $\bar{p}_\Gamma$ replaced by ℓ :

$$B_\ell(q, v) := -(q, \operatorname{div} v) + \langle q - \ell(q), v \cdot n \rangle_{\Gamma_h}.$$

$\operatorname{CNS}^* = \operatorname{CNS}_\ell^*$ with $\ell = \bar{p}_\Gamma$ (global boundary mean). Extend ℓ to $\tilde{p}$ by the same formula.

Theorem 10.26 (Theorem 7.2 on general domains). *Under Assumptions 10.1, 10.4, (M0)–(M2), (D) and (R1), and for any such ℓ , Theorem 7.2 holds verbatim for $\operatorname{CNS}_\ell^*$ (with β from Lemma 10.15). In particular one global correction suffices for well-posedness and for the rate $h_\Gamma^{3/2} + h_\Omega^k$, for any number m of curved components.*

Proof. We go through the steps of Theorem 7.2.

- (1) *Consistency.* For $c \in \mathbb{R}$, $B_\ell(q + c, v) = B_\ell(q, v) - c \int_{\Gamma_h} v \cdot n$, because $\int_{\Omega_h} \operatorname{div} v = \int_{\Gamma_h} v \cdot n$ for $v \in V_h^R$. With $p^* := \tilde{p} - \ell(\tilde{p})$,

$$B_\ell(p^*, v) = -(\tilde{p}, \operatorname{div} v) + \langle \tilde{p}, v \cdot n \rangle + \ell(\tilde{p}) \left(\int_{\Omega_h} \operatorname{div} v - \int_{\Gamma_h} v \cdot n \right) = -(\tilde{p}, \operatorname{div} v) + \langle \tilde{p}, v \cdot n \rangle,$$

which is the pressure form of Lemma 5.1. So (3) holds for all $v \in V_h^R$.

- (2) *Split.* Verbatim.

- (3) *Pressure.* Verbatim: $v \in V_h^0$ vanishes on all of Γ_h , and Lemma 10.15 applies.

- (4) *Velocity.* $B_\ell(e_p, e_h) = \langle e_p, e_h \cdot n \rangle - \ell(e_p) \sum_i \Phi_i(e_h) = \langle e_p, e_h \cdot n \rangle$ by Lemma 10.16(a). Also $\langle \tilde{\xi}, e_h \cdot n \rangle = \tilde{\xi} \sum_i \Phi_i(e_h) = 0$. Only the *total* flux enters, never the individual Φ_i . The rest is verbatim.

- (5) *Absorb.* Verbatim.

- (6) *Uniqueness* (Proposition 10.27). □

Proposition 10.27 (Kernel analysis). *For $\gamma \geq \gamma_2$, the homogeneous $\operatorname{CNS}_\ell^*$ system ($g_I = 0$, $\tilde{f} = 0$) has only the trivial solution. Hence $\operatorname{CNS}_\ell^*$ is uniquely solvable for every $m \geq 1$.*

Proof. Let (u_h, p_h) solve the homogeneous system. By Lemma 4.3 $u_h \in Z_h$, and steps 3–5 with $z = 0$, $\tilde{u} = 0$, $\tilde{p} = 0$ (so $\mathcal{G} = 0$, $E_h = 0$, $\eta = 0$) give $u_h = 0$. Step 3 then gives $\|p_h - \bar{p}_h\| = 0$, where $\bar{p}_h$ is the mean over Ω_h . Since Ω_h is connected (Lemma 10.9), $p_h \equiv c$. Then for all $v \in V_h^R$,

$$0 = B_\ell(c, v) = -c \int_{\Omega_h} \operatorname{div} v + (c - c \ell(1)) \int_{\Gamma_h} v \cdot n = -c \int_{\Gamma_h} v \cdot n.$$

The edge bubble $b = \beta_e n_e$ on any single $e \in \mathcal{E}_h^\Gamma$ has $\int_{\Gamma_h} b \cdot n \neq 0$, so $c = 0$. The system is square, so uniqueness gives existence. □

Remark 10.28 (The velocity does not depend on ℓ). Let $B_0(q, v) := -(q, \operatorname{div} v) + \langle q, v \cdot n \rangle_{\Gamma_h}$ be the naive form of Proposition 7.1. Then $B_\ell(q, v) = B_0(q, v) - \ell(q) \int_{\Gamma_h} v \cdot n$, so $\operatorname{CNS}_\ell^*$ is the naive system plus the rank-one term $-\ell(p_h) \int_{\Gamma_h} v \cdot n$. For a constant c and $v \in V_h^R$, $B_0(c, v) = -c \int_{\partial\Omega_h} v \cdot n + c \int_{\Gamma_h} v \cdot n = 0$. Hence, if (u_h, p_h) solves $\operatorname{CNS}_\ell^*$ and ℓ' is another normalisation, then $(u_h, p_h + c)$ with $c := \ell(p_h) - \ell'(p_h)$ solves $\operatorname{CNS}_{\ell'}^*$, because $\ell'(p_h + c) = \ell(p_h)$ and the continuity row does not involve ℓ . So the discrete velocity and the scalar $\lambda_h := \ell(p_h)$ do not depend on the normalisation; the pressures differ by constants. The naive system is solvable if and only if $\lambda_h = 0$: if $\lambda_h = 0$, (u_h, p_h) solves it; conversely, if (u, p) solves it, then $(u, p - \ell(p))$ solves $\operatorname{CNS}_\ell^*$, and uniqueness gives $\lambda_h = \ell(p - \ell(p)) = 0$.

Remark 10.29 (Per-component correction is inconsistent). Let $m \geq 2$ and replace $\langle q - \bar{p}_\Gamma(q), v \cdot n \rangle$ by $\sum_i \langle q - \ell_i(q), v \cdot n \rangle_{\Gamma_{h,i}}$ with $\ell_i(q) := |\Gamma_{h,i}|^{-1} \int_{\Gamma_{h,i}} q$. This is *not* of the form CNS_ℓ^* . For $v \in V_h^R$,

$$B_{\text{per}}(q, v) = B_{\bar{p}_\Gamma}(q, v) - \sum_i (\ell_i(q) - \bar{p}_\Gamma(q)) \Phi_i(v).$$

Adding a constant to q does not change the sum, so no choice of pressure constant makes the exact pair consistent. The residual on Z_h is

$$\mathcal{I}(v) = \sum_i (\ell_i(\tilde{p}) - \bar{p}_\Gamma(\tilde{p})) \Phi_i(v) = \sum_i (\bar{p}_i - \bar{p}) \Phi_i(v) + O(h_\Gamma^2) \|v\|_{L^1(\Gamma_h)}.$$

Test with $v = v_\alpha$ from the proof of Lemma 10.16(b), $\alpha_i := \bar{p}_i - \bar{p}$, $A^2 := \sum_i |\Gamma_i| \alpha_i^2$. Then $\mathcal{I}(v_\alpha) = -A^2 + O(h_\Gamma^2 + h^k) A^2$, and $\|v_\alpha\| \leq (\mu/h)^{1/2} A(1 + o(1)) + CA$ (using $|\alpha| \leq CA$). Hence, if $A > 0$, the dual norm of $\mathcal{I}$ with respect to $\|\cdot\|$ is at least $c(h/\mu)^{1/2} A$ for $h \leq h_0$, because $h/\mu = h_\Gamma/\gamma \leq h_\Gamma/\gamma_0$ is small. We do *not* prove that the discrete error of the per-component method is actually of this size. Unlike GS, its pressure does not drop out on Z_h , so the lower-bound argument of Theorem 10.21 does not apply directly. It is, however, not a consistent method of order $h_\Gamma^{3/2}$ when the component means of p differ. The per-component method is uniquely solvable for γ large: the extra term is absorbed in step 4 like $\langle \xi - \bar{\xi}, e_h \cdot n \rangle$, since $|\ell_i(\xi - \bar{\xi})| \leq |\Gamma_{h,i}|^{-1/2} \|\xi - \bar{\xi}\|_{\Gamma_{h,i}}$, and the kernel argument of Proposition 10.27 goes through.

10.7. The penalty threshold. Proposition 8.1 is local to a boundary star and uses only that the angle of Ω_h at a vertex of Γ_h is $\pi + O(h_\Gamma)$, which is Lemma 10.8(a). It holds verbatim. No computations have been run for general domains.

10.8. Summary of dependencies and scope.

- $\Gamma \in C^3$, $\Sigma \neq \emptyset$, Ω connected: Lemmas 10.7–10.12, Section 10.3, Lemmas 5.1 and 5.2, Theorem 10.21 (upper bound), Theorem 10.26.
- Additionally (R2) ($\Gamma \in C^{k+3}$, $p|_\Gamma \in C^{k+1}$): everything using v_φ , namely the lower bound in Theorem 10.21, Theorems 10.22 and 10.23, Corollary 10.24, Lemma 10.16(b), Proposition 10.25, and the lower bound in Remark 10.29.
- New geometric constants: $\|\varkappa\|_{C^1}$, the tube radius r , the graph scale a_0 , and the h -independent cover of Lemma 10.14. The discrete constants $C_I, c_1, M, \gamma_0, \gamma_2$ depend on these only through κ and β .
- Inherited without re-proof, exactly as in Sections 4 and 6: the reduction of the Guzmán–Scott discrete inf-sup constant to the continuous one plus a Poincaré constant, and the local nodal structure and interpolation estimate of the Morgan–Scott C^1 interpolant.
- Not covered: the identification of the H^1 constant (Section 6.1), the lower bound (Section 7.1), the single-triangle witness (Section 8.1), Clough–Tocher meshes (Section 9), and the case $\Sigma = \emptyset$ or components that mix curved and straight pieces. We make no claim about these on general domains.

11. STEADY NAVIER–STOKES

Gjerde and Scott’s application is cylinder drag, which is a Navier–Stokes computation [8]. This section extends Theorems 6.1–7.2 and Corollary 6.7 to

$$-\nu \Delta u + (u \cdot \nabla)u + \nabla p = f, \quad \text{div } u = 0 \quad \text{in } \Omega, \quad u|_\Gamma = 0, \quad u|_{\partial R} = g.$$

All results of this section assume a discrete stability property of the linearisation at the exact solution, (L) for GS or (L*) for CNS*. We prove these only under a small-data condition (SD) (Lemma 11.7); deriving them from nonsingularity of a large-data solution branch is open (Remark 11.14). The Stokes estimates are reused, in particular steps 4–5 of the proof of Theorem 6.4.

The only new ingredients are a fixed-point argument around the linearisation at the exact solution, and a check that the Stokes estimates are stable under lower-order perturbations that are bounded in the H^1 dual norm.

11.1. What Gjerde–Scott print. The arXiv version of [8] (arXiv:2306.12362) states the continuous problem $-\nu\Delta u + u \cdot \nabla u + \nabla p = 0$. The drag functional (their (7)) contains the standard convective term $(u \cdot \nabla u) \cdot v$. The discrete Nitsche system (27)–(28) is printed for the Stokes operator only. No discrete convective form is printed, and in particular there is no skew-symmetrisation and no inflow/outflow boundary term. The nonlinear solve is described only as the iterated penalty method. We therefore analyse both natural choices below, and show that the conclusions are the same for both.

11.2. Methods. Let $\langle \cdot, \cdot \rangle$, N_h , V_h^R , Z_h , W_h , Π_h and $\|\cdot\|$ be as in Section 2. For $w, u, v \in H^1(\Omega_h)^2$ define

$$c_1(w; u, v) := ((w \cdot \nabla)u, v), \quad c_s(w; u, v) := \frac{1}{2} [((w \cdot \nabla)u, v) - ((w \cdot \nabla)v, u)].$$

Here c denotes either one.

Definition 11.1 (GS and CNS* for Navier–Stokes). Find $u_h \in V_h$ with $u_h|_{\partial R} = g_I$ and $p_h \in \Pi_h$ such that, for all $v \in V_h^R$ and $q \in \Pi_h$,

$$\nu N_h(u_h, v) + c(u_h; u_h, v) - (p_h, \operatorname{div} v) [+ (p_h - \bar{p}_\Gamma(p_h), v \cdot n)] = (\tilde{f}, v), \quad (q, \operatorname{div} u_h) = 0.$$

Without the bracket this is GS; with it, it is CNS*.

Remark 11.2 (Penalty convention). Multiplying N_h by ν makes the wall penalty $\nu\mu/h$. If instead one uses $\nu a_h - \nu \langle \partial_n u, v \rangle - \nu \langle u, \partial_n v \rangle + (\hat{\mu}/h) \langle u, v \rangle$ with an unscaled $\hat{\mu}$, this equals νN_h with $\mu = \hat{\mu}/\nu$. Every statement below holds with $\nu\mu$ replaced by $\hat{\mu}$. In particular the leak velocity below is then $(h/\hat{\mu})(p - \bar{p})$, and coercivity needs $\hat{\mu} \geq \nu\mu_0$.

11.3. Assumptions.

Assumption 11.3.

- (NS1) (u, p) is a smooth solution on $\bar{\Omega}$, $\tilde{f}$ is a smooth extension of f , and (M0)–(M2) and (D) hold.
 (NS2) The extensions are as in Section 2: $\tilde{u} = \operatorname{curl} \tilde{\psi}$ (so $\operatorname{div} \tilde{u} = 0$ on a neighbourhood of $\bar{\Omega}$) and $\tilde{p}$ is smooth. The defect

$$F := \tilde{f} + \nu \Delta \tilde{u} - (\tilde{u} \cdot \nabla) \tilde{u} - \nabla \tilde{p}$$

vanishes on Ω , is supported in $\bar{S}_h$, and $\|F\|_\infty \leq C$.

Stability is assumed in one of two forms, (L) or (L*) below. Lemma 11.7 shows that the standard small-data condition implies both. Let $\mathfrak{A}$ be the linearisation at $\tilde{u}$:

$$\mathfrak{A}(w, v) := \nu N_h(w, v) + c(w; \tilde{u}, v) + c(\tilde{u}; w, v).$$

Set $\|\ell\|_{*,X} := \sup_{0 \neq v \in X} \ell(v) / \|v\|$ for $X = Z_h$ or $X = V_h^R$.

- (L) There is $\beta_L > 0$, independent of h , with $\beta_L \|\delta\| \leq \|\mathfrak{A}(\delta, \cdot)\|_{*,Z_h}$ for all $\delta \in Z_h$.
 (L*) There is $\beta_* > 0$, independent of h , such that for every linear functional ℓ on V_h^R the problem

$$(\delta, \xi) \in Z_h \times \Pi_h : \quad \mathfrak{A}(\delta, v) + B^*(\xi, v) = \ell(v) \quad \forall v \in V_h^R,$$

with $B^*(q, v) = -(q, \operatorname{div} v) + \langle q - \bar{p}_\Gamma(q), v \cdot n \rangle$, has a unique solution, and $\beta_* \|\delta\| \leq \|\ell\|_{*,V_h^R}$.

(L) and (L*) are the discrete forms of “ u lies on a nonsingular branch” for GS and CNS* respectively; see Remark 11.14.

11.4. Tools.

Lemma 11.4 (Uniform Sobolev constants). *There are C_1 and N , independent of h , such that*

- (a) $\|v\|_{H^1(\Omega_h)} \leq C_1 \|\nabla v\|$ for all $v \in H^1(\Omega_h)^2$ with $v|_{\partial R} = 0$;
- (b) $|c(w; u, v)| \leq N \|w\|_{H^1(\Omega_h)} \|u\|_{H^1(\Omega_h)} \|\nabla v\|$ for all $w, u \in H^1(\Omega_h)^2$ and all $v \in H^1(\Omega_h)^2$ with $v|_{\partial R} = 0$, for $c = c_1$ and $c = c_s$.

The same holds in three dimensions.

Proof. (a) is Lemma 3.2. For (b), let Φ_h be the bi-Lipschitz map of Lemma 3.2 with $\|D\Phi_h - I\|_\infty \leq Ch_\Gamma$. The embedding $H^1(\Omega) \subset L^4(\Omega)$ on the fixed Lipschitz domain Ω ($d = 2, 3$) transfers to $\|w\|_{L^4(\Omega_h)} \leq S \|w\|_{H^1(\Omega_h)}$ with S independent of h . Then $|c_1(w; u, v)| \leq \|w\|_{L^4} \|\nabla u\| \|v\|_{L^4}$ and $|((w \cdot \nabla)v, u)| \leq \|w\|_{L^4} \|\nabla v\| \|u\|_{L^4}$. Finish with (a). $\square$

Lemma 11.5 (Convective boundary identities). *Let $w \in H^1(\Omega_h)^2$ with $\operatorname{div} w = 0$, and $u, v \in H^1(\Omega_h)^2$ with $v|_{\partial R} = 0$. Then*

$$c_1(w; u, v) - c_s(w; u, v) = \frac{1}{2} \langle (w \cdot n)u, v \rangle, \quad c_1(w; v, v) = \frac{1}{2} \langle w \cdot n, |v|^2 \rangle, \quad c_s(w; v, v) = 0.$$

Proof. $((w \cdot \nabla)u, v) + ((w \cdot \nabla)v, u) = \int_{\Omega_h} w \cdot \nabla(u \cdot v) = \int_{\partial\Omega_h} (w \cdot n)(u \cdot v) - \int_{\Omega_h} (\operatorname{div} w)(u \cdot v)$. The last integral vanishes, and so does the part of the boundary integral on ∂R , since $v = 0$ there. The outward normal of Ω_h on Γ_h is n . $\square$

So $c_1(u_h; v, v)$ vanishes for all v only if $u_h \cdot n = 0$ on Γ_h . Nitsche's method never enforces this, and for GS the normal trace is the leak $(h/\nu\mu)(p - \bar{p})$ of Corollary 11.9. Section 11.7 shows that this term nevertheless does not affect any result.

Lemma 11.6 (Error equation). *For $v \in V_h^R$,*

$$\nu N_h(\tilde{u}, v) + c(\tilde{u}; \tilde{u}, v) - (\tilde{p}, \operatorname{div} v) + \langle \tilde{p}, v \cdot n \rangle = (\tilde{f}, v) + \mathcal{G}_\nu(v) + \kappa_c(v),$$

where

$$\mathcal{G}_\nu(v) := \nu \left[\langle (\nabla \tilde{u})^\top n, v \rangle - \langle \tilde{u}, \partial_n v \rangle + \frac{\mu}{h} \langle \tilde{u}, v \rangle \right] - (F, v), \quad \kappa_{c_1} = 0, \quad \kappa_{c_s}(v) = -\frac{1}{2} \langle (\tilde{u} \cdot n) \tilde{u}, v \rangle.$$

Moreover $|\mathcal{G}_\nu(v)| \leq C(1 + \gamma^{1/2})h_\Gamma^{3/2} \|v\|$ on V_h^R , $|\mathcal{G}_\nu(v)| \leq Ch_\Gamma^{3/2} \|\nabla v\|$ on V_h^0 , and $|\kappa_{c_s}(v)| \leq Ch_\Gamma^4 \|\nabla v\|$.

Proof. Follow Lemma 5.1: $\nu a_h(\tilde{u}, v) = -\nu(\Delta \tilde{u}, v) + \nu \langle D(\tilde{u})n, v \rangle$ and $-\nu \Delta \tilde{u} = \tilde{f} - F - (\tilde{u} \cdot \nabla)\tilde{u} - \nabla \tilde{p}$. This gives the identity for c_1 . For c_s , apply Lemma 11.5 with $w = u = \tilde{u}$. The bound on $\mathcal{G}_\nu$ is Lemma 5.2, since F enters only through Lemma 3.3, which uses only $\|F\|_\infty \leq C$ and $\operatorname{supp} F \subset \bar{S}_h$. For κ_{c_s} , use $\|\tilde{u}\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^2$ (Lemma 3.3) and Lemma 3.2. $\square$

Lemma 11.7 (Small data implies (L) and (L*)). *Let $\gamma \geq \gamma_0$ and*

$$\lambda := \frac{NC_1 \|\tilde{u}\|_{H^1(\Omega_h)}}{\nu c_1} \leq \frac{1}{4}. \quad (\text{SD})$$

Then (L) holds with $\beta_L = \nu c_1/2$. There is $\gamma'_2 \geq \gamma_0$, independent of ν and h , such that for $\gamma \geq \gamma'_2$ (L) holds with $\beta_* = \nu/C_*$, where C_* is independent of ν and h .*

(SD) follows for small h from $NC_1 \|u\|_{H^1(\Omega)} < \nu c_1/4$, since $\|\tilde{u}\|_{H^1(\Omega_h)} \rightarrow \|u\|_{H^1(\Omega)}$. This is the usual small-data (low Reynolds number) condition. It is stated with the *discrete* Nitsche coercivity constant c_1 of Lemma 4.2, so it is more restrictive than the continuous one.

Proof. (1) (L). For $\delta \in V_h^R$, Lemma 11.4 gives $|c(\delta; \tilde{u}, \delta)| + |c(\tilde{u}; \delta, \delta)| \leq 2NC_1 \|\tilde{u}\|_{H^1} \|\nabla \delta\|^2 \leq 2\lambda \nu c_1 \|\delta\|^2$. With Lemma 4.2, $\mathfrak{A}(\delta, \delta) \geq \frac{1}{2} \nu c_1 \|\delta\|^2$.

- (2) *Pressure bound for (L^*) .* Let (δ, ξ) solve the problem in (L^*) and $v \in V_h^0$. Then $B^*(\xi, v) = -(\xi - \bar{\xi}, \operatorname{div} v)$, so $(\xi - \bar{\xi}, \operatorname{div} v) = \mathfrak{A}(\delta, v) - \ell(v)$. On V_h^0 ,

$$N_h(\delta, v) = a_h(\delta, v) - \langle \delta, \partial_n v \rangle \leq C_N \|\delta\| \|\nabla v\|, \quad C_N := 2 + C_I(c_0 \gamma_0)^{-1/2},$$

the convective terms are at most $2\lambda \nu c_1 \|\delta\| \|\nabla v\|$, and $\|v\| \leq (1 + C_I^2)^{1/2} \|\nabla v\|$. Lemma 4.5 gives

$$\beta \|\xi - \bar{\xi}\| \leq \nu(C_N + c_1/2) \|\delta\| + (1 + C_I^2)^{1/2} \|\ell\|_{*, V_h^R}.$$

- (3) *Velocity bound.* Test with $v = \delta \in Z_h$. Since $\operatorname{div} \delta = 0$ and $\int_{\Gamma_h} \delta \cdot n = 0$, we have $B^*(\xi, \delta) = \langle \xi - \bar{\xi}, \delta \cdot n \rangle$. As in step 4 of Theorem 7.2, its modulus is at most $C_I(c_0 \gamma)^{-1/2} \|\xi - \bar{\xi}\| \|\delta\|$. So

$$\frac{1}{2} \nu c_1 \|\delta\| \leq \|\ell\|_{*, V_h^R} + C_I(c_0 \gamma)^{-1/2} \beta^{-1} [\nu(C_N + c_1/2) \|\delta\| + (1 + C_I^2)^{1/2} \|\ell\|_{*, V_h^R}].$$

Choose γ'_2 with $C_I(c_0 \gamma'_2)^{-1/2} \beta^{-1} (C_N + c_1/2) \leq c_1/4$. The factor ν cancels, so γ'_2 does not depend on ν . This gives $\|\delta\| \leq C_* \nu^{-1} \|\ell\|_{*, V_h^R}$.

- (4) *Existence and uniqueness.* If $\ell = 0$, then $\delta = 0$, then ξ is constant by step 2, and $\xi = 0$ by the edge bubble of Lemma 4.3. The system is square, because $\dim Z_h + \dim \Pi_h = \dim V_h^R$ by Lemma 4.3. $\square$

11.5. The printed method. Let $z \in W_h$ attain the bound of Lemma 4.6, so $\|\tilde{u} - z\| \leq C\varepsilon_h$. With the lift E from the proof of Theorem 6.6(ii) and Lemma 11.4(a),

$$\|\tilde{u} - z\|_{H^1(\Omega_h)} \leq C(\varepsilon_h + h^{k+1/2}) \leq C\varepsilon_h. \quad (8)$$

Put $\mathcal{R}(v) := -\langle \tilde{p} - \bar{p}, v \cdot n \rangle$, as in Corollary 5.3, where $\bar{p}$ is the mean over Γ of the Navier–Stokes pressure p . Let $v_\varphi \in Z_h$ be the field of Section 6, built from this p . Write $G = \|p - \bar{p}\|_{L^2(\Gamma)}$ and

$$X := h/\mu + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h.$$

Theorem 11.8 (GS for Navier–Stokes). *Assume (NS1)–(NS2), (L) and $\gamma \geq \gamma_0$, for either convective form. There are $c_* > 0$ and C , depending on ν , β_L , N , C_1 , the mesh constants and the solution (and on $\|p - \bar{p}\|_{W^{1,\infty}(\Gamma)}$), such that if $X \leq c_*$, then:*

- (a) *Existence and local uniqueness.* GS has a solution (u_h, p_h) . It is the only one with $u_h \in z - (\delta_0 + \{\|\cdot\| \leq r\})$, where $\delta_0 \in Z_h$ is the linearised error defined in (10) and $r = CX$.
- (b) *H^1 rate 1.* $\|\nabla(\tilde{u} - u_h)\| \leq C[h/\mu + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$.
- (c) *Energy rate 1/2.* $\|\tilde{u} - u_h\| \leq \nu^{-1}(h/\mu)^{1/2}G + CX$.
- (d) *Quadratic closeness to the linearisation.* $\|(\tilde{u} - u_h) - (\tilde{u} - z) - \delta_0\| \leq CX^2$.

Proof. (1) *Reformulation.* Subtracting the GS equations from Lemma 11.6, and using $\operatorname{div} v = 0$ and $\int_{\Gamma_h} v \cdot n = 0$ for $v \in Z_h$, the error $e := \tilde{u} - u_h$ satisfies

$$\nu N_h(e, v) + c(\tilde{u}; \tilde{u}, v) - c(u_h; u_h, v) = \mathcal{R}(v) + \mathcal{G}_\nu(v) + \kappa_c(v) \quad \forall v \in Z_h.$$

By bilinearity, $c(\tilde{u}; \tilde{u}, v) - c(\tilde{u} - e; \tilde{u} - e, v) = c(e; \tilde{u}, v) + c(\tilde{u}; e, v) - c(e; e, v)$, so

$$\mathfrak{A}(e, v) = \mathcal{R}(v) + \mathcal{G}_\nu(v) + \kappa_c(v) + c(e; e, v).$$

Write $u_h = z - \delta$ with $\delta \in Z_h$, so $e = e(\delta) := (\tilde{u} - z) + \delta$. Then u_h solves the GS momentum equation on Z_h if and only if

$$\mathfrak{A}(\delta, v) = \ell_0(v) + c(e(\delta); e(\delta), v) \quad \forall v \in Z_h, \quad \ell_0 := \mathcal{R} + \mathcal{G}_\nu + \kappa_c - \mathfrak{A}(\tilde{u} - z, \cdot). \quad (9)$$

Given such $u_h \in W_h$, the functional $v \mapsto \nu N_h(u_h, v) + c(u_h; u_h, v) - (\tilde{f}, v)$ vanishes on $Z_h = \ker(\operatorname{div}|_{V_h^R})$. Since $\operatorname{div} V_h^R = \Pi_h$ (Lemma 4.3), there is a unique $p_h \in \Pi_h$ with $(p_h, \operatorname{div} v)$ equal to it on V_h^R . So solving (9) is equivalent to solving GS.

(2) *Linearised error.* By (L), $\mathfrak{A}$ is invertible on Z_h (a square system). Let

$$\delta_0 := \mathfrak{A}^{-1}\ell_0 = \delta_R + \delta_G, \quad \mathfrak{A}(\delta_R, \cdot) = \mathcal{R}, \quad \mathfrak{A}(\delta_G, \cdot) = \mathcal{G}_\nu + \kappa_c - \mathfrak{A}(\tilde{u} - z, \cdot) \quad \text{on } Z_h. \quad (10)$$

(3) *Bound on δ_G .* Since $|\mathfrak{A}(\tilde{u} - z, v)| \leq \nu M \|\tilde{u} - z\| \|v\| + 2N \|\tilde{u}\|_{H^1} \|\tilde{u} - z\|_{H^1} \|\nabla v\|$, Lemma 11.6 and (8) give $\|\delta_G\| \leq C\beta_L^{-1}[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$.

(4) *Ansatz for δ_R .* Set $\delta_R = (h/\nu\mu)v_\varphi + r$. Since $\nu N_h \cdot (h/\nu\mu) = (h/\mu)N_h$,

$$\mathfrak{A}(r, v) = m(v) + \ell(v) - \frac{h}{\nu\mu} [c(v_\varphi; \tilde{u}, v) + c(\tilde{u}; v_\varphi, v)],$$

where m and ℓ are exactly the functionals of step 3 of Theorem 6.4. Steps 4–5 of that proof give $|m(v)| + |\ell(v)| \leq [C(h/\mu + h_\Gamma^{3/2} + h^k) + C(h/\mu)h^k h_\Gamma^{-1/2}] \|\nabla v\|$. The convective term is at most $C(h/\mu) \|\nabla v\|$ by Lemma 11.4, since $\|v_\varphi\|_{H^1} \leq C$. Using $(h/\mu)h^k h_\Gamma^{-1/2} \leq Ch^k$, $\|\nabla v\| \leq \|v\|$ and (L),

$$\|r\| \leq C\beta_L^{-1}(h/\mu + h_\Gamma^{3/2} + \varepsilon_h).$$

(5) *Consequences for δ_0 .* Since $\|\nabla v_\varphi\| \leq C$ and $\|v_\varphi\| \leq (\mu/h)^{1/2}(G + Ch_\Gamma^2 + Ch^k) + C$ (step 2 of the lower bound in Theorem 6.1),

$$\|\nabla \delta_0\| \leq CX, \quad \|\delta_0\| \leq \nu^{-1}(h/\mu)^{1/2}G + CX. \quad (11)$$

(6) *Contraction.* Let $\Phi(\delta) := \mathfrak{A}^{-1}[\ell_0 + c(e(\delta); e(\delta), \cdot)]$ on Z_h . Set $A := \|\tilde{u} - z\|_{H^1} + C_1 \|\nabla \delta_0\|$, so $A \leq CX$ by (8) and (11). Consider the ball $B := \{\delta \in Z_h : \|\delta - \delta_0\| \leq A/C_1\}$. For $\delta \in B$, Lemma 11.4(a) gives $\|e(\delta)\|_{H^1} \leq A + C_1 \|\delta - \delta_0\| \leq 2A$.

- By (L) and Lemma 11.4(b), $\|\Phi(\delta) - \delta_0\| \leq \beta_L^{-1}N(2A)^2$, which is at most A/C_1 if $A \leq \beta_L/(4NC_1)$.
- For $\delta_1, \delta_2 \in B$ with $d := \delta_1 - \delta_2$, we have $c(e_1; e_1, v) - c(e_2; e_2, v) = c(d; e_1, v) + c(e_2; d, v)$. Hence

$$\|\Phi(\delta_1) - \Phi(\delta_2)\| \leq \beta_L^{-1}N \cdot 4A \cdot C_1 \|d\| \leq \frac{1}{2} \|d\| \quad \text{if } A \leq \beta_L/(8NC_1).$$

Both conditions follow from $X \leq c_*$. Banach's fixed-point theorem gives a unique $\delta \in B$ with $\Phi(\delta) = \delta$, which proves (a), and $\|\delta - \delta_0\| \leq 4\beta_L^{-1}NA^2 \leq CX^2$, which proves (d).

(7) (b) and (c). From $e = (\tilde{u} - z) + \delta_0 + (\delta - \delta_0)$, (11), (d), $\|\tilde{u} - z\| \leq C\varepsilon_h$ and $X^2 \leq X$. $\square$

Corollary 11.9 (Exact leading term; sharpness). *Assume in addition that $G > 0$, $h_\Gamma \rightarrow 0$, $h/\mu \rightarrow 0$, $\gamma h_\Gamma \rightarrow 0$ and $\varepsilon_h/(h_\Gamma/\gamma)^{1/2} \rightarrow 0$ (the hypotheses of Corollary 6.7). Then*

$$\tilde{u} - u_h = \frac{h}{\nu\mu} v_\varphi + \vartheta_h, \quad \|\vartheta_h\| = o((h/\mu)^{1/2}), \quad \|\vartheta_h\|_{\Gamma_h} = o(h/\mu),$$

and consequently

$$\frac{\|\tilde{u} - u_h\|_{L^2(\Gamma_h)}}{(h/\nu\mu)G} \rightarrow 1, \quad \frac{\|\tilde{u} - u_h\|}{\nu^{-1}(h/\mu)^{1/2}G} \rightarrow 1, \quad \|\nabla(\tilde{u} - u_h)\| \geq c \frac{h}{\nu\mu} G - Ch^{k+1/2} \quad \text{for small } h.$$

So $u_h \cdot n \approx (h/\nu\mu)(p - \bar{p})$ on Γ_h , with the Navier–Stokes pressure, and, if moreover μ is bounded (for instance fixed γ and bounded ρ), the H^1 rate 1 of Theorem 11.8(b) is sharp.

Proof. (1) Set $\vartheta_h := (\tilde{u} - z) + \delta_G + r + (\delta - \delta_0)$. Steps 3–4 and 6 of the proof of Theorem 11.8 give $\|\vartheta_h\| \leq C[h/\mu + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h] + CX^2 \leq CX$.

(2) Under the hypotheses, $X/(h/\mu)^{1/2} \rightarrow 0$. Indeed,

$$\frac{h/\mu}{(h/\mu)^{1/2}} \rightarrow 0, \quad \frac{(1 + \gamma^{1/2})h_\Gamma^{3/2}}{(h_\Gamma/\gamma)^{1/2}} = (\gamma^{1/2} + \gamma)h_\Gamma \rightarrow 0 \quad (\gamma \geq \gamma_0), \quad \frac{\varepsilon_h}{(h/\mu)^{1/2}} \rightarrow 0.$$

Hence $\|\vartheta_h\| = o((h/\mu)^{1/2})$ and $\|\vartheta_h\|_{\Gamma_h} \leq (h/\mu)^{1/2} \|\vartheta_h\| = o(h/\mu)$.

- (3) Step 4 of the proof of Corollary 6.7 gives $\|v_\varphi\|_{\Gamma_h} = G(1 + o(1))$ and $\|v_\varphi\|^2 = (\mu/h)\|v_\varphi\|_{\Gamma_h}^2 + O(1)$. This yields the two limits.
- (4) For the H^1 lower bound, use $\|e\|_{\Gamma_h} \leq C_{\text{tr}}(\|\nabla e\| + \|\nabla E\|)$ with $\|\nabla E\| \leq Ch^{k+1/2}$, where E is the lift from the proof of Theorem 6.6(ii). If μ is bounded, $h^{k+1/2} = o(h/\mu)$ and the term is absorbed. $\square$

Remark 11.10. The slip and energy constants are universal ($= 1$), exactly as for Stokes. The H^1 constant $\lim\|\nabla(\tilde{u} - u_h)\| \nu \mu / (hG)$, if it exists, is not universal. It now depends on the *Oseen* response: r solves the linearised problem, including $c(v_\varphi; \tilde{u}, \cdot) + c(\tilde{u}; v_\varphi, \cdot)$, which is of the same order as ∇v_φ . So the Stokes value $K = \|\nabla U\|/G$ of Corollary 6.11 (2.7565 for test P) is not expected to carry over; we have not identified the Navier–Stokes analogue.

Remark 11.11 (Cylinder drag forces $G > 0$). Under (L) and the hypotheses of Corollary 11.9, the following holds. For flow past the cylinder, the pressure drag is $D_p = \int_\Gamma p n_1$, up to the orientation of n . Since $\int_\Gamma n_1 = 0$, $|D_p| = |\int_\Gamma (p - \bar{p}) n_1| \leq |\Gamma|^{1/2} G$, so $G \geq |D_p|/\sqrt{2\pi}$. In Gjerde–Scott’s own application, the printed method therefore always leaks, at velocity $(h/\nu\mu)(p - \bar{p})$, and its asymptotic H^1 rate at fixed μ is 1, not 3/2. With their $\mu = 10^6$ the leak is small in absolute terms, which is consistent with their accurate drag values. The effect of the leak on the computed drag functional is not analysed here.

11.6. The pressure-consistent method.

Theorem 11.12 (CNS* for Navier–Stokes: Scott’s rate). *Assume (NS1)–(NS2) and (L*); by Lemma 11.7, (SD) and $\gamma \geq \gamma'_2$ suffice. Let $Y := (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h$. There are c_*, C such that if $Y \leq c_*$, then CNS* has a solution, whose velocity is unique in the $\|\cdot\|$ -ball of radius CY about $z - \delta_0$, where δ_0 is defined in step 2 below. It satisfies*

$$\|\tilde{u} - u_h\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h].$$

Hence, for fixed $k \geq 4$, $\gamma \in [\gamma'_2, \gamma_{\max}]$ and $h_\Gamma \geq h_\Omega^{2(\sigma_k - k)}$,

$$\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k).$$

Proof. (1) *Error equation.* Let $p^* := \tilde{p} - \bar{p}_\Gamma(\tilde{p})$, $\eta := p^* - \pi_h p^*$ (with π_h the L^2 projection onto Π_h), and write $p_h = p^* - \eta - \xi$ and $u_h = z - \delta$. As in step 1 of Theorem 7.2 and step 1 above, (u_h, p_h) solves CNS* if and only if $(\delta, \xi) \in Z_h \times \Pi_h$ solves

$$\mathfrak{A}(\delta, v) + B^*(\xi, v) = \ell_0(v) + c(e(\delta); e(\delta), v) \quad \forall v \in V_h^R, \quad \ell_0 := \mathcal{G}_\nu + \kappa_c - \mathfrak{A}(\tilde{u} - z, \cdot) - B^*(\eta, \cdot).$$

The continuity row holds because $\text{div } u_h = \text{div } z - \text{div } \delta = 0$.

- (2) *Size of ℓ_0 .* We have $(\eta, \text{div } v) = 0$ for all $v \in V_h^R$, because $\text{div } V_h^R \subset \Pi_h$. Hence $|B^*(\eta, v)| \leq 2\|\eta\|_{\Gamma_h} (h/\mu)^{1/2} \|v\| \leq Ch_\Gamma^k \|v\|$. With Lemma 11.6 and (8), $\|\ell_0\|_{*, V_h^R} \leq CY$. Let (δ_0, ξ_0) be the solution of (L*) with data ℓ_0 ; then $\|\delta_0\| \leq \beta_*^{-1} CY$.
- (3) *Fixed point.* Let $\Phi(\delta)$ be the Z_h -component of the (L*) solution with data $\ell_0 + c(e(\delta); e(\delta), \cdot)$. Since $\|c(e; e, \cdot)\|_{*, V_h^R} \leq N\|e\|_{H^1}^2$, step 6 of the proof of Theorem 11.8 applies verbatim, with β_L replaced by β_* and $A := \|\tilde{u} - z\|_{H^1} + C_1\|\delta_0\| \leq CY$.
- (4) *Conclusion.* This gives the unique fixed point and $\|\tilde{u} - u_h\| \leq \|\tilde{u} - z\| + \|\delta_0\| + CY^2 \leq CY$. The H^1 statement follows as in Theorem 7.2, using the lift E for the L^2 part. $\square$

11.7. Is an inflow/outflow correction needed? The next lemma quantifies the convective boundary term.

Lemma 11.13 (Wall Péclet number). *Let $\mu \geq \mu_0$, $w \in V_h$ with $\operatorname{div} w = 0$, and $v \in V_h^R$. Then $\nu N_h(v, v) + c_s(w; v, v) \geq \nu c_1 \|v\|^2$, and*

$$\nu N_h(v, v) + c_1(w; v, v) \geq (\nu c_1 - \tfrac{1}{2} \operatorname{Pe}_\Gamma(w)) \|v\|^2, \quad \operatorname{Pe}_\Gamma(w) := \frac{h}{\mu} \|(w \cdot n)_-\|_{L^\infty(\Gamma_h)},$$

where $(w \cdot n)_- := \max(-w \cdot n, 0)$ is the inflow part. Here n points out of Ω_h (into P_h).

Proof. By Lemma 11.5, $c_1(w; v, v) = \frac{1}{2} \langle w \cdot n, |v|^2 \rangle \geq -\frac{1}{2} \|(w \cdot n)_-\|_\infty \|v\|_{\Gamma_h}^2$, and $\|v\|_{\Gamma_h}^2 \leq (h/\mu) \|v\|^2$. $\square$

Consequences:

- (1) *The results above need no correction, for either form.* (L) and (L*) linearise at $\tilde{u}$, whose normal trace on Γ_h is $O(h_\Gamma^2)$. For c_1 versus c_s , $\mathfrak{A}$ changes by $\frac{1}{2} \langle (\tilde{u} \cdot n) \delta, v \rangle + \frac{1}{2} \langle (\delta \cdot n) \tilde{u}, v \rangle = O(h_\Gamma^2) \|\delta\|_{\Gamma_h} \|v\|_{\Gamma_h}$ (Lemma 11.5). So (L) for one form implies (L) for the other with $\beta_L - Ch_\Gamma^2 h/\mu$. The discrete leak enters only through the quadratic term $c(e; e, v)$, which the contraction controls by continuity alone (Lemma 11.4); no sign is needed.
- (2) *The leak does make $c_1(u_h; u_h, u_h) \neq 0$.* For GS, Corollary 11.9 gives $u_h \cdot n = (h/\nu\mu)(p - \bar{p}) + o(h/\mu)$ in $L^2(\Gamma_h)$, with n pointing into the cylinder. The discrete wall is porous: fluid leaves Ω_h where $p > \bar{p}$ (the front stagnation region) and re-enters where $p < \bar{p}$. The leak law $(\nu\mu/h) u_h \cdot n \approx p - \bar{p}$ is a Darcy-type condition with permeability $h/\nu\mu$. If the leading term also held in $L^\infty(\Gamma_h)$ (Corollary 11.9 is an L^2 statement; an L^∞ version would need an inverse estimate and is not proved here), the inflow part would give $\operatorname{Pe}_\Gamma(u_h) \approx (h/\mu)(h/\nu\mu) \|(p - \bar{p})_-\|_\infty = O((h/\mu)^2)$. The penalty would then dominate the inflow term by a factor of order $(h/\mu)^{-2}$. Formally, the kinetic-energy flux $\frac{1}{2} \int_{\Gamma_h} (u_h \cdot n) |u_h|^2$ is cubic in a boundary trace of size $h_\Gamma^2 + h/\mu$, which is negligible against the penalty dissipation $(\nu\mu/h) \|u_h\|_{\Gamma_h}^2 \approx (h/\nu\mu) G^2$. Point 1 is what the proofs actually use; point 2 is only the physical picture.
- (3) *When a correction would matter.* Lemma 11.13 degrades only when $\operatorname{Pe}_\Gamma(w) \gtrsim \nu c_1$ for iterates w far from $\tilde{u}$. This happens on coarse meshes, at large Reynolds number with fixed μ , or in Picard or iterated-penalty sweeps that start far from the solution. There c_s (for which $c_s(w; v, v) = 0$ identically), or an upwind inflow term $-\langle (w \cdot n)_- u, v \rangle$ in the style of Nitsche methods for Oseen problems (consistent to $O(h_\Gamma^4)$ at $\tilde{u}$), restores the energy inequality. For the asymptotic statements above neither is needed. c_s is also the natural choice for proving existence for arbitrary data, which we do not attempt: that would need a discrete Hopf extension of g_I inside W_h .

Remark 11.14 (What a Brezzi–Rappaz–Raviart version needs). (SD) is sufficient but not necessary for (L) and (L*). To derive (L) from continuous nonsingularity (the linearised Navier–Stokes operator at u is an isomorphism on the divergence-free subspace of H_0^1) one would run a Schatz/Gårding argument. Write $\mathfrak{A} = \nu N_h + K_h$ with K_h of lower order. It would then suffice that the discrete Nitsche–Stokes solution operators S_h (on Z_h , respectively $Z_h \times \Pi_h$) converge to the continuous one S on the moving domains $\Omega_h \rightarrow \Omega$, in operator norm on the compact set $\{K\delta : \|\nabla \delta\| \leq 1\}$. For data that are only in $L^{3/2}$ or H^{-1+s} , this requires Theorems 6.1 and 7.2 without a rate but with rough data. There the pressure is only in $W^{1,3/2}$ near Γ , and Lemma 3.5 must be replaced by an $o(1)$ statement for $\nabla w \in H^s$. We expect this to go through for CNS*, and for GS in the energy norm (whose inconsistency is $(h/\mu)^{1/2} \|p - \bar{p}\|_{L^2(\Gamma)} \rightarrow 0$), but we have not written it out. It is the only gap between the small-data theorems above and a nonsingular-branch theorem.

12. THREE DIMENSIONS

This section asks which of Theorems 6.1–7.2, Corollary 6.7, Proposition 8.1 and Section 11 survive when the curved wall is a surface approximated by an inscribed triangulated polyhedron.

The answer (Theorem 12.13) is that all of them except the necessity half, Proposition 8.1(b), carry over, with the same rates, under one stability hypothesis, (IS3). (IS3) is known only for split meshes (Alfeld/barycentric, $k \geq 3$), modulo the uniformity of the published constants over the perturbed domains (item 3 of Section 12.7). Only a spherical obstacle is treated. Three things change:

- the odd-in- s cancellation of Lemma 3.5 fails on non-equilateral faces (Lemma 12.2(d));
- the vertex terms of Lemma 6.3 become edge terms (Lemma 12.11);
- the Morgan–Scott stream-function construction of v_φ is replaced by an interpolate-and-correct construction (Lemma 12.10).

The first change turns out to be harmless, because the error analysis only needs a penalty-weighted bound (Lemma 12.6); this is also true in 2D. The necessity half of Proposition 8.1 is not proved in 3D.

12.1. Setting. Let $R = (-L, L)^3$ with $L > 1$, let D be the open unit ball, $\Omega = R \setminus \overline{D}$ and $\Gamma = \partial D$. Let P_h be a convex polyhedron with triangular faces whose vertices lie on Γ . Then $P_h \subset D$ and $\Omega \subset \Omega_h := R \setminus P_h$. $\mathcal{T}_h$ is a conforming tetrahedral mesh of Ω_h , and $\mathcal{F}_h^\Gamma$ is its set of faces on $\Gamma_h = \partial P_h$. For simplicity we assume that each $F \in \mathcal{F}_h^\Gamma$ is a whole face of P_h (this holds for an Alfeld refinement of such a mesh; for splits that subdivide boundary faces see Remark 12.12). Put

$$h := \max_T \text{diam } T, \quad h_\Gamma := \max_{F \in \mathcal{F}_h^\Gamma} \text{diam } F, \quad \rho := h / \min_{F \in \mathcal{F}_h^\Gamma} \text{diam } F, \quad \gamma := \mu h_\Gamma / h.$$

The spaces $V_h, V_h^R, V_h^0, \Pi_h, Z_h, W_h$, the forms a_h, N_h , and the methods GS and CNS* are defined exactly as in Section 2, with n the unit normal of Γ_h pointing into P_h . The extension $\tilde{u}$ is built as in two dimensions, with a vector potential in place of the stream function: since $\text{div } u = 0$ and $\int_\Gamma u \cdot n = 0$, on a spherical shell $\{1 < |x| < 1 + r\}$ we have $u = \text{curl } \psi$ for a smooth vector potential ψ ; we extend ψ smoothly across Γ to $\tilde{\psi}$ and set $\tilde{u} := \text{curl } \tilde{\psi}$ there, so that $\text{div } \tilde{u} = 0$ on a neighbourhood of $\overline{\Omega}$. The energy norm is

$$\|v\|^2 := \|\nabla v\|_{\Omega_h}^2 + \frac{\mu}{h} \|v\|_{\Gamma_h}^2 + \sum_{F \in \mathcal{F}_h^\Gamma} \text{diam } F \|\partial_n v\|_{L^2(F)}^2.$$

Assumption 12.1.

- (M0³) $\mathcal{T}_h$ is shape regular and $c_0 h_\Gamma \leq \text{diam } F \leq h_\Gamma$ for all $F \in \mathcal{F}_h^\Gamma$. Hence the faces of P_h are shape regular and $\rho/h \leq 1/(c_0 h_\Gamma)$.
- (IS3) $k \geq 3$, and there is $\beta > 0$, independent of h , such that for every $q \in \Pi_h \cap L_0^2(\Omega_h)$ there exists $v \in V_h^0$ with $\text{div } v = q$ and $\|\nabla v\| \leq \beta^{-1} \|q\|$.
- (D) is as in Assumption 2.1.

Status of (IS3). In 2D, (IS3) is Lemma 4.5 (Guzmán–Scott, $k \geq 4$, general meshes under (M1)–(M2)). In 3D the known results for the Scott–Vogelius pair $P_k - P_{k-1}^{\text{disc}}$ are these.

- *Alfeld (barycentric) refinements* of an arbitrary shape-regular mesh: stable for $k \geq 3$ [19], and for $k \geq d$ in dimension d [10]. Farrell, Mitchell and Scott [6] summarise this as “ $k \geq d$ for Alfeld splits”.
- *Worsey–Farin splits*: lower degrees [9], with a robustness caveat at nearly singular vertices, as tabulated in [6, Table 2.1].
- *Freudenthal meshes*: $k \geq 6$ [20], conjectured $k \geq 4$ in [6]. This is irrelevant here, because a Freudenthal mesh cannot be fitted to an inscribed polyhedron.
- *General tetrahedral meshes*: we know of no unconditional result for the full pair with P_{k-1}^{disc} pressures. Neilan [14] constructs smooth three-dimensional Stokes pairs, for $k \geq 6$, whose pressure spaces carry extra continuity, under a mesh condition; we do not claim that this settles (IS3) for the Scott–Vogelius pair.

So the unconditional 3D theory below is for Alfeld-refined meshes with $k \geq 3$. Uniformity of β in h has two parts. The domain part, a Bogovskii constant uniform over the perturbed domains Ω_h , is proved in Lemma 12.5. The discrete part, that the constant of the split-element proof depends only on shape regularity and the continuous inf-sup constant, is how these results are stated. We have not re-audited those proofs line by line for hidden dependence on the domain; see item 3 of Section 12.7.

12.2. Geometry of inscribed triangles.

Lemma 12.2 (Faces). *Let $F \in \mathcal{F}_h^\Gamma$ lie in the plane $\{y \cdot \nu_F = d_F\}$, where $\nu_F = -n_F$ is the unit normal pointing away from the centre. Let r_F be the circumradius of F and $o_F = d_F \nu_F$ its circumcentre; then $d_F = (1 - r_F^2)^{1/2}$ and $r_F \leq Ch_\Gamma$. For $x \in F$ let $x^* = x/|x|$ be its radial projection. Then:*

- (a) $\text{dist}(x, \Gamma) \leq 1 - d_F = r_F^2/2 + O(r_F^4) \leq Ch_\Gamma^2$.
- (b) $n(x^*) - n_F = -(x - o_F) + O(h_\Gamma^2)$, where $n(x^*) = -x^*$. The leading term is tangential to F , linear in x , and vanishes at the circumcentre.
- (c) The area Jacobian of $x \mapsto x^*$ from F to Γ is $d_F/|x|^3 = 1 + O(h_\Gamma^2)$.
- (d) $|F|^{-1} \int_F (n(x^*) - n_F) dA = -(x_c - o_F) + O(h_\Gamma^2)$, where x_c is the centroid. The leading term vanishes if and only if F is equilateral.

Proof. Write $x = o_F + y$ with $y \perp \nu_F$ and $|y| \leq r_F$. Then $|x|^2 = d_F^2 + |y|^2 \in [d_F^2, 1]$, which gives (a). For (b),

$$n(x^*) - n_F = -\frac{o_F + y}{|x|} + \nu_F = -\frac{y}{|x|} + \nu_F \left(1 - \frac{d_F}{|x|}\right),$$

with $|x|^{-1} = 1 + O(r_F^2)$ and $0 \leq 1 - d_F/|x| \leq 1 - d_F$. For (c), central projection onto the unit sphere has area element $(x^* \cdot \nu_F)|x|^{-2}$. (d) follows by integrating (b); the centroid and circumcentre of a triangle coincide exactly when it is equilateral. $\square$

Computation: `notes/v3/face_mean_check.py`. For three faces inscribed in the unit sphere at circumradius $r_F = 0.2, 0.1, 0.05$, the tangential part of the face mean of $n(x^*) - n_F$ is 0 (equilateral face) and $0.105 r_F, 0.122\sqrt{2} r_F$ (two non-equilateral faces), matching $|x_c - o_F|$.

For a general smooth strictly convex D the same holds, with o_F replaced by the point at which the gradient of the quadratic part of Γ , written as a local graph over F , equals the gradient of its linear interpolant. Only (a), (c) and $|n(x^*) - n_F| \leq Ch_\Gamma$ are used below.

In 2D the circumcentre of a chord is its midpoint, so the leading term is odd about the midpoint. That is the cancellation in Lemma 3.5. In 3D it is lost on every non-equilateral face.

Proposition 12.3 (Transposed-gradient term). *For $v \in V_h^R$:*

- (i) On F , $(\nabla \tilde{u})^\top n_F = \sigma n(x^*) + O(h_\Gamma^2)$ with $\sigma := \partial_n u(x^*) \cdot (n_F - n(x^*)) = \partial_n u(x^*) \cdot (x - o_F) + O(h_\Gamma^2)$. So the load is normal, of size $O(h_\Gamma)$, and has nonzero face means in general.
- (ii) $|\langle (\nabla \tilde{u})^\top n, v \rangle| \leq Ch_\Gamma \|v\|_{L^1(\Gamma_h)} \leq Ch_\Gamma (h/\mu)^{1/2} \|v\| = C\gamma^{-1/2} h_\Gamma^{3/2} \|v\|$.
- (iii) In the $\|\nabla v\|$ -dual norm on V_h^R the rate is generically only h_Γ . For smooth ψ on Γ , let v be the Lagrange interpolant of $\chi\psi n$, with a fixed cutoff χ . Then

$$\langle (\nabla \tilde{u})^\top n, v \rangle = \sum_{F \in \mathcal{F}_h^\Gamma} |F| \psi(x_F^*) \partial_n u(x_F^*) \cdot (x_c - o_F) + O(h_\Gamma^2),$$

while $\|\nabla v\| \leq C$. This is $\Theta(h_\Gamma)$ whenever $h_\Gamma^{-1} \sum_F |F| \psi \partial_n u \cdot (x_c - o_F) \not\rightarrow 0$. We do not exhibit a specific mesh family for which this holds; the point is only that the two-dimensional cancellation is not available.

Proof. Lemma 3.4 is dimension-free: $\nabla u = \partial_n u \otimes n$ on Γ and $\partial_n u \cdot n = 0$. So $(\nabla u)^\top n_F = n(\partial_n u \cdot n_F) = n(\partial_n u \cdot (n_F - n))$. Taylor-expand $\nabla \tilde{u}$ from x^* to x (distance $O(h_\Gamma^2)$) and apply Lemma 12.2(b); this gives (i). (ii) follows from $|\sigma| \leq Ch_\Gamma$, Cauchy–Schwarz on Γ_h ($|\Gamma_h| \leq C$) and $\|v\|_{\Gamma_h}^2 \leq (h/\mu)\|v\|^2$. For (iii), use $v \cdot n(x^*) = \psi + O(h_\Gamma)$ and Lemma 12.2(d). $\square$

12.3. Tools.

Lemma 12.4 (Dimension-independent tools). *Under $(M0^3)$ and (D) :*

- (a) Uniform Korn, trace and Friedrichs (Lemma 3.2) hold in $3D$.
- (b) Extension (Lemma 3.3): $\|\tilde{u}\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^2$ and $|(F, v)| \leq Ch_\Gamma^2 \|\nabla v\|$ for $v \in V_h^R$.
- (c) Inverse traces and coercivity (Lemmas 4.1 and 4.2) hold verbatim, with $|e|$ replaced by $\text{diam } F$ and with the same $\mu_0 = 4\kappa\rho C_I^2$, c_1 and M .
- (d) Divergence range (Lemma 4.3): under $(IS3)$, $\text{div } V_h^R = \Pi_h$ and $W_h \neq \emptyset$, and discrete velocities are pointwise divergence-free.

Proof. (a) Let $\rho_h(\omega) = d_F/(\omega \cdot \nu_F)$ be the radial function of Γ_h on the cone over F . Then $\|\rho_h - 1\|_\infty \leq Ch_\Gamma^2$, and ρ_h is Lipschitz with $|\nabla_\omega \rho_h| = d_F|\nu_F - (\omega \cdot \nu_F)\omega|/(\omega \cdot \nu_F)^2 \leq Ch_\Gamma$. The map $\Phi_h(r, \omega) = (r + \chi(r)(\rho_h(\omega) - 1), \omega)$ then satisfies $\|D\Phi_h - I\|_\infty \leq Ch_\Gamma$, and steps 3–6 of Lemma 3.2 apply unchanged.

- (b) The first bound is Lemma 12.2(a). For the second, apply a one-dimensional Poincaré inequality along normal segments across S_h , whose thickness is at most Ch_Γ^2 . This gives $\|v\|_{L^2(S_h)} \leq Ch_\Gamma(\|v\|_{\Gamma_h} + h_\Gamma\|\nabla v\|)$. Then $|(F, v)| \leq \|F\|_\infty |S_h|^{1/2} \|v\|_{L^2(S_h)}$ with $|S_h| \leq Ch_\Gamma^2$, and (a) finishes.
- (c) The proofs use only scaling and shape regularity.
- (d) $\text{div } V_h^0 = \Pi_h \cap L_0^2$ is $(IS3)$. The cubic face bubble $b_F \geq 0$ of the tetrahedron on $F \in \mathcal{F}_h^\Gamma$ gives $b_F n_F \in V_h^R$ (since $k \geq 3$) and $\int \text{div}(b_F n_F) = \int_F b_F \neq 0$. The rest is as in Lemma 4.3. $\square$

Lemma 12.5 (Uniform continuous inf-sup constant). *The Bogovskii (continuous inf-sup) constant of Ω_h is bounded independently of h .*

- Proof.* (1) Fix $\alpha < \pi/2$, $r_b > 1$, $r_B > 0$ with $(r_b - r_B)\cos(\alpha + \arcsin(r_B/r_b) + \alpha_0) > 1$, where $\alpha_0 > 0$ is small. Cover S^2 by finitely many caps of half-angle α about directions ω_j , such that adjacent caps overlap in a cap of fixed size. Let W_j be the corresponding cones, $c_j := r_b \omega_j$ and $B_j := B(c_j, r_B) \subset W_j \cap R$.
- (2) Let $x \in W_j \cap \Omega_h$. Since $0 \in P_h$ (for small h_Γ), the ray from 0 through x leaves P_h through a face F that meets W_j , and x lies on the outer side of the plane of F . Because $|\nu_F - \omega| \leq Ch_\Gamma$ for directions ω hitting F , the angle between ν_F and ω_j is at most $\alpha + Ch_\Gamma$.
 - (3) Hence every $y \in B_j$ satisfies $y \cdot \nu_F \geq (r_b - r_B)\cos(\alpha + \arcsin(r_B/r_b) + Ch_\Gamma) > 1 \geq d_F$. So y is also on the outer side of that plane. The segment $[y, x]$ therefore stays on the outer side of the plane, so it misses the convex set P_h , and it lies in the convex set $W_j \cap R$.
 - (4) Thus $W_j \cap \Omega_h$ is star-shaped with respect to B_j , uniformly in h . Galdi’s Lemma III.3.4 [7] gives a Bogovskii operator on $\Omega_h = \bigcup_j (W_j \cap \Omega_h)$ with a bound independent of h . The diameters, ball radii and overlaps are fixed. $\square$

Lemma 12.6 (Geometric consistency). *With $\mathcal{G}$ as in Lemma 5.1 (which is dimension-free), $|\mathcal{G}(v)| \leq C(1 + \gamma^{1/2})h_\Gamma^{3/2}\|v\|$ for $v \in V_h^R$, and $|\mathcal{G}(v)| \leq Ch_\Gamma^{3/2}\|\nabla v\|$ for $v \in V_h^0$.*

Proof. Take the four terms in turn.

- The transposed term is at most $C\gamma^{-1/2}h_\Gamma^{3/2}\|v\| \leq C\gamma_0^{-1/2}h_\Gamma^{3/2}\|v\|$ by Proposition 12.3(ii).
- $|\langle \tilde{u}, \partial_n v \rangle| \leq Ch_\Gamma^2 \cdot C_I(c_0 h_\Gamma)^{-1/2} \|\nabla v\|$.
- $\frac{\mu}{h} |\langle \tilde{u}, v \rangle| \leq (\mu/h)^{1/2} Ch_\Gamma^2 \|v\| = C\gamma^{1/2} h_\Gamma^{3/2} \|v\|$.
- $|(F, v)| \leq Ch_\Gamma^2 \|\nabla v\|$.

On V_h^0 the first and third terms vanish. $\square$

Remark 12.7 (The 2D cancellation is not needed for any error bound). The proof of Lemma 12.6 works verbatim in 2D. So Lemma 3.5 (the odd-in- s cancellation) is not needed for Lemma 5.2 or Theorem 7.2; it only gives the sharper $\|\nabla v\|$ -dual bound. Likewise, in step 4 of Theorem 6.4 the $O(h_\Gamma)$ tangential load $(p - \bar{p})(n_e - n(x^*))$ can be bounded in the energy dual norm by $C_p h_\Gamma (h/\mu)^{1/2}$. Since step 6 tests with r_h in $\|\cdot\|$, this suffices.

Lemma 12.8 (Approximation in W_h). *Under $(M0^3)$ and $(IS3)$, $E_h := \inf_{z \in W_h} \|\tilde{u} - z\| \leq C\varepsilon_h^{(3)}$, where*

$$\varepsilon_h^{(3)} := (1 + (\gamma h_\Gamma)^{1/2})h^k + h^{k+1}(h_\Gamma^{-1/2} + (\mu/h)^{1/2}).$$

For bounded γ the flux term is at most Ch^k if $h_\Gamma \geq h^2$. If g_I is flux-corrected, it reduces to the contribution of Γ_h , which is below h^k for bounded γ . The lower bound $\|\tilde{u} - z\| \geq (\mu/h)^{1/2}|m_R|/|\Gamma_h|^{1/2}$ after Lemma 4.6 holds verbatim.

Proof. As for Lemma 4.6, with three changes.

- *Net flux.* Estimate it crudely by pointwise interpolation error: $|m| \leq \int_{\partial R} |g_I - g| + \int_{\Gamma_h} |w_h - \tilde{u}| \leq Ch^{k+1}$. No extra quadrature exactness of Lagrange interpolation on triangles is used; unlike the 1D Newton–Cotes case, there is none in general (for $k = 2$ the induced rule gives $1/24$ for $\int x^3 = 1/20$ on the reference triangle).
- *Bubble.* $b := \sum_F b_F n_F$ with cubic face bubbles. Then $\int_{\Gamma_h} b \cdot n = c_b |\Gamma_h|$ with $c_b > 0$, $\|\nabla b\| + \|\operatorname{div} b\| \leq Ch_\Gamma^{-1/2}$ (there are $O(h_\Gamma^{-2})$ bubbles, each of H^1 -seminorm squared $O(h_\Gamma)$), $\|b\|_{\Gamma_h} \leq C$, and $\sum_F \operatorname{diam} F \|\partial_n b\|_F^2 \leq Ch_\Gamma^{-1}$.
- *Correction.* Use $(IS3)$ in place of Lemma 4.5. $\square$

12.4. The leak field without stream functions.

Lemma 12.9 (A divergence-free field with normal trace $p - \bar{p}$). *Let $q := (p - \bar{p})|_\Gamma$ and let ϕ_0 solve $\Delta_{S^2} \phi_0 = q$ on the unit sphere; this is solvable since $\int_\Gamma q = 0$. Let $\chi \in C^\infty$ with $\chi \equiv 1$ on $[\frac{1}{2}, 1 + r_1]$ and $\chi \equiv 0$ on $[1 + 2r_1, \infty)$, where $1 + 2r_1 < L$. In spherical coordinates set*

$$w := \chi(r) r^{-2} q(\omega) e_r - \chi'(r) r^{-1} \nabla_{S^2} \phi_0(\omega).$$

Then w is smooth on $\{r > \frac{1}{2}\} \supset \Omega_h$, $\operatorname{div} w = 0$, $w \equiv 0$ near ∂R , and $w|_\Gamma = q e_r = -(p - \bar{p})n$.

Proof. $\operatorname{div}(f(r)q e_r) = r^{-2}(r^2 f)'q$, which equals $r^{-2}\chi'q$ for $f = \chi r^{-2}$. A tangential field $g(r)r^{-1}\nabla_{S^2}\phi_0$ has divergence $g r^{-2}\Delta_{S^2}\phi_0 = g r^{-2}q$. With $g = -\chi'$ the two cancel. At $r = 1$, $\chi = 1$ and $\chi' = 0$. $\square$

Lemma 12.10 (Discrete leak field). *Let I_h be Lagrange interpolation and b as in Lemma 12.8. Set*

$$v_\varphi := I_h w - ab - v_0, \quad a := \frac{\int_{\Gamma_h} I_h w \cdot n}{c_b |\Gamma_h|},$$

where $v_0 \in V_h^0$ is given by $(IS3)$ with $\operatorname{div} v_0 = \operatorname{div}(I_h w - ab)$. Then $v_\varphi \in Z_h$ and

$$\begin{aligned} \|v_\varphi - w\|_{L^\infty(\Gamma_h)} &\leq Ch_\Gamma^{k+1}, & \|\nabla(v_\varphi - w)\| &\leq Ch^k, \\ \|\partial_n(v_\varphi - w)\|_{\Gamma_h} &\leq Ch^k h_\Gamma^{-1/2}, & \sum_F \operatorname{diam} F \|\partial_n(v_\varphi - w)\|_F^2 &\leq Ch^{2k}. \end{aligned}$$

These bounds replace $\|v_\varphi - w\|_{W^{1,\infty}} \leq Ch^k$ (Morgan–Scott) in every use in Sections 6 and 11. The same construction works in 2D and removes the dependence on the C^1 interpolant there.

- Proof.* (1) *The mean.* Since $\operatorname{div} w = 0$ on Ω_h and $w = 0$ near ∂R , $\int_{\Gamma_h} w \cdot n = 0$. So $|a| \leq C \|I_h w - w\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^{k+1}$, because boundary tetrahedra have diameter at most Ch_Γ . By the choice of a , $\operatorname{div}(I_h w - ab) \in \Pi_h \cap L_0^2$.
- (2) *The correction.* Its norm satisfies $\|\operatorname{div}(I_h w - ab)\| \leq Ch^k + C|a|h_\Gamma^{-1/2} \leq Ch^k$. So $\|\nabla v_0\| \leq Ch^k$. Since $v_0 = 0$ on Γ_h , the inverse trace inequality gives $\|\partial_n v_0\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2} h^k$ and $\sum_F \operatorname{diam} F \|\partial_n v_0\|_F^2 \leq Ch^{2k}$.
- (3) *The remaining terms.* $I_h w - w$ and ab contribute $O(h_\Gamma^{k+1})$ pointwise on Γ_h , $O(h^k)$ in H^1 , and $O(h_\Gamma^k)$ pointwise for ∂_n (for ab this is $|a|h_\Gamma^{-1}$).

The uses are:

- Theorem 6.1 lower bound, steps 1–2 (pointwise trace, and $\|\nabla v_\varphi\| + \sum \operatorname{diam} F \|\partial_n v_\varphi\|^2 \leq C$);
- Theorem 6.4, steps 4–5 ($\|\partial_n v_\varphi\|_{\Gamma_h} \leq C$, and $|\langle v_\varphi - w, \partial_n v \rangle| \leq Ch_\Gamma^{k+1} h_\Gamma^{-1/2} \|\nabla v\|$);
- Theorem 6.6(i), steps 1–5;
- Corollary 6.7, step 4;
- Theorem 11.8, step 5.

Each needs only the listed bounds. $\square$

12.5. Normal-load cancellation with edge terms.

Lemma 12.11 (3D analogue of Lemma 6.3). *Let $S \in W^{1,\infty}$ on a neighbourhood of Γ_h , with $|S - (S \cdot n_F)n_F| \leq Ch_\Gamma$ on every $F \in \mathcal{F}_h^\Gamma$. (For instance $S = w$ of Lemma 12.9, since $w(x) \parallel n(x^*)$ up to $O(h_\Gamma^2)$ and $|n(x^*) - n_F| \leq Ch_\Gamma$.) Then for $v \in Z_h$,*

$$|\langle S, \partial_n v \rangle| \leq C(\|v\|_{\Gamma_h} + h_\Gamma^{1/2} \|\nabla v\|) \leq C \|\nabla v\|,$$

with C depending on $\|S\|_{W^{1,\infty}}$, k and $(M\mathcal{O}^3)$.

Proof. (1) *Tangential part.* $|\langle S - (S \cdot n_F)n_F, \partial_n v \rangle| \leq Ch_\Gamma \|\partial_n v\|_{\Gamma_h} \leq Ch_\Gamma^{1/2} \|\nabla v\|$, by Lemma 12.4(c).

- (2) *Normal part.* On F , v is a polynomial on the adjacent tetrahedron with $\operatorname{div} v = 0$ and constant n_F . So $\partial_n v \cdot n_F = -\operatorname{div}_F v_F$, where v_F is the part of v tangential to F and div_F is the in-plane divergence. With $\sigma_F := S \cdot n_F$ and $\nu_{\partial F}$ the in-plane outward conormal,

$$\int_F \sigma_F \partial_n v \cdot n_F = \int_F \nabla_F \sigma_F \cdot v_F - \int_{\partial F} \sigma_F v \cdot \nu_{\partial F}.$$

The first term is at most $C\|v\|_{L^1(F)}$.

- (3) *Edge terms.* Each edge E of the closed triangulated surface Γ_h is shared by exactly two faces F, F' . Since v is continuous across E , the two contributions combine to $\int_E v \cdot (\sigma_F \nu_{E,F} + \sigma_{F'} \nu_{E,F'})$. With t_E a unit tangent oriented so that $\nu_{E,F} = t_E \times n_F$, we have $\nu_{E,F'} = -t_E \times n_{F'}$. Hence $\nu_{E,F} + \nu_{E,F'} = t_E \times (n_F - n_{F'}) = O(h_\Gamma)$ and $\sigma_F - \sigma_{F'} = S \cdot (n_F - n_{F'}) = O(h_\Gamma)$. So the coefficient is $O(h_\Gamma)$.
- (4) *Summing.* The inverse inequality $\|v\|_{L^1(\partial F)} \leq C(\operatorname{diam} F)^{-1} \|v\|_{L^1(F)}$ for $v|_F \in P_k$ (scaling and shape regularity) gives

$$\sum_E h_\Gamma \|v\|_{L^1(E)} \leq C \sum_F \|v\|_{L^1(F)} = C \|v\|_{L^1(\Gamma_h)} \leq C \|v\|_{\Gamma_h}.$$

- (5) Finish with the uniform trace inequality. $\square$

Remark 12.12. If boundary faces of $\mathcal{T}_h$ subdivide the faces of P_h (Worsey–Farin splits, or Alföld refinements of a mesh whose boundary faces already subdivide the faces of P_h), the extra edges lie inside a plane. There $n_F = n_{F'}$ and the conormals are opposite, so their contributions in step 3 cancel exactly. Lemma 12.2 is applied to the faces of P_h , which must then be shape regular of size $\asymp h_\Gamma$.

12.6. Main results in three dimensions.

Theorem 12.13. *Assume $(M0^3)$, $(IS3)$ and (D) , and let (u, p) be a smooth Stokes solution on $\bar{\Omega}$. Then Theorems 6.1, 6.4, 6.6 and 7.2, Corollaries 6.2 and 6.7, Proposition 7.1 and Proposition 8.1(a) hold verbatim in 3D, with ε_h replaced by $\varepsilon_h^{(3)}$ and v_φ by the field of Lemma 12.10. In particular:*

- (i) *GS converges at rate h/μ in H^1 and $(h/\mu)^{1/2}$ in energy. To leading order its error is $(h/\mu)v_\varphi$ with $v_\varphi|_{\Gamma_h} \approx -(p - \bar{p})n$, with universal constant 1 in the slip and energy norms.*
- (ii) *CNS* is well posed for $\gamma \geq \gamma_2$ and satisfies $\|\tilde{u} - u_h\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h^{(3)}]$. For bounded γ and $h_\Gamma \geq h^2$ this gives $\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h^k)$.*

On Alfeld-refined meshes this holds for every $k \geq 3$, subject to the uniformity caveat in item 3 of Section 12.7.

Proof. The 2D proofs use the following facts, each replaced as indicated.

- Lemma 3.1 is replaced by Lemma 12.2: sagitta $O(h_\Gamma^2)$, $|n(x^*) - n_F| \leq Ch_\Gamma$, Jacobian $1 + O(h_\Gamma^2)$. Hence $|\mathcal{R}(v)| \leq (G + Ch_\Gamma^2)(h/\mu)^{1/2}\|v\|$ and $\mathcal{R}(v_\varphi) = G^2 + O(h_\Gamma^2 + h^k)$. Note $v_\varphi \cdot n_F = -(p - \bar{p})(x^*)n \cdot n_F + O(h_\Gamma^{k+1})$ with $n \cdot n_F = 1 + O(h_\Gamma^2)$.
- Lemmas 3.2, 3.3, 4.1, 4.2 and 4.3 are replaced by Lemma 12.4; Lemmas 4.5 and 4.6 by (IS3), Lemma 12.5 and Lemma 12.8; Lemma 5.2 by Lemma 12.6; Lemma 6.3 by Lemma 12.11; and the Morgan–Scott v_φ by Lemma 12.10.
- Corollary 5.3, Proposition 7.1 and the consistency and uniqueness steps of Theorem 7.2 are dimension-free. They use $\int_{\partial\Omega_h} v \cdot n = \int \operatorname{div} v$ and a boundary bubble.
- *Step 4 of Theorem 6.4.* On F , $(\bar{p} - \bar{p})n_F + v_\varphi = (p - \bar{p})(x^*)(n_F - n(x^*)) + O(h_\Gamma^2) + O(h_\Gamma^{k+1})$. This is $O(h_\Gamma)$ with nonzero face means, so we use the energy-dual bound (Remark 12.7): $|m(v)| \leq C_p(h_\Gamma(h/\mu)^{1/2} + h_\Gamma^2 + h^k)\|v\|$. Step 6 then gives $\|r_h\| \leq C_p(h/\mu + h_\Gamma^2) + C\varepsilon_h^{(3)}$, using $h_\Gamma(h/\mu)^{1/2} \leq \frac{1}{2}(h_\Gamma^2 + h/\mu)$. The conclusion of Theorem 6.4, and the bound on $\|r_h\|$ in Corollary 6.7, are unchanged or better.
- *Step 2 of Theorem 6.6(i).* Here $|\mathcal{G}(v_\varphi)| \leq C(h_\Gamma + \gamma h_\Gamma^3 + \gamma h_\Gamma h^k)$, with h_Γ in place of $h_\Gamma^{3/2}$ by Proposition 12.3(i). The proof only needs this to be at most δG^2 , which follows from the hypothesis $h_\Gamma \leq h_0(G)$ and the X -condition. The penalty part $\tilde{u} \cdot v_\varphi = O(h_\Gamma^4 + h_\Gamma^2 h^k)$ is unchanged: $\tilde{u} \approx \pm d \partial_n u(x^*)$ is tangential and v_φ is normal to Γ up to $O(d)$.
- *Flux proviso.* The condition $h_\Gamma \geq h_\Omega^{2(\sigma_k - k)}$ of Theorem 7.2 becomes $h_\Gamma \geq h^2$, since the 3D flux exponent is $k + 1$. $\square$

Corollary 12.14 (Navier–Stokes in 3D). *Lemma 11.7, Theorems 11.8 and 11.12, Corollary 11.9 and Lemma 11.13 hold in 3D under the assumptions of Theorem 12.13 together with (NS1)–(NS2) (with $\tilde{u} = \operatorname{curl} \tilde{\psi}$ for a vector potential $\tilde{\psi}$, which exists because $\int_\Gamma u \cdot n = 0$), with $\varepsilon_h^{(3)}$ in place of ε_h .*

Proof. Section 11 reuses the Stokes estimates, in particular steps 4–5 of the proof of Theorem 6.4, together with Lemma 11.4, which holds for $d = 3$ because $H^1 \subset L^4$. Replace the Stokes lemmas by Theorem 12.13 and its ingredients. In step 5 of Theorem 11.8 use $\|v_\varphi\|_{H^1} \leq C$ from Lemma 12.10. In 3D the functional m of step 4 of Theorem 6.4 is bounded only in the energy-dual norm (proof of Theorem 12.13); this suffices, because (L) and (L*) are stated in the energy-dual norms $\|\cdot\|_{*,Z_h}$ and $\|\cdot\|_{*,V_h^R}$, so step 4 of Theorem 11.8 needs only an energy-dual bound on m . $\square$

12.7. What is not proved in 3D.

- (1) *Penalty necessity* (Proposition 8.1(b)). Scaling gives the same threshold form in 3D: under dilation by ℓ , a and $\langle \partial_n v, v \rangle$ scale like ℓ and $\|v\|_{\Gamma_h}^2$ like ℓ^2 , so $N_h(v, v) = \ell[A - 2B + (\mu\ell/h)C]$.

- But no 3D witness (a divergence-free $v \in Z_h$ supported in a boundary vertex or face star with $2B > A$) has been computed or certified. Sufficiency $\mu \geq 4\kappa C_I^2 \rho$ holds.
- (2) *(IS3) on general tetrahedral meshes.* It is unknown in 3D for unsplit meshes, as far as we know. The theory is unconditional only on split meshes.
 - (3) *Uniformity of the split-mesh inf-sup constant over Ω_h .* The domain part is Lemma 12.5. We rely on the published statements for the claim that the discrete part depends only on shape regularity; we have not re-audited Zhang’s or Guzmán–Neilan’s proofs for hidden domain dependence.
 - (4) *Sharpness of $h_\Gamma^{3/2}$ for CNS*.* Not proved in 3D. In 2D it is Theorem 7.12, unconditionally for $k \geq 7$ and under (M3) for $4 \leq k \leq 6$; that proof uses the 2D chord geometry and 2D boundary stars.
 - (5) *Non-convex D and other domains.* Not treated in 3D, because $\Omega \subset \Omega_h$ is used. In 2D, Section 10 treats general smooth, possibly nonconvex, multiply connected domains.

13. COMPUTATIONS

13.1. Code and meshes. Except in Section 13.8, we use P_4 Scott–Vogelius elements with Nitsche’s method on the inscribed polygon and strong Dirichlet data on the square $R = (-2.5, 2.5)^2$. Divergence is enforced exactly: all ten P_3 moments per triangle.

Saddle-point systems are solved by SuperLU [12] with partial pivoting and three steps of iterative refinement. Over the 148 production solves (74 cases, two methods each), the relative residual is at most $3.8 \cdot 10^{-15}$ and the discrete divergence at most $7 \cdot 10^{-11}$ (median $9 \cdot 10^{-12}$).

For the zero-pressure-gradient benchmark of [5], the code gives the coercivity threshold $\mu^* = 60.343$ at $N = 20$ (`results/pod/thresh/N20.L2.5_m5.json`), against the 60.338 reported there.

The vertices of Γ_h are the radial projections of N equispaced points on the square, so the boundary chords are *not* equal (except for $N = 8$): $\max |e| / \min |e|$ is 1.43, 1.72, 1.87 and 1.97 at $N = 16, 32, 64$ and 256. The ratio $h/h_\Gamma = h_\Omega / \max |e|$ stays in 3.28–3.48, while $\rho = h / \min |e|$ of Section 2 is 4.70, 5.73 and 6.35 at $N = 16, 32$ and 64. h_Ω decreases from 1.505 to 0.109 for $N = 16 \rightarrow 256$; since h/h_Γ is nearly constant, rates in h_Ω and h_Γ coincide. Penalty thresholds are reported as $\gamma^* = \mu^* h_\Gamma / h$, which is the γ of Section 2. Rates in the tables are computed between consecutive runs of the full sequence $N = 16, 24, 32, 48, 64, 96, 128, 192, 256$, some of which are omitted from the tables.

13.2. Test problems.

- A** Scott’s benchmark: the shear flow $u = (1 - r^{-2})(-y, x)$ with $p \equiv 0$, so $G = 0$.
- B_a** $\psi = (r^2 - 1)^2(x + y^2/2)/10$ and $p = a(x^2y - y + x/2)$, so $G = a(7\pi/8)^{1/2}$: $G = 1.658$ for $a = 1$ and 165.8 for $a = 100$.
- P** Pure pressure: $u \equiv 0$, $f = \nabla p$, with p as in B_1 . CNS* reproduces this exactly ($u_h = 0$ and $p_h = p - \bar{p}_\Gamma(p)$, since $p \in \Pi_h$); its computed velocity is below 10^{-13} in H^1 . The GS output is therefore *exactly* the response to the missing traction.
- C** ψ as in B, with the non-polynomial pressure $p = e^{x/2} \cos y$. This exercises the term $p^* - \pi_h p^*$ of Theorem 7.2.

Tests A and B were run with $\mu \in \{100, 1000, 10000\}$ and $N \leq 192$, plus $N = 256$ for $(A, 100)$ and $(B_1, 100)$, on a 32-core machine (36 minutes in total). Test P (all three μ) and test C ($\mu = 100$) were run with $N \leq 96$ on a 2-core machine.

13.3. Theorem 7.2: CNS*. Table 4 gives the CNS* errors. At $N = 256$ the energy-norm rate is 1.52 and the slip rate is 2.02; the slip carries an extra factor $(h/\mu)^{1/2}$. The H^1 rates decrease monotonically toward 3/2, the geometric term of Lemma 3.5. The h_Ω^4 term is invisible at these levels because the geometric term dominates.

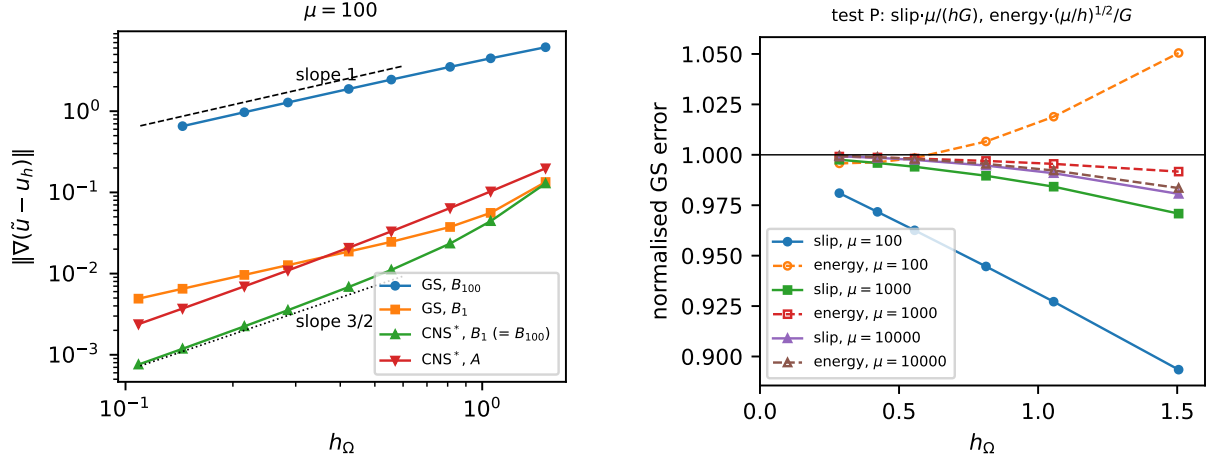


FIGURE 1. Left: H^1 errors at $\mu = 100$. Once $G > 0$, GS eventually has slope 1, while CNS* tends to slope 3/2 independently of the pressure. Right: the normalised slip and energy errors of GS on test P approach 1 (Corollary 6.7); at $N = 96$ the energy ratios are within 0.5% of 1 and the slip ratios within 2%.

TABLE 4. CNS* attains Scott's rate $h_\Gamma^{3/2}$ (Theorem 7.2). H^1 error, $\mu = 100$.

N	h_Ω	A	rate	B_1	rate
32	0.813	$6.41 \cdot 10^{-2}$	1.80	$2.34 \cdot 10^{-2}$	2.43
64	0.422	$2.08 \cdot 10^{-2}$	1.69	$6.83 \cdot 10^{-3}$	1.76
96	0.285	$1.09 \cdot 10^{-2}$	1.64	$3.54 \cdot 10^{-3}$	1.68
128	0.215	$6.96 \cdot 10^{-3}$	1.60	$2.24 \cdot 10^{-3}$	1.63
192	0.144	$3.71 \cdot 10^{-3}$	1.58	$1.19 \cdot 10^{-3}$	1.60
256	0.109	$2.39 \cdot 10^{-3}$	1.56	$7.58 \cdot 10^{-4}$	1.57

The error is independent of the pressure.

- For polynomial pressures this is structural ($p \in \Pi_h$). B_1 and B_{100} agree to all printed digits wherever both were run ($N \leq 192$, every μ).
- With the non-polynomial pressure of test C, the CNS* H^1 error differs from that of B_1 by a relative $7.9 \cdot 10^{-7}$, $3.9 \cdot 10^{-8}$ and $9.7 \cdot 10^{-9}$ at $N = 16, 64$ and 96 . This is the size of the contribution $\langle p^* - \pi_h p^*, v \cdot n \rangle$, which is of higher order, as Theorem 7.2 predicts.

13.4. Theorems 6.1–6.4: the printed method. Table 5 compares the H^1 rates of GS and CNS* on B_{100} .

For the non-polynomial test C at $\mu = 100$, the GS H^1 rate from $N = 64$ to 96 is already 1.12, against 1.68 for CNS*. For the small pressure B_1 at $\mu = 1000$, GS still shows 1.51 at $N = 192$. The $(h/\mu)G$ term dominates only once $(h/\mu)G \gtrsim h_\Gamma^{3/2}$, which always happens eventually because $h_\Gamma^{3/2}$ decays faster. Larger μ or smaller G delays the crossover but does not remove it.

13.5. Corollary 6.7: the leading term. Table 6 and Figure 1 (right) show the normalised GS error on test P, and Figure 2 shows the leak field itself.

TABLE 5. GS eventually loses Scott's rate once $G > 0$. H^1 rate between $N = 128$ and $N = 192$, test B₁₀₀.

μ	GS	CNS*
100	0.99	1.60
1000	1.00	1.56
10000	1.06	1.58

TABLE 6. The leading term of Corollary 6.7. Normalised GS error on test P ($G = 1.658$), where $e = \tilde{u} - u_h$.

N	μ	$\ \nabla e\ \mu/(hG)$	$\ e\ _{\Gamma_h} \mu/(hG)$	$\ e\ (\mu/h)^{1/2}/G$
16	100	2.464	0.893	1.050
32	100	2.612	0.945	1.007
64	100	2.684	0.972	0.996
96	100	2.708	0.981	0.996
96	1000	2.770	0.998	0.999
96	10000	2.790	0.9993	0.9994

- At $N = 32$, $\mu = 100$, the normal trace of u_h on Γ_h matches the prediction $(h/\mu)(p - \bar{p})$ with relative $L^2(\Gamma_h)$ misfit 0.055. The maximum of the tangential trace is about 12% of that of the normal one, and the tangential trace oscillates at the element scale.
- The slip and energy ratios approach 1: at $N = 96$ the energy ratios are within 0.5% of 1 for every μ . The defect in the slip ratio shrinks with h/μ : at $N = 96$ it is 0.019, 0.0025 and 0.0007 for $\mu = 10^2$, 10^3 and 10^4 .
- The H^1 constant is not covered by the corollary. For $\mu = 100$ it rises from 2.46 to 2.71; for $\mu = 10^3$ and 10^4 it falls, from 2.78 to 2.77 and from 3.06 to 2.79. Section 6.1 identifies the limit as $K = \|\nabla U\|/G = 2.7565$ for this box, proves convergence to it as $h \rightarrow 0$ at fixed μ (Corollary 6.11; on these meshes h/h_Γ is nearly constant, so fixed μ means fixed γ and bounded ρ), and compares it pointwise with these runs. The $\mu = 100$ values approach it from below; the larger- μ values approach it from above, in a regime ($\gamma h_\Gamma \gg 1$) that the proof does not cover (Remark 6.12).
- On test B, GS itself (B₁, $\mu = 100$, $N = 256$) gives slip and energy ratios 0.995 and 1.001.
- For $\mu = 10^4$ with the small pressure B₁, $\gamma h_\Gamma \approx 120 \gg G$ at $N = 192$, so the hypothesis of the corollary fails. The GS energy ratio there is about 7.8, dominated by the geometric error rather than the traction response.
- The difference $u_{\text{GS}} - u_{\text{CNS}^*}$ is not a clean proxy for the traction response on test B. It also contains the difference between the two methods on the velocity part, which is $O(h_\Gamma^{3/2})$ and non-zero even for test A. Test P removes this.

13.6. Proposition 8.1: the penalty study. The coercivity threshold μ^* on Z_h is the smallest μ for which the unpivoted symmetric factorisation of $A_\mu + rB^\top M^{-1}B$ has only positive pivots; we locate it by bisection. Table 7 and Figure 3 show the results.

Under refinement at fixed h/h_Γ , γ^* increases by 3.4 ($N = 8 \rightarrow 16$), 1.8 ($16 \rightarrow 32$) and 0.95 ($32 \rightarrow 64$). The increments roughly halve. The patch computations of Section 8.1 (limit geometry, windows of up to 32 boundary edges) predict a limit $\gamma_\infty^* \approx 21.4$; this is an extrapolation, not a proof.

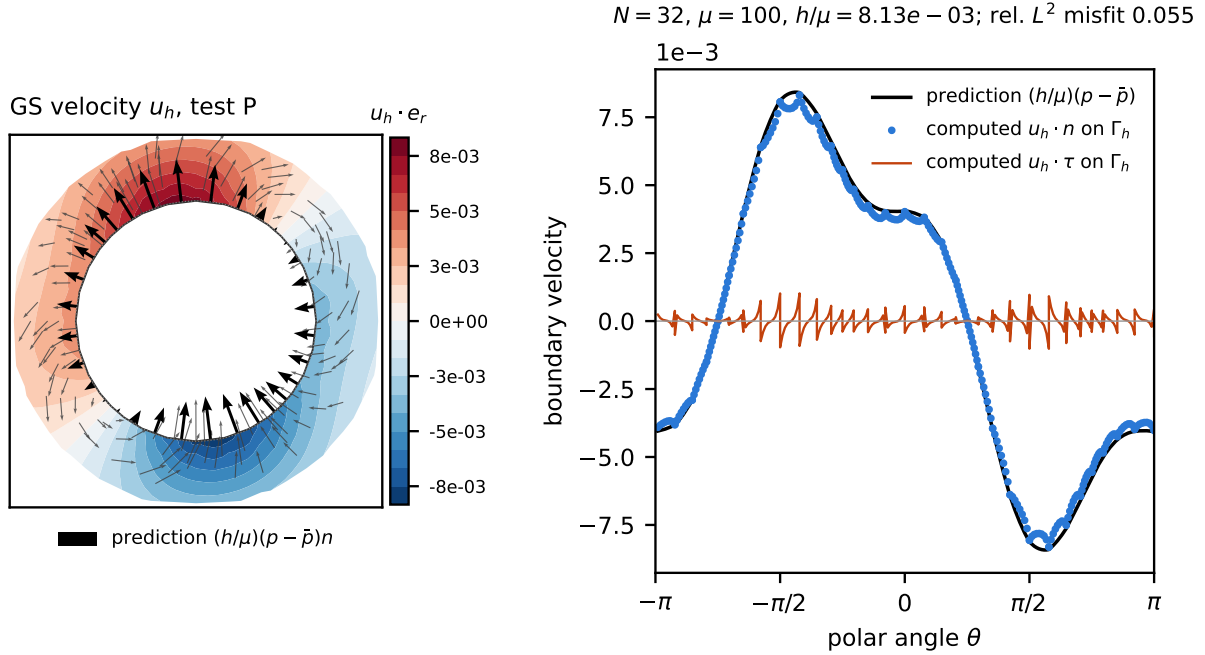


FIGURE 2. The leak of the printed method on test P ($u = 0$, so $u_h = -(\tilde{u} - u_h)$), $N = 32$, $\mu = 100$. Left: radial velocity $u_h \cdot e_r$ near the wall (colour) and u_h (thin arrows); thick arrows show the prediction $(h/\mu)(p - \bar{p})n$ of Corollary 6.7 at the chord midpoints. Right: normal and tangential traces of u_h along Γ_h against $(h/\mu)(p - \bar{p})$.

In part (b) of Table 7, h/h_Γ is varied by coarsening the far field while the vertices of Γ_h are held fixed and the boundary layer approximately fixed. Then μ^* is linear in h/h_Γ over more than a decade, with a constant slope of about 18–20. The vertices of Γ_h do not depend on L , so $\max|e|/\min|e|$ is fixed at fixed N , and μ^* is equally linear in ρ , with the slope divided by that ratio. The threshold is thus set locally at Γ_h ($\mu/h \gtrsim c/h_\Gamma$), and the scaling with h/h_Γ (hence with ρ) is the cost of normalising the penalty by the global h .

In practical terms:

- for $k = 4$ a safe choice is $\gamma = \mu h_\Gamma/h \geq 25$, that is, $\mu \geq 25 h/h_\Gamma$; since $\rho \geq h/h_\Gamma$, $\mu \geq 25\rho$ is also safe;
- a fixed μ fails once the far field is coarsened beyond $h/h_\Gamma \approx \mu/20$;
- the threshold grows quickly with k , and is larger on Clough–Tocher refinements (Section 13.8), so these values do not transfer to other elements.

The exact certificate $\lambda \geq 9$ on $\mathcal{S}_3$ and $\mathcal{S}'_4$ is a local lower bound, consistent with the global $\gamma^* \approx 18$ –21; Section 8.1 accounts for the gap.

13.7. Lemma tests. We compute exact discrete dual norms, $\sup_v \ell(v)/\|\nabla v\|$ over Z_h or V_h^R , by a saddle-point solve with the pointwise divergence constraint. Table 8 gives the rates for $N = 48 \rightarrow 64$.

Computation: code/lemma_tests.py 16,24,32,48,64; output logs/lemma_tests.log.

13.8. Higher degree and Clough–Tocher refinements. We repeated the study with $k = 5, 6$ on the same meshes, and with $k = 2, 3$ on their Clough–Tocher refinements (Section 9), using a degree-general version of the solver (code/svn_k.py). The refinement leaves Γ_h , the square and

TABLE 7. Coercivity threshold μ^* and $\gamma^* = \mu^* h_\Gamma / h$ ($k = 4$). The row h/h_Γ is $h_\Omega / \max |e|$; the $\rho = h / \min |e|$ of Section 2 is larger by the factor $\max |e| / \min |e|$ (for example $\rho = 4.70, 5.73, 6.35$ at $N = 16, 32, 64$ in part (a)).

<i>(a) Refinement at fixed $h/h_\Gamma \approx 3.3$.</i>								
N	8	12	16	20	24	32	48	64
h/h_Γ	3.41	3.02	3.28	3.17	3.30	3.33	3.37	3.40
μ^*	50.3	51.8	59.3	60.3	63.3	66.0	69.1	70.7
γ^*	14.7	17.1	18.1	19.1	19.2	19.9	20.5	20.8

<i>(b) Varying h/h_Γ through the outer box $(-L, L)^2$.</i>						
	L	2.5	5	10	20	40
$N = 16$	h/h_Γ	3.28	5.90	11.3	22.2	43.9
	μ^*	59.3	112.1	208.0	411.5	809.3
	γ^*	18.1	19.0	18.4	18.5	18.4
$N = 32$	h/h_Γ	3.33	5.79	10.8	21.1	41.5
	μ^*	66.0	113.9	218.0	422.2	835.7
	γ^*	19.9	19.7	20.1	20.0	20.1

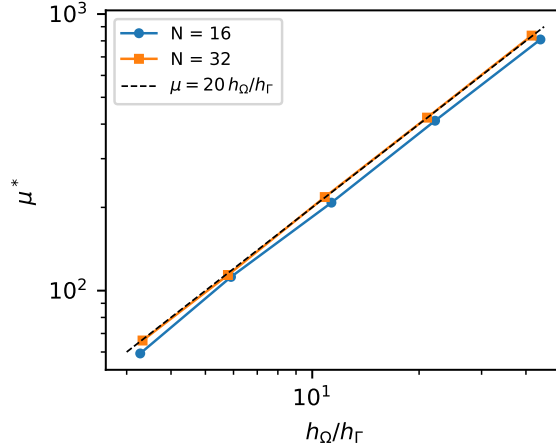


FIGURE 3. Coercivity threshold μ^* against h/h_Γ (Table 7(b)).

TABLE 8. Exact discrete dual norms; rate in h_Γ for $N = 48 \rightarrow 64$ (positive means decay).

Functional	Space	Rate
$\langle S, \partial_n v \rangle, S$ normal	Z_h	+0.05 (bounded, ≈ 1.75)
$\langle S, \partial_n v \rangle, S$ normal	V_h^R	-0.52
$\langle S, \partial_n v \rangle, S$ tangential	Z_h	-0.52
$\langle (\nabla \tilde{u})^\top n, v \rangle$, tests A / B	V_h^R	1.57 / 1.56 (Lemma 3.5: 3/2)

$h = h_\Omega$ unchanged, so the h values are those of the standard meshes. At $k = 4$ with SuperLU the new code is bit-identical to `svn.py`, and agrees with the stored production records to $2.6 \cdot 10^{-10}$ relative. SuperLU runs out of memory at the target sizes, so these runs use MKL PARDISO. The

TABLE 9. Coercivity threshold $\gamma^* = \mu^* h_\Gamma / h$ for higher degrees and on Clough–Tocher (CT) refinements. The $k = 4$ row is Table 7(a).

	$N = 8$	16	32	64	128
$k = 4$	14.7	18.1	19.9	20.8	–
$k = 5$	25.7	30.3	32.1	33.0	–
$k = 6$	–	45.2	47.4	48.3	–
CT, $k = 2$	5.7	9.1	10.5	11.2	11.5
CT, $k = 3$	–	31.6	33.9	34.6	–
CT, $k = 4$	–	61.9	65.5	–	–

TABLE 10. Normalised GS error on test P at the finest level run, $e = \tilde{u} - u_h$.

Element	μ	N	$\ \nabla e\ \mu / (hG)$	$\ e\ _{\Gamma_h} \mu / (hG)$	$\ e\ (\mu/h)^{1/2} / G$
$k = 4$	1000	96	2.770	0.998	0.999
$k = 5$	1000	128	2.768	0.9983	0.9994
$k = 6$	1000	96	2.769	0.9976	0.9993
CT, $k = 2$	100	256	2.737	0.9928	0.9974
CT, $k = 2$	1000	128	2.751	0.9982	0.9992
CT, $k = 3$	1000	128	2.763	0.9983	0.9996

saddle-point matrix is ill-conditioned, and for robustness of PARDISO we regularise the pressure block by 10^{-12} times the pressure mass matrix. The result agrees with SuperLU to $\leq 10^{-9}$ relative ($k = 4$ and $k = 6$, $N \leq 32$). Over the 260 solves (390 records, including 130 derived records for D) the relative residual of the unregularised system is at most $2.2 \cdot 10^{-12}$ and the discrete divergence at most 10^{-9} . One side effect: CNS* on test P returns $\|\nabla u_h\| \approx 4 \cdot 10^{-11}$, the regularisation floor, instead of round-off.

Thresholds. Table 9 gives γ^* . It grows quickly with k and is larger on the Clough–Tocher refinements, whose boundary triangles are three times thinner while h is the macro size. In every case the measured value exceeds the corresponding certified local constant (Proposition 8.1 for $k \geq 4$, Proposition 9.10 on the refinements), which is consistent with those constants being local lower bounds. γ^* still drifts upward with N (+7% to +24% from $N = 16$ to 64). In particular $\mu = 100$ is *below* the threshold for $k = 5$ ($N \geq 32$), for $k = 6$ and for Clough–Tocher $k = 3$ (all N). The $\mu = 100$ runs in those cases are erratic, as they should be (for instance the CNS* H^1 error for $k = 5$ on B_1 jumps between 0.05 and 3.7), and they are not covered by the theory. We therefore report $\mu = 300$ and $\mu = 1000$ for them.

The leak (test P). Table 10 shows that the slip and energy constants of Corollary 6.7 tend to 1 for every element, with rates 1 (slip), $1/2$ (energy) and 1 (H^1). The H^1 constants are close to the $K = 2.7565$ of Section 6.1, which does not depend on the element; this agreement is observed, not proved, for these elements. On test B_1 the leak part $D := u_{GS} - u_{CNS^*}$ has the same normalised slip and energy ratios (0.986–1.003 and 0.991–1.006 at the finest levels), H^1 rate 1.01–1.15 and energy rate 0.49–0.52. Its normalised H^1 constant is still drifting down at the finest levels (3.03 for $k = 5$, $\mu = 1000$).

Rates. Table 11 gives H^1 rates at the finest pair of levels.

- CNS* tends to rate $3/2$ for $k = 5, 6$ on B_1 (1.58–1.61, energy rate 1.53–1.54), non-increasing from $N = 32$ on. The CNS* errors for $k = 4, 5, 6$ are almost equal (for example $1.05 \cdot 10^{-2}$,

TABLE 11. H^1 rates at the finest pair of levels (energy-norm rate of CNS* in parentheses). $D = u_{\text{GS}} - u_{\text{CNS}^*}$ is the leak part; it is not reported for test A, where $G = 0$.

Element	Test	μ	N	GS	CNS*	D
$k = 5$	B ₁	300	96 $\rightarrow$ 128	1.20	1.59 (1.53)	1.03
$k = 5$	B ₁	1000	96 $\rightarrow$ 128	1.53	1.58 (1.54)	1.13
$k = 6$	B ₁	300	64 $\rightarrow$ 96	1.27	1.61 (1.54)	1.04
$k = 6$	B ₁	1000	64 $\rightarrow$ 96	1.57	1.61 (1.54)	1.15
CT, $k = 3$	A	1000	96 $\rightarrow$ 128	1.56	1.56 (1.55)	–
CT, $k = 2$	A	100	192 $\rightarrow$ 256	1.78	1.78 (1.55)	–
CT, $k = 2$	B ₁	100	192 $\rightarrow$ 256	1.91	1.91 (1.91)	1.01
CT, $k = 3$	B ₁	1000	96 $\rightarrow$ 128	2.83	2.83 (1.64)	1.05

$9.02 \cdot 10^{-3}$, $8.47 \cdot 10^{-3}$ at $N = 64$, $\mu = 1000$): the geometric term dominates. On Clough–Tocher $k = 3$, test A, the H^1 rate is 1.56 at $N = 128$.

- GS *tends to rate 1 when $G > 0$* . For $k = 5$, $\mu = 300$, the GS H^1 rate on B₁ decreases monotonically, 1.63, 1.58, 1.48, 1.38, 1.28, 1.20 for $N = 24, \dots, 128$, while its leak part D already has rate 1.03. At $\mu = 1000$ the crossover comes later (1.53 at $N = 128$), exactly as for $k = 4$.
- CNS* *is pressure-independent* for $k = 5, 6$: on test P it returns $u_h = 0$ to the regularisation floor at every N and every $\mu \geq \mu^*$. By linearity this is the statement that $u_h(B_a) = u_h(B_1)$ for every a .

Caveats for Clough–Tocher $k = 2, 3$.

- (1) CNS* *on test P is not exactly zero*. The pressure $p = x^2y - y + x/2$ is cubic, so $p \notin \Pi_h$ for $k \leq 3$, and the term $\eta = p^* - \pi_h p^*$ of Theorem 7.2 is present. The spurious velocity is small and converges at a rate close to $k + \frac{1}{2}$: $\|\nabla u_h\| = 1.4 \cdot 10^{-6}$ at $N = 256$ for $k = 2$ (rate 2.56), and $4.7 \cdot 10^{-9}$ at $N = 128$ for $k = 3$ (rate 3.53). So for $k \leq 3$ the pressure-independence of CNS* holds only up to a term of order $h^{k+1/2}$ that scales with the amplitude of p . It is below $h_\Gamma^{3/2}$ asymptotically, but for B₁₀₀ it would be 100 times larger.
- (2) *On test B₁ the rate 3/2 is not yet visible*. The H^1 error is dominated by interpolation of the steep velocity (u grows like r^5 , since ψ grows like r^6 , on the square of side 5): for $k = 2$ it is 0.41 at $N = 256$, against $7.6 \cdot 10^{-4}$ for $k = 4$, with rate 1.91 still rising towards 2. For $k = 3$ the rate is 2.83, close to 3. The runs are pre-asymptotic and the crossover to $h_\Gamma^{3/2}$ is out of reach. The leak part D is on theory (rate 1.01, ratios 1.0009 and 1.0056 at $N = 256$ for $k = 2$).
- (3) On test A (bounded velocity) the energy-norm rate is already 1.55 for $k = 2$, while the H^1 rate (1.78 at $N = 256$) is still decreasing.

Memory limits (a shared 5.8 GB budget) prevented $k = 6$ at $N = 128$ and Clough–Tocher $k = 3$ at $N = 192$.

APPENDIX A. REPRODUCIBILITY

All code, raw outputs and logs are in the repository that accompanies this paper. Paths below are relative to its root; commands are run from `code/`. `code/regen_logs.sh` regenerates every log in `logs/` with the commands below, and `make report` checks that all three reports are reproduced byte-for-byte.

Code (`code/`).

- `svn.py`: the solver. It contains the mesh generator, the P_4 space, assembly of N_h , B and the CNS* coupling, the SuperLU saddle-point solver (`solve_lu`), error norms, and the exact solutions for tests A, B, P and C.
- `job.py`: one study or threshold job per process.
- `run_all.py`: memory- and CPU-aware scheduler for the full campaign.
- `report.py` and `report_local.py`: produce the tables.
- `plots.py`: produces the figures.
- `penalty.py`: computes the coercivity threshold μ^* .
- `lemma_tests.py`: computes the dual norms of Table 8.
- `lamstar_exact.py` and `lamstar_exact_v2.py`: the exact certificates of Proposition 8.1.
- `lamstar.py`: the floating-point λ^* on other fans.

Added in this version:

- `svn_k.py`: degree-general solver (P_k velocities, $k \geq 2$; standard or Clough–Tocher mesh; SuperLU or PARDISO). `job_k.py` runs one study or threshold job, `penalty_k.py` computes μ^* , `report_v3.py` writes the tables, and `validate_k.py` checks $k = 4$ against `svn.py` and PARDISO against SuperLU. The scheduler is `results/v3/sched.py`, with job list `results/v3/jobs.txt`.
- `lamstar_exact_ct.py`: the exact local-surjectivity ranks and penalty certificates on Clough–Tocher stars (Section 9).
- `lamstar_family.py`: the single-triangle forms, the shape-family certificates and the patch thresholds (Section 8.1).
- `h1_limit.py`: the annulus bracket, the series solution and the finite element comparison for the H^1 constant (Section 6.1).
- `lower_bound_tests.py`: the (M3) star constants, the assembled test field and the dual norms (Section 7.1).
- `plot_leak.py`: Figure 2.
- `notes/v3/face_mean_check.py`: the face-mean check of Lemma 12.2(d).

Where each number comes from.

Result	Command	Output
Tables 4, 5, 7	SVN_OUT=./results/pod python3 run_all.py, then python3 report.py ./results/pod	results/pod/report.txt
Table 6, test C	SVN_OUT=./results/local python3 job.py study N P1 MU and SVN_OUT=./results/local python3 job.py study N C 100, then python3 report_local.py	results/local/report_local.txt
Certificate for S_3	python3 lamstar_exact.py	logs/lamstar_exact_ S3_S4singular.log
Certificate for S'_4 Table 8	python3 lamstar_exact_v2.py python3 lemma_tests.py 16,24,32,48,64	logs/lamstar_exact_S4prime.log logs/lemma_tests.log
Floating-point λ^*	python3 -W ignore::DeprecationWarning lamstar.py	logs/lamstar_float.log
Figures 1, 3 Figure 2 Tables 9–11	python3 plots.py python3 plot_leak.py SVN_SOLVER=pardiso python3 ../results/v3/sched.py ../results/v3/jobs.txt LOG (runs job_k.py; output root results/v3 by default, or SVN_OUT), then python3 report_v3.py	paper/figures/h1_rates.pdf etc. paper/figures/leak_field.pdf results/v3/report_v3.txt
Validation of <code>svn.k.py</code> Lemma 9.3 ($k = 3$), Table 3	python3 validate_k.py python3 lamstar_exact_ct.py	results/v3/validation.txt logs/lamstar_exact_ct.log
Theorem 8.2, Proposition 8.3 Patch thresholds (Section 8.1)	python3 lamstar_family.py bubble; ... certify python3 lamstar_family.py mesh; ... limit; ... limitstar	logs/lamstar_family_bubble.log, ..._certify.log logs/lamstar_family_mesh.log, ..._limit.log, ..._limitstar.log
Level-set and tall-triangle values of λ_T (Section 8.1)	python3 lamstar_family.py levels	logs/lamstar_family_levels.log
H^1 constant (Section 6.1) (M3) and Tables 1, 2	python3 h1_limit.py annulus and ... mps (series log); ... fem python3 lower_bound_tests.py with stars 8,12,...,1024, field 16,...,512, global 16,...,64	logs/h1_limit_series.log, ..._fem.log logs/lower_bound_stars.log, ..._field.log, ..._global.log
Lemma 12.2(d)	python3 notes/v3/ face_mean_check.py (from the root)	standard output

Resources. The full A/B campaign (74 study jobs up to $N = 256$ and 16 threshold jobs) took 36 minutes on 32 cores with 125 GB of memory. The largest single job ($N = 256$, both methods) takes about 23 minutes; the scheduler reserved up to 55 GB for it. Every job with $N \leq 96$ runs on a 2-core, 7 GB machine.

The runs of Section 13.8 used a 2-core machine with a 5.8 GB memory limit shared with other jobs; the largest single job used 3.6 GB (peak RSS is recorded in each record). Each job took at most about 2.5 minutes.

The environment was Python 3.11 with NumPy 2.4, SciPy 1.17 (SuperLU) and SymPy 1.14. The runs of Section 13.8 also used MKL PARDISO through `pypardiso`.

USE OF GENERATIVE-AI TOOLS DECLARATION

The author used Anthropic’s Claude Opus 5.5 as a research-support tool. It assisted with the following:

- developing and checking the proofs;
- implementing, running and validating the finite element code and the exact-arithmetic certificates;
- independently reviewing the arguments as a referee;
- preparing the manuscript and the accompanying repository.

The author directed the project and retains authorship of the work. Responsibility for all mathematical statements, computations, references and conclusions rests with the author.

REFERENCES

1. J. H. Argyris, I. Fried, and D. W. Scharpf, *The TUBA family of plate elements for the matrix displacement method*, Aeronaut. J. R. Aeronaut. Soc. **72** (1968), 701–709.
2. Douglas N. Arnold and Jinshui Qin, *Quadratic velocity/linear pressure Stokes elements*, Advances in Computer Methods for Partial Differential Equations VII (R. Vichnevetsky, D. Knight, and G. Richter, eds.), IMACS, New Brunswick, NJ, 1992, TODO verify pages and editors, pp. 28–34.
3. Philippe G. Ciarlet, *The finite element method for elliptic problems*, North-Holland, Amsterdam, 1978, TODO verify that the HCT interpolation estimate cited is in Section 6.1.
4. R. W. Clough and J. L. Tocher, *Finite element stiffness matrices for analysis of plate bending*, Proceedings of the Conference on Matrix Methods in Structural Mechanics (Wright-Patterson Air Force Base, Ohio), 1965, TODO verify pages, pp. 515–545.
5. Claudemir de Souza Cavalcante, *Scott–Vogelius–Nitsche convergence for a zero-pressure-gradient problem on a polygonally approximated boundary*, Preprint, Zenodo, 2026.
6. Patrick E. Farrell, Lawrence Mitchell, and L. Ridgway Scott, *Two conjectures on the Stokes complex in three dimensions on Freudenthal meshes*, SIAM J. Sci. Comput. (2024), arXiv:2211.05494. TODO verify year, volume and pages.
7. Giovanni P. Galdi, *An introduction to the mathematical theory of the Navier–Stokes equations. steady-state problems*, 2 ed., Springer, New York, 2011.
8. Ingeborg G. Gjerde and L. Ridgway Scott, *Verification and validation of cylinder drag: Pressure and stress approximations on curved boundaries*, Adv. Comput. Sci. Eng. **2** (2024), no. 3, 295–319, arXiv:2306.12362.
9. Johnny Guzmán, Anna Lischke, and Michael Neilan, *Exact sequences on Worsey–Farin splits*, Math. Comp. **91** (2022), 2571–2608, TODO verify volume and pages.
10. Johnny Guzmán and Michael Neilan, *Inf-sup stable finite elements on barycentric refinements producing divergence-free approximations in arbitrary dimensions*, SIAM J. Numer. Anal. **56** (2018), no. 5, 2826–2844.
11. Johnny Guzmán and L. Ridgway Scott, *The Scott–Vogelius finite elements revisited*, Math. Comp. **88** (2019), 515–529.
12. Xiaoye S. Li, *An overview of SuperLU: Algorithms, implementation, and user interface*, ACM Trans. Math. Software **31** (2005), no. 3, 302–325.
13. John Morgan and Ridgway Scott, *A nodal basis for C^1 piecewise polynomials of degree $n \geq 5$* , Math. Comp. **29** (1975), 736–740.
14. Michael Neilan, *Discrete and conforming smooth de Rham complexes in three dimensions*, Math. Comp. **84** (2015), no. 295, 2059–2081, TODO verify issue number and pages. Its three-dimensional Stokes pairs use pressure spaces with extra continuity; verify what, if anything, is proved there for the full P_{k-1}^{disc} Scott–Vogelius pair.
15. Joachim Nitsche, *Über ein Variationsprinzip zur Lösung von Dirichlet-Problemen bei Verwendung von Teilräumen, die keinen Randbedingungen unterworfen sind*, Abh. Math. Sem. Univ. Hamburg **36** (1971), 9–15.
16. L. Ridgway Scott, *Relaxing gradient constraints for exactly divergence-free spaces (zero gradient prize)*, Ridgway Scott Foundation, <https://people.cs.uchicago.edu/~ridg/prizes/zerogradprize.pdf>, Prize problem PPL 115.

17. L. Ridgway Scott and Michael Vogelius, *Norm estimates for a maximal right inverse of the divergence operator in spaces of piecewise polynomials*, RAIRO Modél. Math. Anal. Numér. **19** (1985), 111–143.
18. L. Ridgway Scott and Shangyou Zhang, *Finite element interpolation of nonsmooth functions satisfying boundary conditions*, Math. Comp. **54** (1990), no. 190, 483–493.
19. Shangyou Zhang, *A new family of stable mixed finite elements for the 3D Stokes equations*, Math. Comp. **74** (2005), no. 250, 543–554, TODO verify issue number.
20. ———, *Divergence-free finite elements on tetrahedral grids for $k \geq 6$* , Math. Comp. **80** (2011), no. 274, 669–695, TODO verify issue number.

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